

## CHAPTER 8

# THE MAGNETOTELLURIC METHOD

*K. Vozoff\**

### SUMMARY

In the magnetotelluric (MT) method, natural electromagnetic fields are used to investigate the electrical conductivity structure of the earth. Natural sources of MT fields above about 1 Hz are thunderstorms worldwide, from which lightning radiates fields which propagate to great distances. At frequencies below 1 Hz, the bulk of the signal is due to current systems in the magnetosphere set up by solar activity.

In both cases the electromagnetic (EM) fields at the surface of the earth behave almost like plane waves, with most of their energy reflected but with a small amount propagating vertically downward into the earth. The amplitude, phase, and directional relationships between electric (**E**) and magnetic (**H** or **B**) fields on the surface depend on the distribution of electrical conductivity in the subsurface. By use of computed models, field measurement programs can be designed to study regions of interest within the earth from depths of a few tens of meters to the upper mantle.

Equipment to carry out the measurements consists of magnetometers for the frequency range of interest; pairs of electrodes separated by suitable spacings to sense the electric field variations; plus amplifiers, filters, and suitable digital recording and processing systems to permit the signals to be captured and analyzed. The magnetometers in particular must have very low noise and great stability because those signals are so weak.

Once the signals have been recorded they must be processed and analyzed. Processing is normally done in the frequency domain because the theory is simpler than in time domain. Hence processing begins with Fourier transformation, from which earth impedances to the incident waves as functions of frequency, direction, and position are computed. Processing in many

systems is now done in real time, while the signals are being acquired.

These computed impedances are next interpreted in terms of electrical conductivity versus position and depth. Numerical models for one-dimensional, two-dimensional, and three-dimensional structures are used for this last step. Interpretation is the most difficult part of the method because information is seldom complete and the models are never complex enough to represent a real earth. For that reason, and to use the MT data to best advantage, other available external data such as well logs, seismic, and any other electrical data are commonly utilized to help with interpretation.

The main shortcoming of the method is the difficulty of obtaining data in electrically noisy areas or where the surface is unstable.

The strengths of the method are its unique capability for exploration from very shallow depths to very great depths without artificial power sources and with little or no environmental impact. At high frequencies, audio frequency MT (AMT) has been used to map groundwater and major base metal deposits at depths from 50–100 m to several kilometers. However, the major application of MT is to petroleum exploration in areas where reflection seismology is very expensive or ineffective, such as in extreme terrain and beneath volcanics. Another major successful application has been to geothermal exploration.

Major research problems at the moment concern interpretation in areas of complex three-dimensional structure, and improvements in production rate.

### INTRODUCTION

This material provides seismologists and geologists with a review aimed at increasing understanding and

\*Centre for Geophysical Exploration Research, Macquarie University, Sydney, N.S.W. 2109, Australia.

effectiveness of MT procedures and with a list of sources where detailed information may be found. The review summarizes the past seven or eight years' research for the field geophysicist who has not had time to keep up, or does not see the more academic journals.

The enormous growth in the use of the MT method since the late 1960s has been a pleasant surprise. MT seems to have offered help of the right kind at the right time for many commercial purposes. As exploration was forced into more difficult areas, and with the volatility in resource prices, the importance of MT in conjunction with other techniques has tended to grow continuously. Its usefulness in poor seismic areas and its negligible environmental impact are integral parts of effective exploration at minimum cost.

With all the new data, problems emerged which had barely been anticipated. Fortunately some qualified research people became interested in solving these problems at about the same time. As a result, the literature must now be one to two orders of magnitude larger than in 1970. Included are new applications, new field data, new instruments and field computers, remote referencing and robust estimation, 3-D modeling and inversion, and a much broader understanding of both modeling and inversion (and their limitations) generally. Our knowledge of factors that control resistivity at depth improved, and we now better understand the complementary roles of natural and applied field methods. To cover these factors in one chapter is difficult; several books on the subject are now in the planning stages or in print.

My 1972 review (Vozoff, 1972) reflected the situation in 1970. The present update proved a much larger project because many of the most important developments are unpublished, and thus are not generally available. The Society of Exploration Geophysicists Magnetotellurics Reprint volume (Vozoff, 1986) was one attempt to help.

The following sections summarize the sources, the interaction of their electric and magnetic fields with the complex conducting earth, the equipment with which the measurements are made, field procedures, and data processing and interpretation. A list of references, by no means comprehensive, follows. Some important questions remain to be answered, particularly the proper and practical use of impedances in 3-D structural environments. In such cases the current state of development is indicated.

## SOURCES

The dependence of MT on natural fields is both its major attraction and its greatest weakness. Other EM methods require power supplies and current control

systems. For low frequency and deep penetration these methods can involve large truck-mounted installations and thus restrict access in many circumstances and at times lead to undesirable environmental damage. The geometry of their source fields also complicates interpretation as compared with MT. These sources do not easily provide the kind of tensor data for structural definition that MT does. The other side of this coin can be the cost and frustration of attempting to collect MT data in the presence of artificial EM noise or natural interference such as ground motion.

Because the amplitude and polarization of source fields is obviously important to the success of MT and AMT surveys, a general description is presented. Familiarity with the normal appearance of pulsations will often make recognition of data contaminated by noise possible. With the present generation of MT equipment, loss of survey time while waiting for signal to exceed system noise in the frequency ranges .001–.1 Hz and  $10^3$ – $10^4$  Hz is unusual. However, to get good data near 1 Hz can take a great deal of patience because of a persistent spectral energy low there. Another signal notch at 1–2 kHz can usually be ignored because it is so narrow. Likewise, since the introduction of digital acquisition, magnetic field polarization has not been a problem for tensor surveys below 1 Hz. These surveys, however, can still require special care at audio frequencies. This section gives an overview of signal sources and the factors that affect their intensity and polarization.

Aside from engineering considerations, a curiosity about the nature and sources of pulsations is developed after observing them for many long hours in the field. Examples of pulsations are shown in Figures 3, 8 and 12. Table 1 shows the classification system developed to describe the phenomenon. Natural EM signals come from an enormous variety of processes and from sources ranging from the core of the earth to distant galaxies. Within the frequency range of interest in exploration, say .001– $10^4$  Hz, only two source regions are important. These are the atmosphere and the magnetosphere. Electrical storms in the lower atmosphere are the dominant cause of fields between 1 Hz and 10 kHz, whereas below 1 Hz the fields originate primarily in hydromagnetic waves in the magnetosphere.

The magnetosphere, the region around the earth in which the main magnetic field is trapped by the solar wind (Figure 1), includes the ionosphere and the atmosphere (Rostoker, 1979) and contains gases, especially oxygen and nitrogen, decreasing in density with elevation. These gases are ionized by ultraviolet and other solar radiation, but below about 100 km the high pressure forces the ions to recombine rapidly so there are few of them. Above 100 km, charged particle

density increases rapidly up to about 250 km and then starts to decline again with decreasing pressure and particle density (Figure 2). This region of relatively high conductivity is called the ionosphere.

The ionosphere is an anisotropic electrical conductor, because the magnetic field exerts a force on any moving charge perpendicular to both the direction of charge motion and the magnetic field itself. For that reason an electric field in one direction may set up currents in the other directions (Parkinson, 1983).

One additional property that plays an important part is the existence of EM hydromagnetic waves, or plasma waves, which can only occur in conductive fluids containing magnetic fields. These waves travel at relatively low velocities and can set up resonances along magnetic field lines in the magnetosphere with frequencies in our range of interest (Rostoker, 1979).

This very complex system is constantly buffeted by a solar wind of energetic ions and weak magnetic fields, ejected from the sun. From the impact zone, where a permanent shock wave is maintained, to the far tail, the conductive solar wind sweeps turbulently past the earth, bottling up the magnetic field and injecting erratic bursts of its own ions and fields. Consequently the time behavior of magnetic pulsation fields is essentially chaotic but includes features localized in frequency and/or space as a result of particular local conditions. As well, the earth's main dipole field imposes a global pattern on the behavior. Figure 3 shows typical examples of ultra low frequency (ULF) signals in the electric and magnetic fields.

Long time averaging gives smooth spectra decreasing monotonically with increasing frequency (Figure 4). The level of these spectra will be greatest at times of high solar activity, with typical differences of 20 dB between quiet and active times, depending on frequency.

To reach the earth's surface from the magneto-

sphere, fields must travel through the ionosphere and then through the insulating atmosphere. Vertical **E** and **H** fields do not penetrate the ionosphere and thus the horizontal components are strongly modified (Hughes and Southwood, 1976a, b; Orr, 1984). Additional horizontal ionospheric currents are set up in this interaction and it appears that the pulsations seen at the surface are largely due directly to those currents.

Later sections show that the vertical component of the natural field is important to the interpretation of MT data. Vertical magnetic fields are normally very small, both because horizontal gradients of current density are small, and because induction in the earth itself tends to suppress them. Any vertical components reaching the surface are a result of restricted current systems associated with oscillations localized

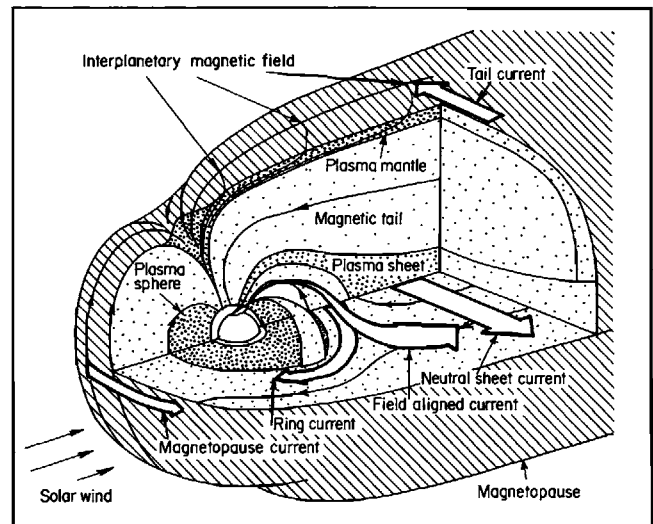


Fig. 1. Major features of the magnetospheric current systems. The earth and atmosphere are the small spherical features in the center. (From Potemra, 1984).

Table 1. Classification system used to describe pulsations.

| Name | Frequency  | Name   | Time Scale                                         |
|------|------------|--------|----------------------------------------------------|
| SHF  | 3–30 GHz   | Pc1    | <b>Period (s)</b><br>$2\pi/\omega = 0.2\text{--}5$ |
| UHF  | 0.3–3 GHz  | Pc2    | $2\pi/\omega = 5\text{--}10$                       |
| VHF  | 30–300 MHz | Pc3    | $2\pi/\omega = 10\text{--}45$                      |
| HF   | 3–30 MHz   | Pc4    | $2\pi/\omega = 45\text{--}150$                     |
| MF   | 0.3–3 MHz  | Pc5    | $2\pi/\omega = 150\text{--}600$                    |
| LF   | 30–300 kHz |        | <b>Rise Time (s)</b>                               |
| VLF  | 3–30 kHz   | Pi1    | $\tau_r = 1\text{--}40$                            |
| ELF  | 3–3000 Hz  | Pi2    | $\tau_r = 40\text{--}150$                          |
| ULF  | <3 Hz      | sc, si | $\tau_r \sim 300\text{--}$                         |

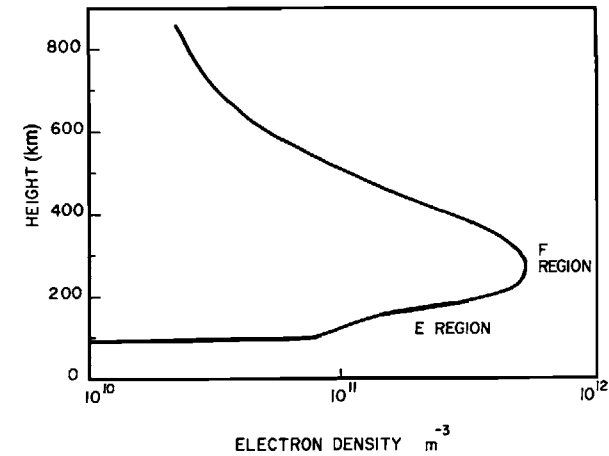
1 GHz =  $10^9$  Hz    1 MHz =  $10^6$  Hz    1 kHz =  $10^3$  Hz  
From Lanzerotti and Southwood (1979)

on particular field lines, as in Figure 5 (Samson and Rostoker, 1972; Southwood and Hughes, 1978). These vertical components are largely cancelled by induction in the earth, but model calculations for 50–60 s periods indicate that the primary vertical component of  $H$ ,  $H_z$  can be as large as 20 percent of horizontal  $H$  at the surface under such conditions. (This fact can confuse interpretation which depends on quantities related to  $H_z$ , and which assumes that  $H_z$  is zero except near a fault or similar feature.) A large vertical electric ( $E_z$ ) pulsation field is also predicted at the surface in these conditions, which are most common on the auroral zones, from 65 to 70 degrees magnetic latitude (Figure 6). It is also expected but not yet reported near the magnetic equator.

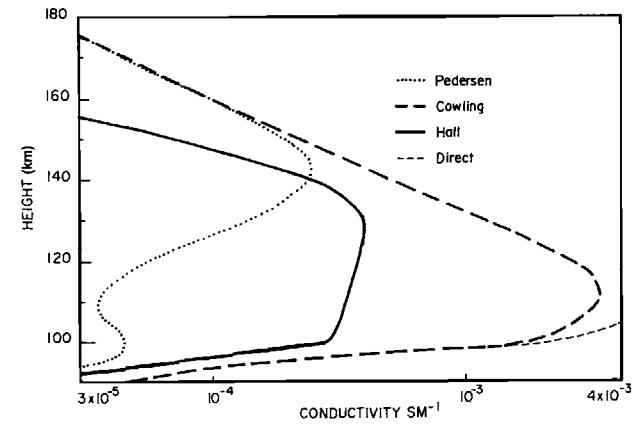
The division of natural EM signal sources near 1 Hz is not rigid. At times, in higher latitudes and remote from storm centers, magnetospheric processes con-

tribute significantly throughout the spectrum. These signals can be heard in audio outputs taken from antennas, and have been given descriptive names such as “dawn chorus,” “hiss,” “whistlers,” “saucers,” and “lions roar.” Their source regions are shown in Figure 7 (Shawan, 1979). At other times the division becomes a gap, especially at lower latitudes, as discussed later.

Polarization in the horizontal plane varies in an organized way with time of day and latitude near the auroral zone (Samson et al. 1971). If polarization is very strong, as is sometimes observed at audio fre-



2 (a)



2 (b)

Fig. 2. (a) Ionospheric electron density at winter noon, midlatitude. (b) Electrical conductivities in the anisotropic ionosphere, mid-day. (After Parkinson, 1983).

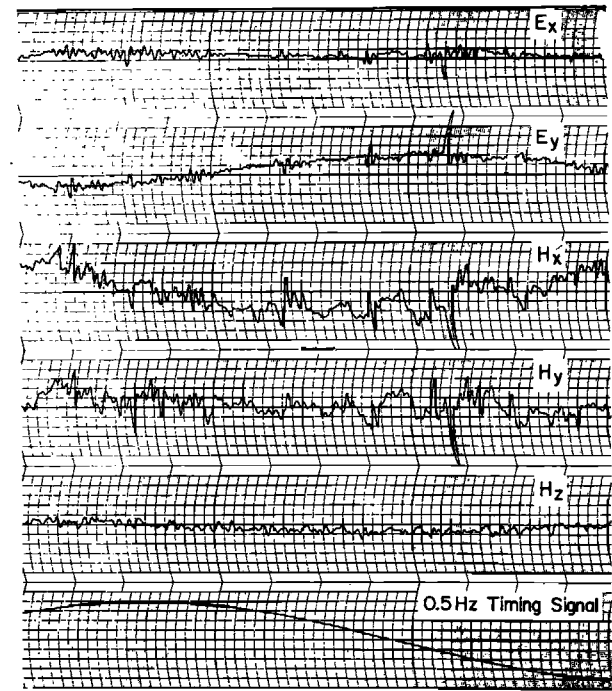


Fig. 3. Typical 5-channel ULF signal set. Sinusoidal time signal (bottom trace) has a period of 2 s.

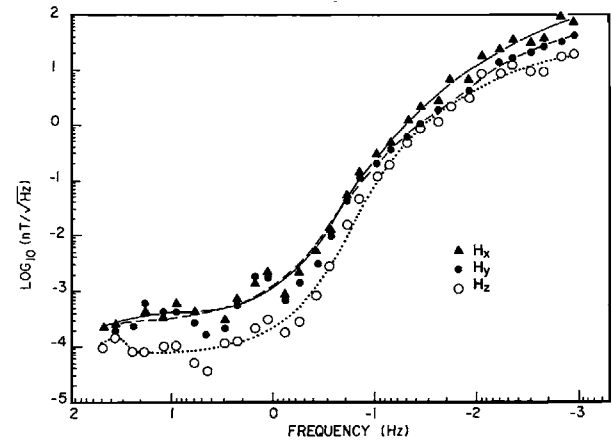


Fig. 4. Typical observed magnetic field amplitude spectrum.

quencies, it can in principle cause problems for MT tensor surveys. However there are no known reports of difficulties on this account at lower frequencies despite the extensive MT work carried out in western Canada, Alaska, and western Siberia. This may be because of the large dynamic range obtained in digital recording.

The Finnish Meteorological Institute has taken a special interest in effects of the electrojet on natural electromagnetic fields of the region. Häkkinen and Pirjola (1986), Pirjola and Häkkinen (1990), and Häkkinen et al. (1990) have shown theoretically that impedances near the electrojet differ significantly from those of plane waves. For their models this difference becomes important at periods greater than 60 s. The fields must be treated as if they are due to a line

source. The group are now applying their theory to data from the EISCAT magnetometer array in northern Scandinavia.

Much knowledge of pulsations has come from studies dealing with high latitudes in and near the auroral zone, yet many areas in which MT surveys are done are tropical or subtropical. Fields weaken very rapidly southward from the auroral zone. Typical results show changes of 20–40 dB in a few hundred kilometers, beyond which the decrease is much more gradual to near the magnetic equator. The few systematic mid-to-low latitude studies also indicate that field variations there have much less spatial structure. It is known that primary  $H_z$  is so small that it is difficult to distinguish it from the  $H_z$  induced by lateral conductivity changes.

The zone within 5 degrees each side of the magnetic equator is known to have special behavior because of the equatorial electrojet, a very intense eastward current on the day side of the earth. This current amplifies the northward component of slower magnetic variations. Hutton (1965) suggested and Carlo et al. (1982), Sarma et al. (1982), and Sastry et al. (1983) confirmed that pulsation amplitudes within the ULF band of MT signal would also be affected. In any case,  $H_z$  should be treated with caution when interpreting MT results from the geomagnetic auroral and equatorial regions.

Above 1 Hz the radiation and propagation of EM signals from lightning discharges explains the character of signal over most of the globe. Lightning gener-

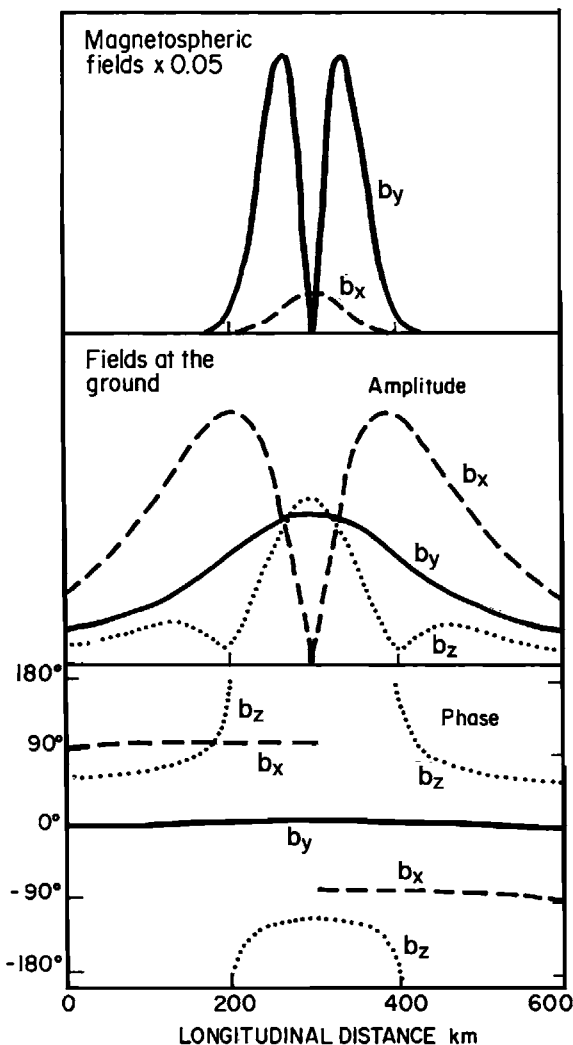


Fig. 5. Theoretical profiles of magnetic field components on the earth's surface from the magnetospheric fields shown at top. The surface fields include a vertical component. (After Southwood and Hughes, 1978.)

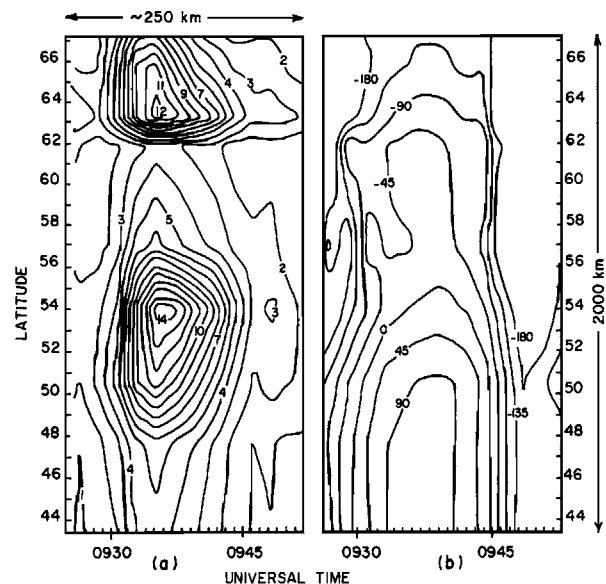


Fig. 6. Latitude-time variations of the  $H_x$  component of a 55 s. Pc4 event observed at ten observatories. (After Orr, 1984.)

ally consists of sequences of discharges called *strokes* (Golde, 1977; Pierce, 1977; Ogawa, 1982; Volland, 1982). The sequences, called *flashes*, include 3–4 strokes on average and last 200–300 ms. The first discharge of each stroke is the cloud-to-ground *leader*, which ionizes a channel for subsequent currents. This is followed by the *ground return* stroke which is the major radiator of EM energy. Typical ground return currents are 20 000 A lasting about 40  $\mu$ s and extending from the surface into the lower part of the responsible thundercloud (Ogawa, 1982).

Estimates of the occurrence frequency of lightning flashes around the world vary between 100 and 1000 per second. The field seen by the AMT system depends on the strengths, path lengths (cloud heights), occurrence frequencies, and distances of the discharges which take place during recording. Signals (called atmospherics or spherics), are therefore largest on summer afternoons in the tropics. The fields die off with distance, very much like those from vertical radio antennas. The vertical electric and transverse magnetic fields are

$$E_z(t) = \frac{M}{4\pi\epsilon_0 r^3} + \frac{dM/dt}{4\pi\epsilon_0 cr^2} + \frac{d^2M/dt^2}{4\pi\epsilon_0 c^2 r} \quad (1a)$$

and

$$B_\phi(t) = \frac{\mu_0(dM/dt)}{4\pi r^2} + \frac{\mu_0(d^2M/dt^2)}{4\pi cr} \quad (1b)$$

where

$c$  = velocity of light

$= 3 \times 10^8$  m/s

$\mu_0$  = permeability of free space

$= 4\pi \times 10^{-7}$  H/m

$\epsilon_0$  = permittivity of free space

$= 8.87 \times 10^{-12}$  F/m.

$r$  is distance from the discharge in meters,  $dM/dt$  is the electric dipole moment in ampere-meters, and  $M$  is thus the net moment of electric charge in coulomb-meters. The vertical electric field  $E$  is in volts per meter, and the transverse magnetic induction  $B$  is in webers per meter.

$M$  is a rapidly time-varying function. The expressions for  $E$  and  $B$  are those of a vertical electric dipole,

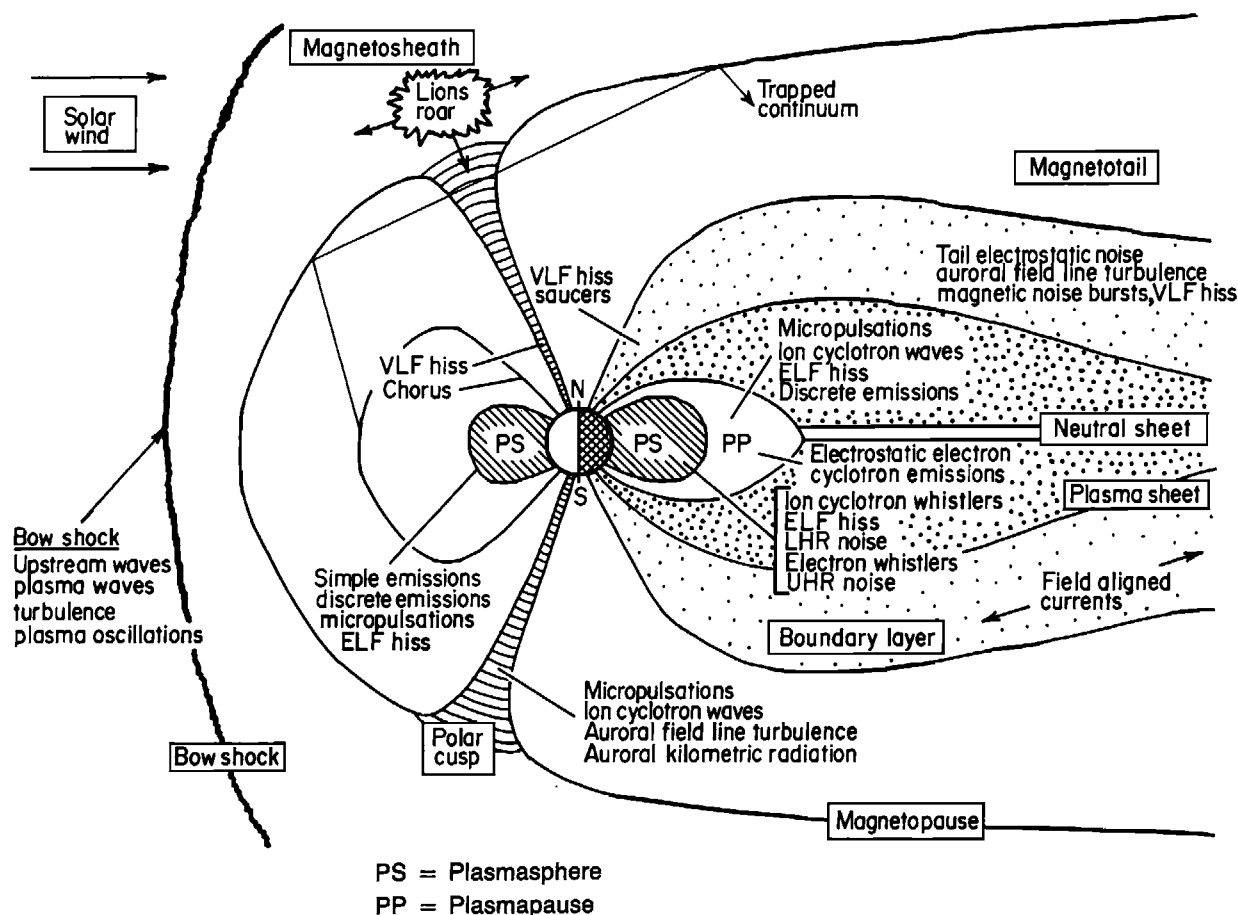


Fig. 7. Source regions in the magnetosphere for the various types of observed atmospherics. (After Shawan, 1979.)

valid beyond a few kilometers out to about 50 km. Beyond this the spatial decay rate depends on frequency. That is, the field behaves approximately as a linear system

$$A(f) = S(f) \times P(r, f) \quad (2)$$

where  $f$  is frequency,  $S(f)$  represents the spectrum of the source, and  $P(r, f)$  is the effect of propagation to distance  $r$ . The general behavior of the signal in various frequency bands at 50 km from the discharge is shown in Figure 8. Figure 9 illustrates how the broadband fields change with distance (note the expanding vertical scale with distance).

These fields can have strong linear polarization if they arrive from a single dominant storm center. Since this polarization can make the MT tensor impedance calculations unstable, special care is sometimes necessary to ensure that signals are captured from more than one source.

Some very large spherics are observed to travel more than once around the earth which suggests that the energy is somehow guided by the earth's surface, since EM waves would normally be expected to travel in straight lines. In fact the insulating shell of air between conductive earth and conductive ionosphere forms a waveguide, partially trapping the fields. Like most waveguides, its response depends on frequency in a way determined by its size, shape, and the nature of its boundaries.

The efficiency with which energy is trapped depends on the sharpness and regularity of the boundaries. In the earth-ionosphere guide the inner surface is a sphere whose surface resistivities range from less than  $0.25 \Omega \cdot \text{m}$  in the oceans to greater than  $10^4 \Omega \cdot \text{m}$  in

extensive shield and permafrost areas. At the ionosphere the resistivity gradually decreases with increasing height over a range of 30 km to a low of about  $500 \Omega \cdot \text{m}$ , depending on local time and solar activity. The base is 20 km higher on the day side than on the night side of the earth. Thus the cavity is not an ideal spherical shell, and its response characteristics are irregular in time and space.

In the majority of arrivals there is little energy around 2 kHz. The exceptions are those whose sources are within a few hundred kilometers. This is clear from the shape of  $P(r, f)$  in Figure 10 which shows very high attenuation ( $>30 \text{ dB}/1000 \text{ km}$ ) in this part of the spectrum, especially in daytime. The reason is that the dominant mode of propagation changes, with the lowest order mode dying out and the next mode having a lower frequency limit near 2 kHz. Phase velocity (Figure 11) is greater for the higher mode, and atmospherics from distant sources show clear dispersion (Figure 12). The spectral gap also shows very obviously in the Fourier transforms of such arrivals.

The earth-ionosphere cavity also can resonate. The fundamental frequency of these "Schumann resonances" is near 8 Hz, with higher modes near 14, 20, and 26

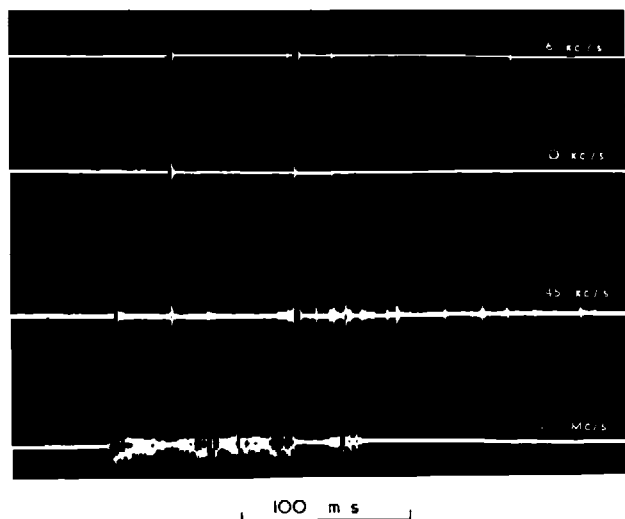


Fig. 8. Atmospherics from a near source as seen through four different band-pass filters. (From Volland, 1982.)

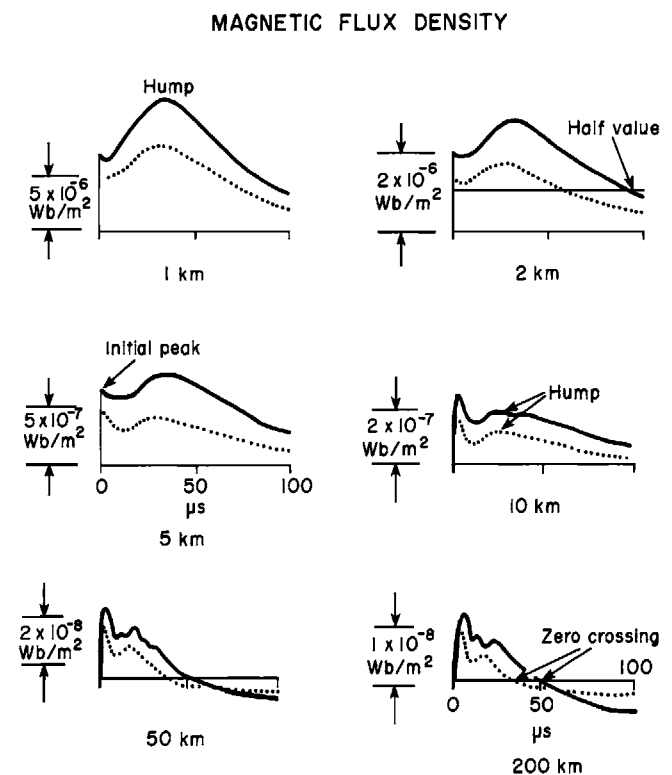


Fig. 9. Spatial variation of the magnetic transients from a discharge. The solid line is from the first discharge; the dotted line is from subsequent discharges. (After Volland, 1982.)

Hz (Polk, 1982). Their  $Q$  is only about 3, but they define the upper limit of the region of weak signal centered around 1 Hz.

Few plots of true spectral density at audio frequencies appear in the literature. Labson et al. (1985) include one collected near Berkeley, California (magnetic latitude 45 degrees) in mid-summer (Figure 13). This is consistent with the lower frequency spectra shown previously. Labson et al. also provide synoptic spectra consisting of two minute averages taken hourly over a 24-hour period in summer (Figure 14) and spot values at 20, 40, 250, and 500 Hz taken at monthly intervals (Figure 15). The latter show a winter minimum which becomes more pronounced as frequency increases. This is compatible with the range in average values between summer and winter for north-

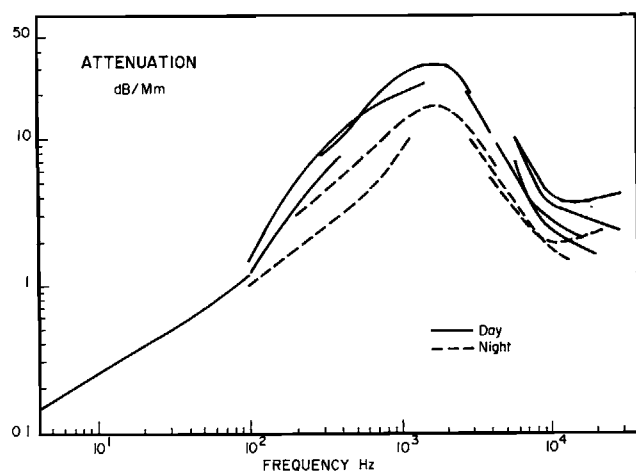


Fig. 10. Attenuation of spherics versus frequency, in dB/1000 km. Peak attenuation is responsible for the weak signal in the 1–2 kHz band. (After Volland, 1982.)

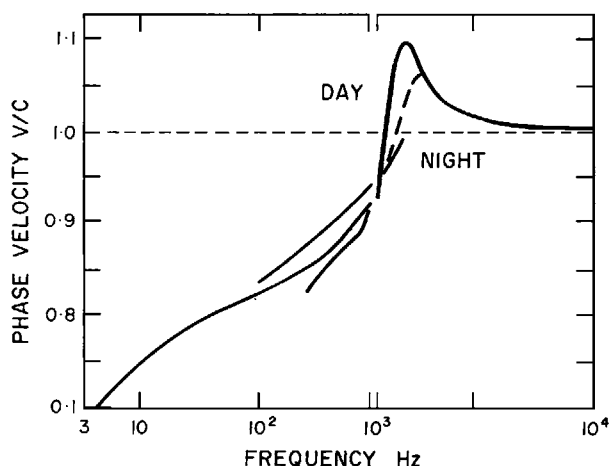


Fig. 11. Normalized phase velocity associated with the attenuation of Figure 10. This describes the frequency dispersion in the spherics. (After Volland, 1982.)

ern California at 10 kHz shown in CCIR Report 322 (CCIR, 1964). Figure 16, from Watt (1967), gives three monthly thunderstorm occurrence frequency for September–November. Figures 17 and 18 (also from Watt, 1967), show two extremes of expected 10 kHz atmospheric noise in  $E_z$  from 0400–0800 GMT in September–November and from 0800–1200 GMT in June–August.

Envelopes of spectral amplitude density for the MT frequency range are presented in Figure 19. They were compiled from many sources, including Clerc and Gilbert (1964), Labson et al. (1985), and numerous unpublished reports and brochures. They are “samples of opportunity” whose statistics are unknown. Except around 3 kHz, the range is in quite good agreement with a similar, after normalization, drawing given in Spaulding (1982, Figure 3). For prediction purposes, an envelope of the electric field spectra can be derived from these spectra for any desired apparent resistivity curve using equation (5).

A full description of spherics amplitudes requires their statistical amplitude distribution, in addition to the averages provided. CCIR (1964), Watt (1967), Volland (1982), and Spaulding (1982) present those statistics and discuss their use for radio communication engineering.

## INTERACTION WITH THE EARTH

### Uniform Earth

To see why the MT fields carry information about the structure of electrical resistivity in the earth we have to look at the way the fields and the earth interact. Here I refer to resistivity structure in the same way as one refers to geologic structure: it is “the general disposition, attitude, arrangement, or relative position” of the conductivity zones of an area (Sheriff, 1984). The relationships between resistivity and petrography are described in Volume I, and in another recent review by Parkhomenko (1982). Keller (1982) provides extensive tables of the electrical properties of rocks and minerals under a wide variety of conditions. Special attention should be drawn to the recent developments in the fundamental theory relating porosity and permeability (Katz and Thompson, 1987). These developments begin to explain Archie’s well-known empirical relationships between porosity and conductivity as well as the apparent absence of permeability from those relationships.

This section describes qualitatively the interaction of the fields with the earth, in order to develop an intuitive appreciation of the processes involved. This description is followed by a more detailed description for some basic structural models. It is assumed throughout that the magnetic permeability of the earth

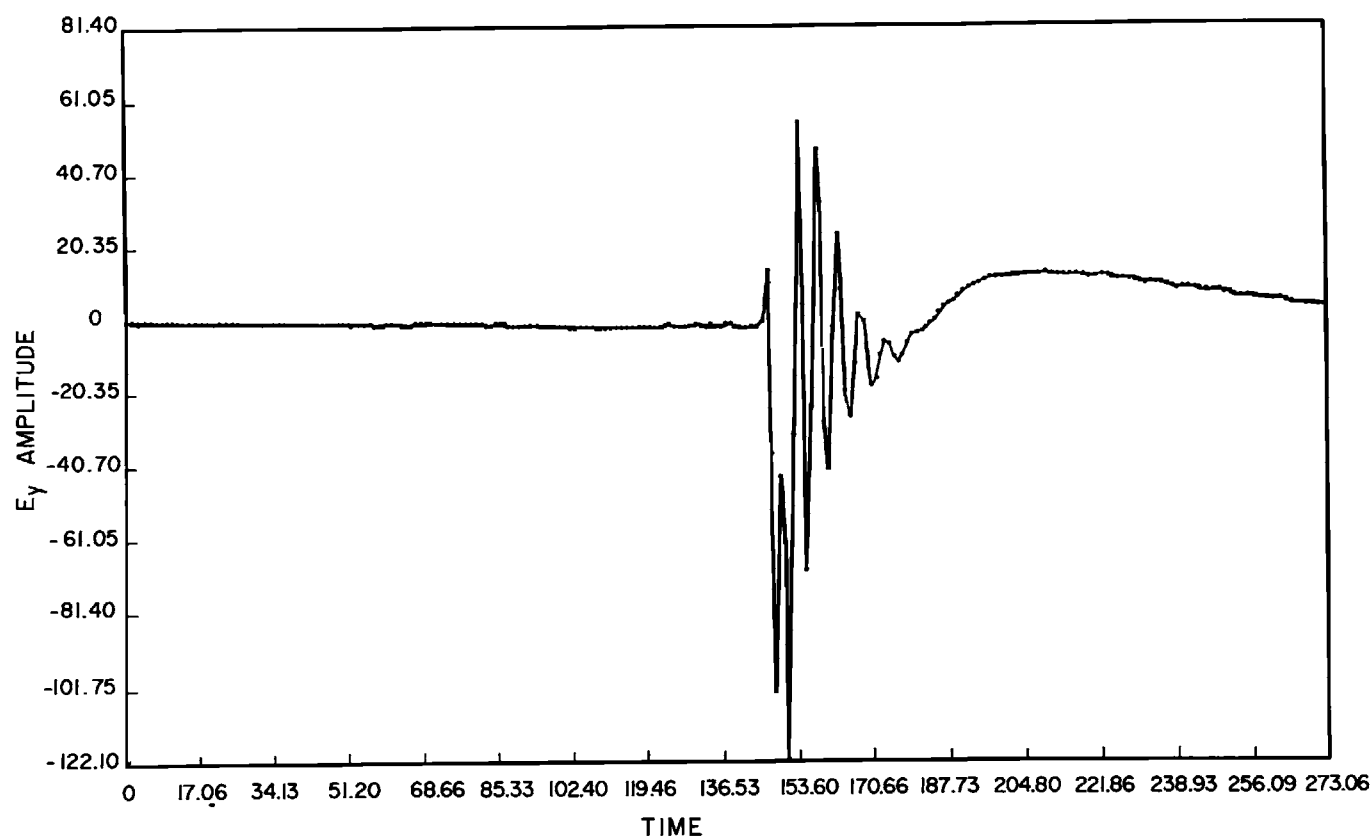


Fig. 12. An observed atmospheric showing dispersion.

is  $\mu = \mu_0 = 4\pi \times 10^{-7}$  H/m and that time variation is as  $e^{i\omega t}$ , where angular frequency  $\omega = 2\pi f$ . It is also assumed that displacement currents can be neglected everywhere in the earth.

The configuration of the EM fields above the surface is slightly different for spherics than for ULF pulsations: the former propagate horizontally while the

latter are assumed to propagate vertically. Because of the very large resistivity contrast between the air and the earth, both signals behave in the same way at the surface and beneath it. They diffuse downward within (at most) 1–2 degrees of vertical, the changing horizontal magnetic field inducing a changing horizontal electric field at right angles, through Faraday's law. The electric field in the conducting earth drives the telluric currents. Excellent general theoretic treatments of the process are found in Stratton (1941) and

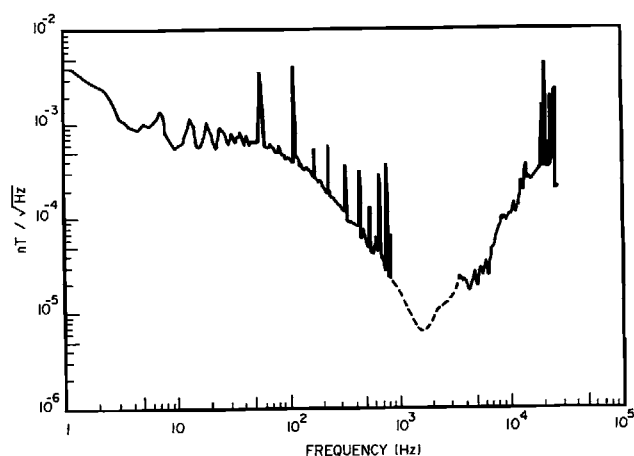


Fig. 13. An observed spectral density, 1 Hz to 30 kHz. (After Labson et al., 1985.)

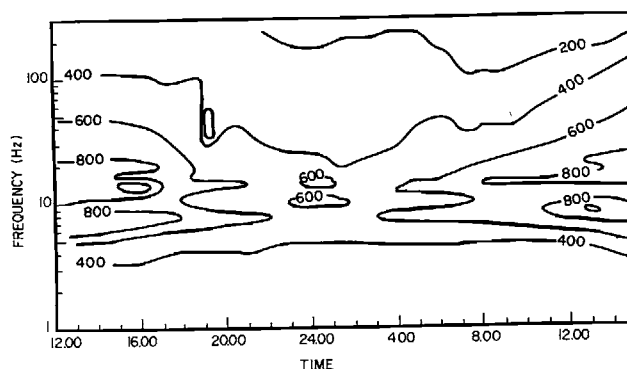


Fig. 14. Variation over a day in audio frequency magnetic field spectral density. (After Labson et al., 1985.)

in Jackson (1975), but some aspects are described here. The fields are assumed uniform enough that they behave like plane waves above and within the earth.

Spherics fields are trapped within the earth-ionosphere waveguide because at each boundary they are almost completely reflected back into it. However, a little energy is lost into the interface on each reflection

and this energy is refracted according to Snell's law, in the same way light is refracted at a velocity boundary. The velocity of EM waves in the conducting earth is far lower than in the near-dielectric air, so that (Figure 20)

$$\sin \theta_1 / \sin \theta_2 = v_1 / v_2 \gg 1. \tag{3}$$

The situation differs from the familiar optics case in that the earth rapidly absorbs the energy and the telluric currents convert it to heat through ohmic ( $I^2R$ ) losses.

The structure of the spherics field is interesting. Just above the surface (Figure 21) it is a transverse magnetic (TM) wave.  $E$  is nearly vertical and  $B$  is horizontal, transverse to the propagation direction, as described in the previous section. The leakage of energy into the surface means the Poynting vector  $S = E \times B$  must have a small real component vertically downward which forces  $E$  to have a small horizontal component in the propagation direction. Hence the total  $E$  vector at the surface has a slight forward tilt in the direction of travel. The size of the horizontal  $E$  component (more correctly, the ratio  $E/H = Z$ ), is determined by resistivity structure and frequency.  $Z$  is

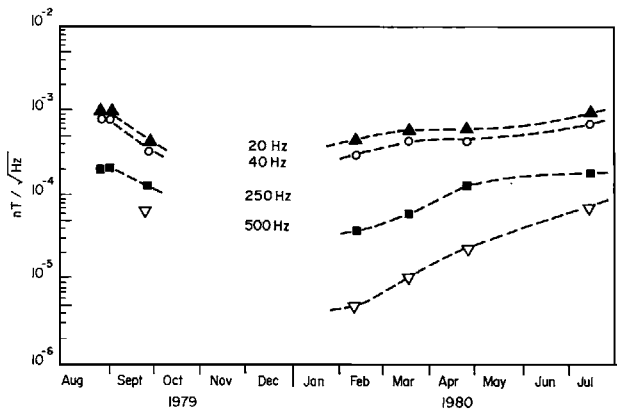


Fig. 15. Variation over a year in magnetic field spectral density at four frequencies. (After Labson et al., 1985.)

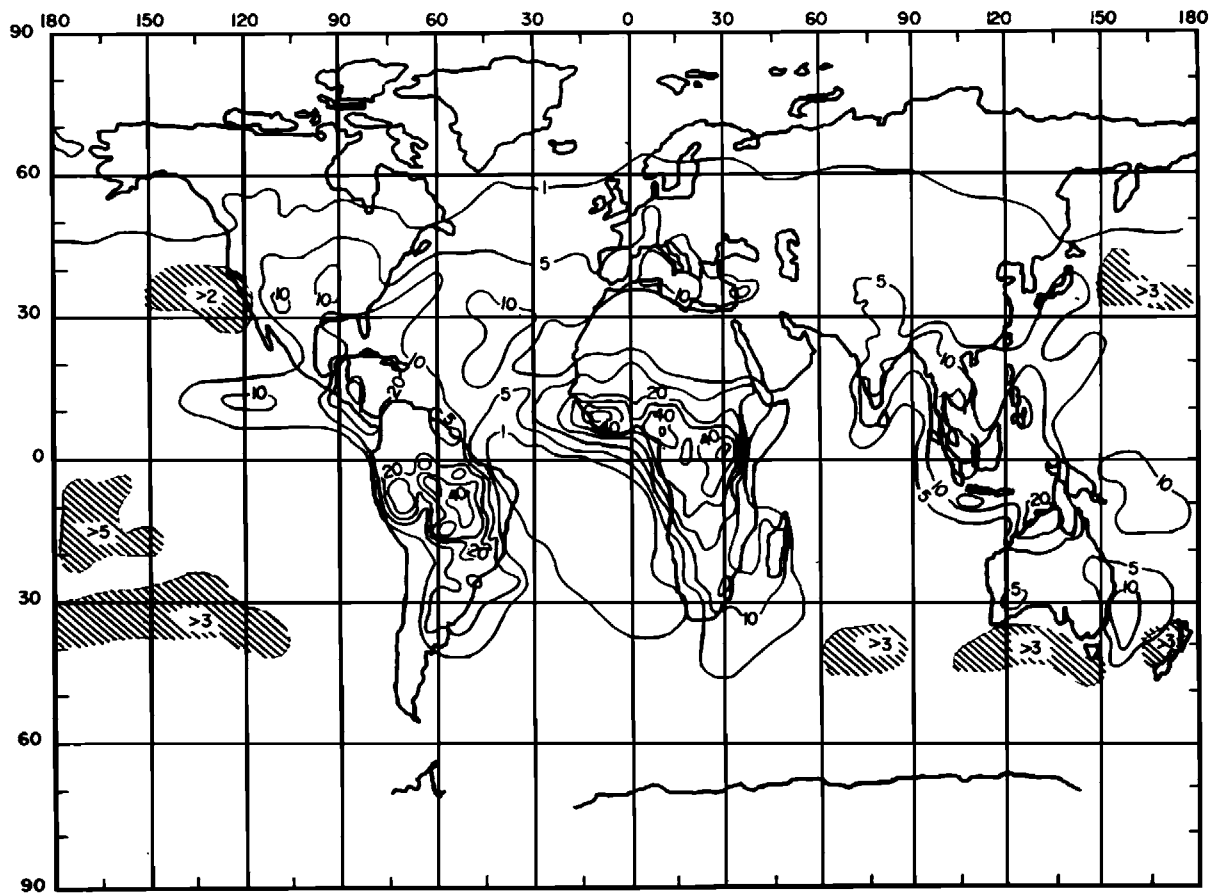


Fig. 16. Frequency of occurrence of thunderstorms, September–October. (After Watt, 1967.)

called impedance. Fields from very low frequency (VLF) transmitters have this same configuration. At least two other, slightly different, impedance definitions,  $E/B$  and  $c = Z/i\mu_0\omega$ , are used by various authors. Their use really makes no difference as long as usage is consistent throughout.

To extract the resistivity information from the fields it is necessary to look at the relations at the surface between the vectors  $\mathbf{E}$  and  $\mathbf{H}$  and the resistivity. This requires the assumption that the fields are plane waves, which is very nearly always in agreement with observation. At each angular frequency  $\omega$ , in (or at the surface of) a *uniform* half-space of conductivity  $\sigma$  (Stratton, 1941, p. 490).

$$\mathbf{H} = \frac{k}{\mu\omega} \mathbf{n} \times \mathbf{E}, \quad (4)$$

where

$$\begin{aligned} k &= (1 - i)\alpha \\ \alpha &= \sqrt{\omega\mu\sigma/2} \\ &= 1/\delta, \end{aligned}$$

$\delta$  is the skin depth in meters,  $k$  is the propagation constant, and  $\mathbf{n}$  is the unit vector pointing vertically downward.

The ratio  $E_x/H_y$  at the surface (the subscripts indicate horizontal cartesian coordinates) is particularly important viz

$$\begin{aligned} E_x/H_y &= \omega\mu/k \\ &= (1 + i)(\omega\mu/2\sigma)^{1/2}. \end{aligned} \quad (5)$$

Frequency will be accurately known since relative time must be precisely maintained while acquiring data. Therefore equation (5) shows directly the relationships between the measured fields and the conductivity. In particular, the ratio of  $E_x$  to  $H_y$  is proportional to  $\sqrt{\rho}$ , where  $\rho = 1/\sigma$ . Setting  $E_x/H_y = Z_{xy}$  we get

$$Z_{xy} = \frac{\omega\mu}{k} = (1 + i)\sqrt{\frac{\rho\omega\mu}{2}} \quad (6)$$

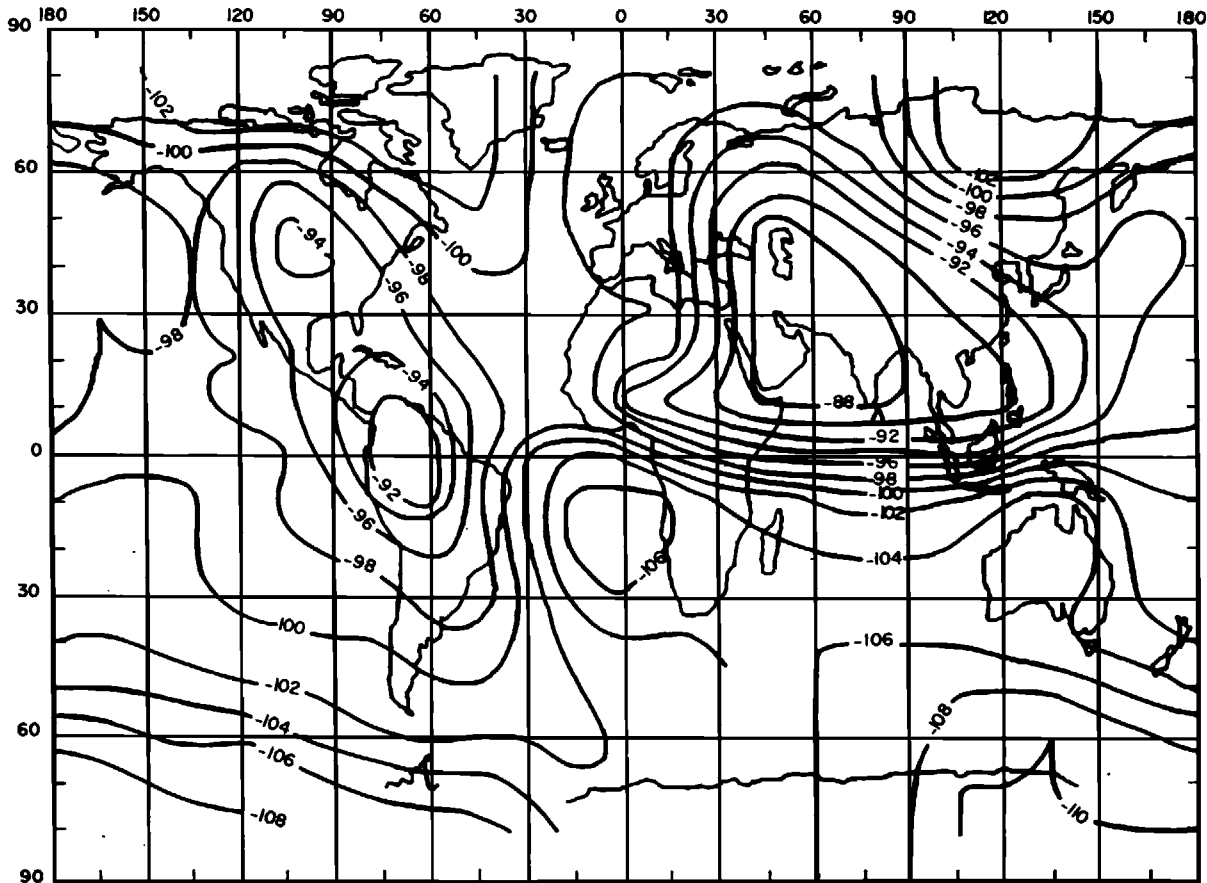


Fig. 17. Expected atmospheric vertical electric field noise at 10 kHz, in dB relative to  $1 \text{ Vm}^{-1}/\sqrt{H_z}$ , 0400–0800 GMT, September–November. (After Watt, 1967.)

or, solving for  $\rho$ ,

$$\rho_{xy} = \frac{Z_{xy} Z_{xy}^*}{\mu \omega} \quad (7)$$

where  $Z^*$  is the complex conjugate of  $Z$ . The phase,  $\phi$ , of  $Z$  is the difference between the phases of  $E$  and  $H$ . From the equation for  $Z_{xy}$  we see that  $E_x$  must lead  $H_y$  in phase by 45 degrees in this (uniform half-space) model.

Equation (7) is usually written

$$\rho_{xy} = \frac{1}{\mu \omega} \left| \frac{E_x}{H_y} \right|^2, \quad (8)$$

and  $\rho$  is given the phase of  $Z$ . Converting  $E_x$  and  $H_y$  to common units (millivolts per kilometer and gammas or nanoteslas, respectively), and looking only at amplitude, then

$$\frac{1}{5f} \left| \frac{E_x}{H_y} \right|^2 \approx \rho_{xy}. \quad (9)$$

Results will be the same at all frequencies and for  $|E_y|/|H_x|$  in this model, but in that case we have to use

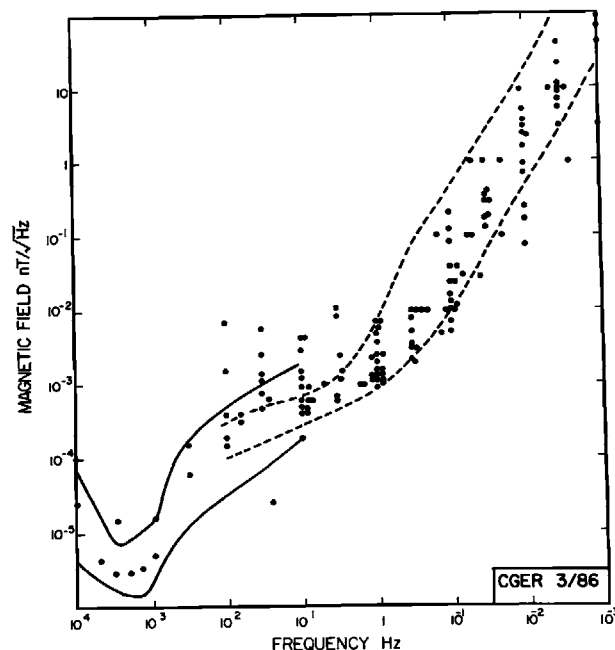


Fig. 19. Envelope of observed horizontal component magnetic field spectral densities.

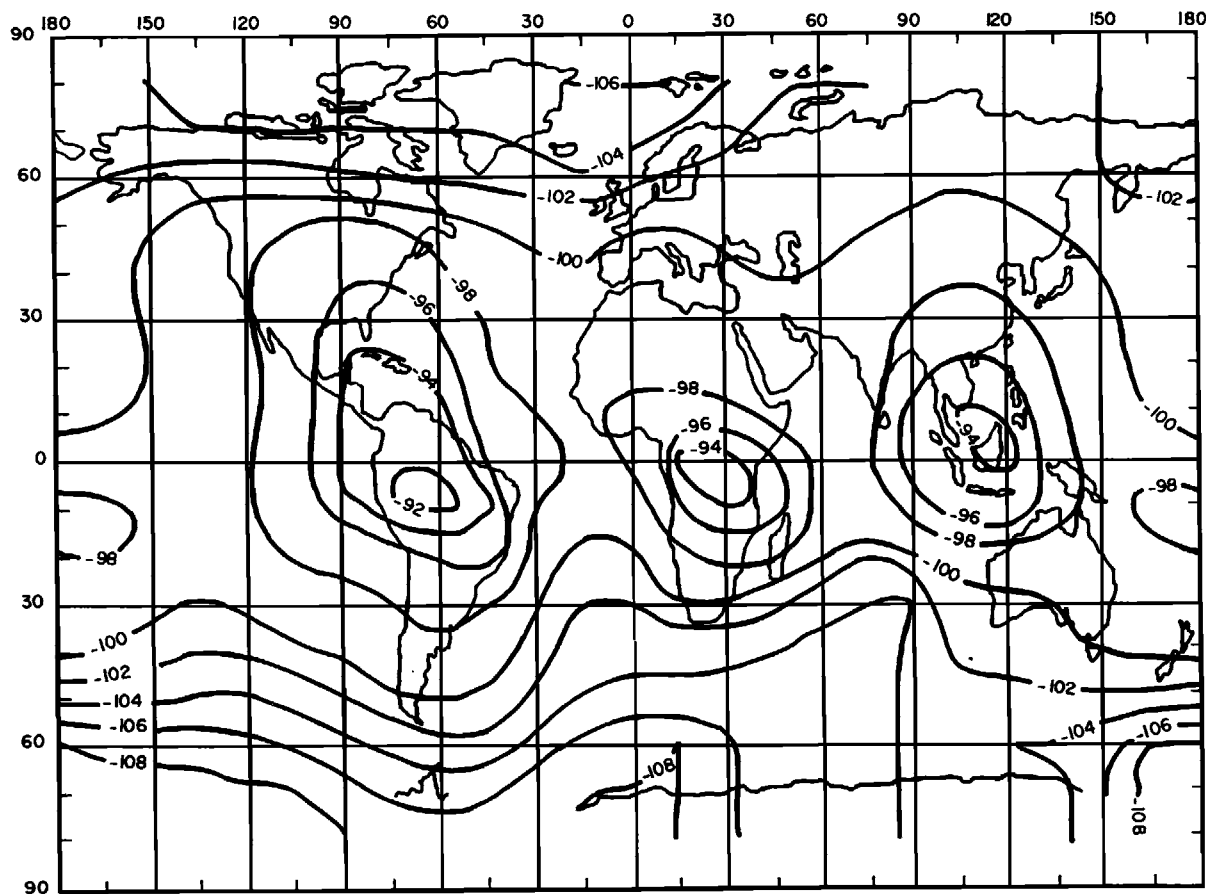


Fig. 18. As Figure 17 but for 0800–1200 GMT, June–August. (After Watt, 1967.)

$(-H_x)$  in order to keep  $\mathbf{S} = \mathbf{E} \times \mathbf{H}$  in the right direction. Thus the phase of  $Z_{yx}$  differs by 180 degrees from the phase of  $Z_{xy}$  in this model. Strictly speaking, there should be a  $B_y$  rather than the  $H_y$  in the denominator of equation (9), but the equation is rarely written that way in practice. As long as the numbers used are in the correct units (nanoteslas, which are units of  $\mathbf{B}$ ), the correct values of resistivity will be obtained.

Two different phase ranges are commonly used, depending upon whether the time variation is  $e^{+i\omega t}$  or  $e^{-i\omega t}$ . The choice is arbitrary, and can make no difference to the physics. If  $\phi_{ij}$  is the phase of  $Z_{ij}$ , then

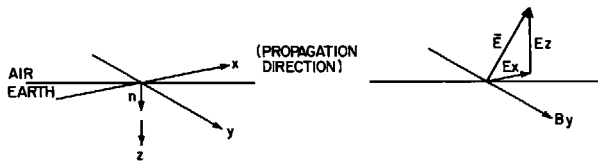
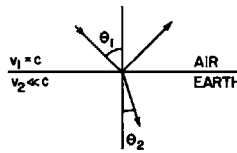


Fig. 20. Refraction of a plane EM wave incident on the earth's surface.

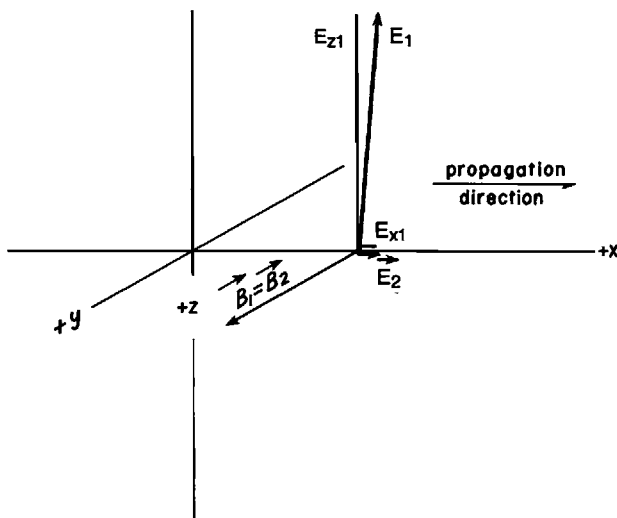


Fig. 21. Fields of a spheric propagating in the  $+x$  direction. The earth's surface is  $z = 0$ . The total electric field above the surface is nearly vertical but has a small tilt in the direction of propagation.

with the plus sign,  $\phi_{xy}$  is in the first quadrant ( $0^\circ$  to  $+90^\circ$ ), and with the minus sign (as in Figure 50) it will be in the fourth quadrant ( $-90^\circ$  to  $0^\circ$ ). In each case,  $\phi_{yx}$  is two quadrants away. The plus sign convention is used in this chapter. In practice, both phases in field results are sometimes presented in the range ( $0^\circ$  to  $90^\circ$ ).

In order to satisfy Maxwell's equations, within the earth  $\mathbf{E}$  and  $\mathbf{B}$  vary as

$$\begin{aligned} A &= A_0 e^{-i(kz - \omega t)} \\ &= A_0 e^{i\omega t} e^{-i\alpha z} e^{-\alpha z} \end{aligned}$$

where  $A_0$  is the surface value. That is, the fields vary as the product of four terms:

1.  $e^{i\omega t}$ , a sinusoidal time variation,
2.  $e^{-i\alpha z}$ , a sinusoidal depth variation,
3.  $e^{-\alpha z}$ , an exponential decay with depth, and
4.  $A_0$ .

Figure 22 shows the  $z$ -dependences of the  $\mathbf{E}$  fields near the interface. From the third term, the amplitudes at  $z = \delta = 1/\alpha$  are  $1/e$  of their surface values, which is why  $\delta$  is called the skin depth. A plot of skin depth versus frequency and resistivity can be found in the Zonge and Hughes CSAMT chapter (this volume). A useful approximation is given by

$$\delta \approx 500 \sqrt{\rho/f} \text{ meters.}$$

### Horizontal Layers

In more complicated models, e.g., horizontal layers or 2-D or 3-D structures, the relations between the  $\mathbf{E}$  and  $\mathbf{H}$  fields also become more involved. In horizontal layers some energy is reflected at each interface, and internal reflection occurs within each layer. The expressions for  $\mathbf{E}$  and  $\mathbf{H}$  include two terms in each layer, of the form

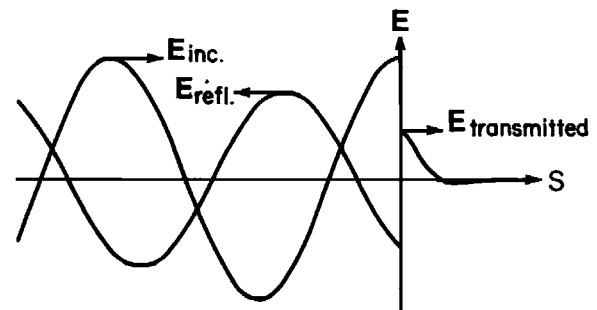


Fig. 22. A "snapshot" of the  $\mathbf{E}$  field of a plane EM wave vertically incident on the earth's surface. Most of the energy is reflected. The small part that is transmitted attenuates exponentially with depth.  $\mathbf{S}$  represents the propagating energy (Poynting vector) into the earth.

$Ae^{+ikz} + Be^{-ikz},$

one for up-going and the other for down-going energy. The fields remain horizontal and at right angles to one another unless there is anisotropy in the horizontal plane. We can still get  $E_x$  and  $H_y$  (or  $E_y$  and  $H_x$ ) on the surface and do the resistivity calculation, but the result depends on frequency and is now an *apparent resistivity*  $\rho_a(f)$ .

As a specific indication of the intuitive behavior of  $\rho_a$ , consider two 2-layer cases (Figure 23). The top layer is the same in both models, but  $\rho_2 = 10\rho_1$  in the one case and  $\rho_2 = (1/10)\rho_1$  in the other. At high frequencies such that the skin depth in the first layer is much less than its thickness,  $\rho_a = \rho_1$  in both cases. As frequency decreases and skin depth increases to the point that it is much greater than  $d$ ,  $\rho_a$  goes to 0.1 in one case and to 10 in the other, and stays there as frequency goes to zero. Note the small undershoot/overshoot in  $\rho_a$  and  $\phi$  as frequency decreases. Also note the phase response occurs at higher frequencies than the apparent resistivity response. Phase is asymptotic to 45 degrees at both high and low frequencies for a finite number of uniform layers.

If thickness varies but  $\rho_1$  and  $\rho_2$  are the same

(Figure 24), then frequency has to go to a lower value in the thicker layer model for the skin depth to exceed thickness and the second layer to become important.

In a three-layer model,  $\rho_a$  is asymptotic to  $\rho_1$  at high frequency and to  $\rho_3$  at low frequency. In between it approaches  $\rho_2$ . How close it gets depends on the thicknesses and resistivities of both layers 1 and 2 (Figure 25). Obviously our ability to resolve several layers depends on their resistivities and thicknesses, on the range of frequencies we record, and on the scatter in the points on the curve.

If  $\rho_a$  is plotted on a log-log scale, its phase is proportional to the slope of the curve but from a baseline at  $-45$  degrees. The relationship is (e.g., Parker, 1983)

$$\phi(f) = \frac{\pi}{4} - \frac{f}{\pi} \int_0^\infty \ln \left( \frac{\rho_a(f')}{\rho_1} \right) \frac{df'}{f'^2 - f^2}.$$

Again,  $\rho_1$  is the high-frequency asymptotic value of  $\rho_a(f)$ . Weidelt (1972), Kunetz (1972), and others show that, as a crude approximation,

$$\phi(f) \approx \frac{\pi}{4} \left[ 1 + \frac{\partial \log \rho_a(f)}{\partial \log f} \right]. \tag{10}$$

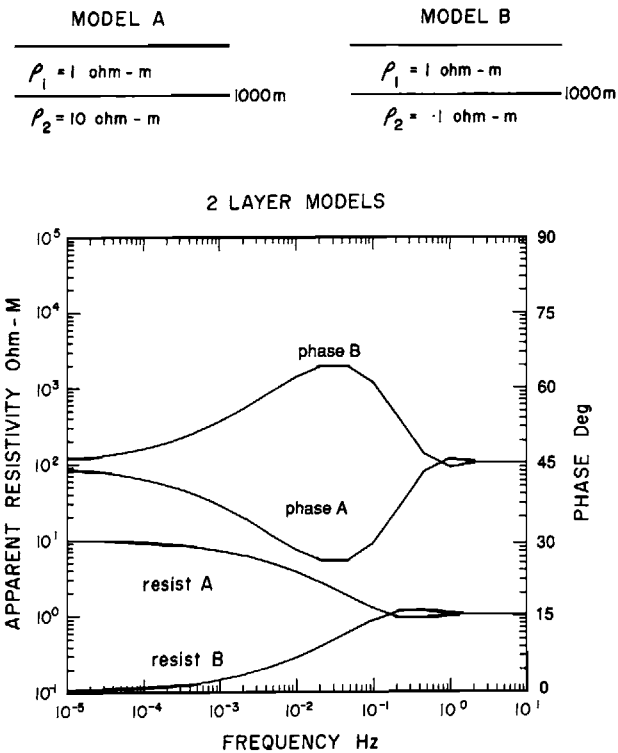


Fig. 23. MT apparent resistivity and phase responses of two-layer models. Model A—resistive basement. Model B—conductive basement.

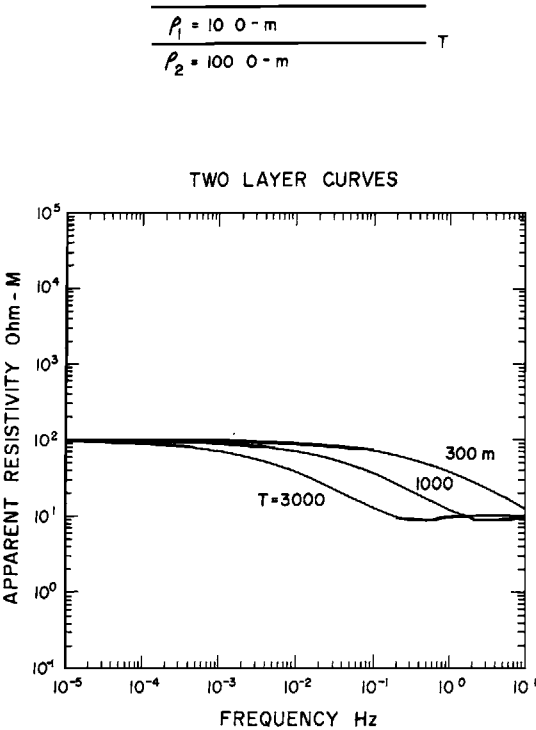


Fig. 24. Changes in MT response with layer thickness  $T$  for a two-layer model.

These are called dispersion relationships. Figure 26, from Weidelt (1972), shows simple examples where the approximation is in error by 10 to 15 degrees.

The net result is that, as frequency decreases, phase anticipates the behavior of  $\rho_a$ . Within a finite frequency range, the phase shows features beyond the reach of the resistivity, so there is an advantage to interpreting them together. For the same reason, features which are too shallow for  $\phi$  may still appear in the highest frequency  $\rho_a$ . The other facet of this behavior is that, as long as  $\rho_a$  is asymptotic to some constant value at low frequency, the phase always returns to  $\pi/4$ . This is different from the apparent resistivity, which can be affected over a very broad frequency range by a superficial irregularity. The importance of this property of the phase is discussed later in this section and in the Interpretation Section.

The terms "penetration depth" and "investigation depth," instead of skin depth, are sometimes used in depth-variable media. These terms have a variety of definitions, as discussed in Spies (1989).

### Anisotropy

In anisotropic media, conductivity varies with the direction of the applied electric field  $\mathbf{E}$ . Conduction in

almost any bedded sediment is easier along the bedding planes than across them: currents can follow the most conductive bands without crossing the resistive ones, and the rock is thus anisotropic. Rocks under stress are also known to become anisotropic. If  $\mathbf{E}$  is applied along the bedding one value of current density  $\mathbf{j}$  (and therefore of  $\sigma = \mathbf{j}/\mathbf{E}$ ) will result, but if  $\mathbf{E}$  is applied across the bedding a smaller value of  $\sigma$  will be indicated. (Units of current density  $\mathbf{j}$  are  $\text{A}/\text{m}^2$ ). If  $\mathbf{E}$  is applied in a direction which is neither parallel nor perpendicular to bedding, then  $\mathbf{j}$  will generally have components in both directions and will not be parallel to the applied  $\mathbf{E}$ .

To some extent all rocks are anisotropic, as can be seen from well logs. Then in three dimensions

$$\mathbf{j} = \sigma \mathbf{E} \quad (11)$$

or

$$\begin{pmatrix} j_x \\ j_y \\ j_z \end{pmatrix} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} \begin{pmatrix} E_x \\ E_y \\ E_z \end{pmatrix} \quad (12)$$

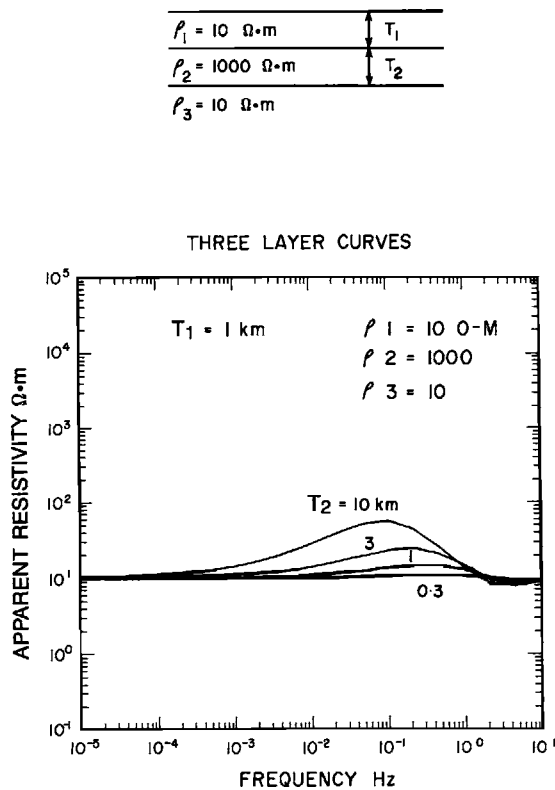


Fig. 25. Changes in MT response with second layer thickness in a three-layer model.

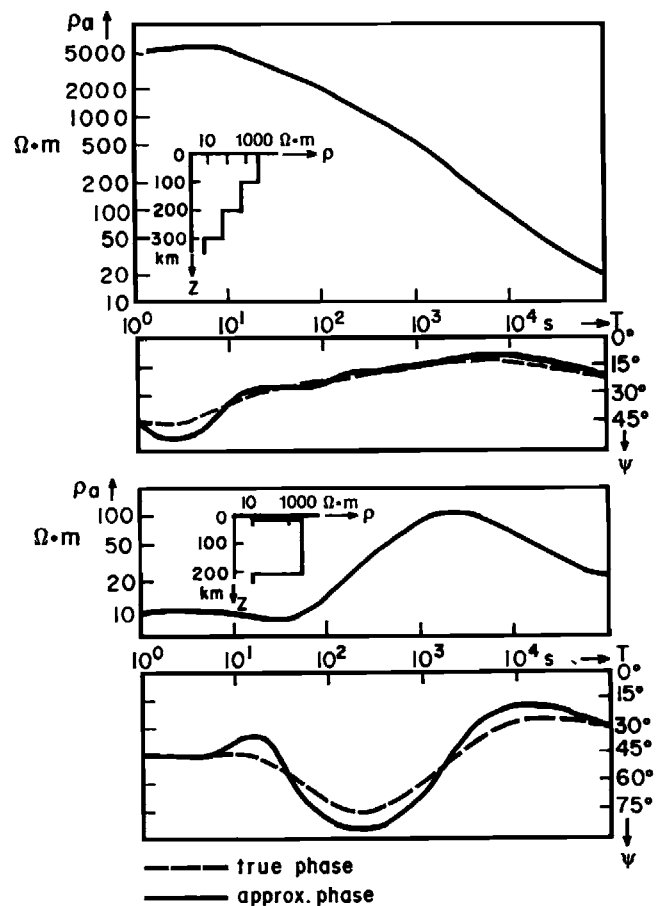


Fig. 26. Phase errors in the approximation to the Hilbert transform (After Weidelt, 1972).

is the general situation. If, for example, we apply a field  $E = E_x$  (with  $H = H_y$ ), and  $E_y = E_z = 0$ , then

$$\begin{aligned} j_x &= \sigma_{xx} E_x \\ j_y &= \sigma_{yx} E_x \\ j_z &= \sigma_{zx} E_x. \end{aligned} \quad (13)$$

The current (and therefore the electric field) will have components along  $x$ ,  $y$ , and  $z$  although the applied electric field is only in the  $x$  direction. Since our assumed incident plane EM wave had its magnetic field in the  $y$  direction here,  $j$  is no longer perpendicular to  $H$ .

The effect on  $\rho_a$  of anisotropy within a horizontal layer is shown in Figure 27, from Vozoff (1972). Apparent resistivity then depends on measurement direction, just as if there were lateral variations. However unlike lateral variations, in this case  $\rho_a$  is the same everywhere on the surface, and no vertical magnetic field is induced. The theory was published in Abramovici (1974). When a geological unit has a different resistivity in the horizontal and vertical directions, which is normal in sediments, MT responds only to the horizontal value because the  $E$  field is horizontal. An independent measurement in which  $E$  has some vertical component is necessary to determine  $\sigma_z$  (Jupp and Vozoff, 1977). This has been used to predict the character of sediments for hydrological purposes (Vozoff et al., 1982).

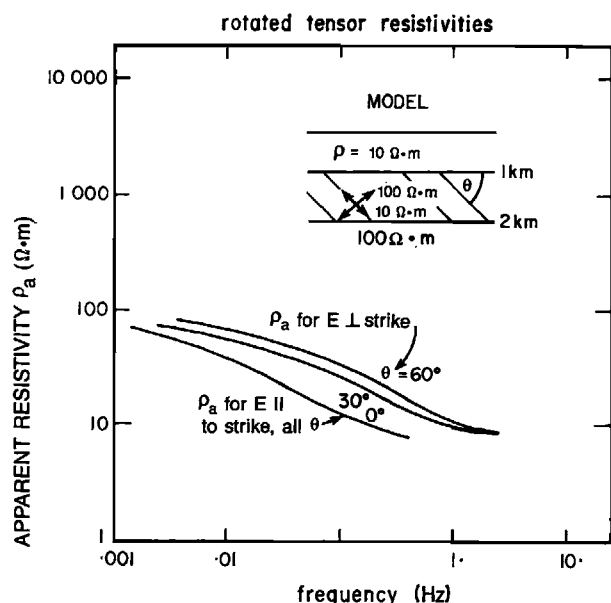


Fig. 27. Effect on apparent resistivity of anisotropy in the second layer of a 3-layer model. Theta is the dip of the smaller ( $10 \Omega \cdot m$ ) principal resistivity axis.

## Inhomogeneity

Anisotropy, one of two possible complications, exists to some extent in all rocks. The other major complication is inhomogeneity, whose description is, after all, the main point of most geophysical surveys. In a 2-D or 3-D structure, the conductivity, currents, and fields vary in  $x$ ,  $y$ , and  $z$  ( $x$  and  $z$  in the 2-D case), and so  $E$  and  $H$  are again usually not at right angles to one another. What effect do the lateral resistivity boundaries have on currents and field? When current encounters a region of discrete or gradational resistivity change, it sets up and maintains a charge distribution in the region (Figure 28). These charges, which are available because of very small differences in integrated current from the surface to  $z = \infty$ , produce their own electric fields that are exactly those needed to satisfy the boundary conditions, and they modify  $E$  and  $j$  in the vicinity. As an example consider the simplest 2-D model, a vertical contact striking in the  $y$  direction at  $x = 0$ , between uniform quarter spaces of resistivities  $\rho_1$  and  $\rho_2$ , with  $\rho_2 > \rho_1$ . A correct guess might be that, for a given magnetic field intensity, the induced current will be different according to whether it has to flow across the fault, or parallel to the fault staying on one side. That is, we expect different values of  $j/H$  and therefore  $E/H$  according to whether  $H$  is parallel or perpendicular to strike. If the two impedances are different, then the apparent resistivities must also differ in the two directions.

The boundary conditions dictate that the current density normal to the boundary must be continuous across the boundaries. The  $E$  field of the resulting charge distribution adds vectorially to the fields in-

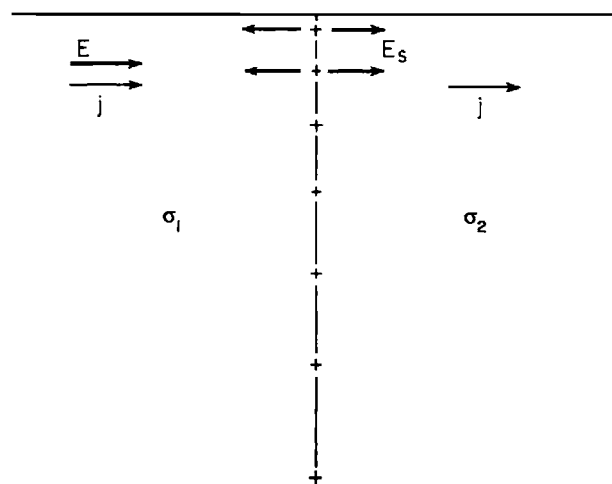


Fig. 28. Charge accumulation due to current crossing a contact,  $\sigma_2 < \sigma_1$ . The  $E$  field of the charges ( $E_s$ ) opposes the applied field on the left of the contact and augments it on the right, so  $E_2 > E_1$ .

duced by  $\mathbf{H}$  in each medium. This *reduces* the current density on the conductive side over what it would be if there were no contrast. Since  $j_x = \sigma E_x$ , the electric field and the impedance are also reduced, making the conductor appear even more conductive. On the resistive side the field *adds* to the initial field, making the region appear even more resistive. These induced charges make it possible to accurately locate contacts and are responsible for many of the characteristic response features of 2-D and 3-D bodies.

When we are far enough away from the contact we expect its effect to vanish. That is,  $\rho_a$  takes on the value appropriate to a uniform half-space. Far enough away is about one skin depth, so that the distance is greater at low frequencies than high frequencies. Figure 29 shows the apparent resistivity, at one frequency, for  $\mathbf{E} \perp$  to strike. This resistivity can be computed for many frequencies and presented as a pseudosection, as in Figure 30a, where the vertical axis is frequency increasing upward. The model is the same as for Figure 29. The effect of the contact is seen at all frequencies, from lowest to highest.

When  $\mathbf{H}$  is perpendicular to strike then  $\mathbf{E}$  and  $\mathbf{j}$  are parallel to strike and no charge is built up on the

contact. The boundary condition just requires that  $\mathbf{E}$  be continuous across the boundary and adjust to the different skin depths on the two sides. Figure 30b shows a pseudosection of  $\rho_a$  for  $\mathbf{E} \parallel$  to strike. Distances over which the adjustment occurs are again related to the skin depths on the two sides. [This general behavior must be distinguished from the specific behavior for which the term "adjustment distance" is used (in the General 3-D Section) to describe large scale effects due to effective anisotropy of the lower crust-upper mantle.)

There is another major difference between the two polarizations: currents are squeezed closer to the surface on the conductive side because the skin depth is smaller there. From the vector equation

$$\nabla \times \mathbf{E} = -\mu \partial \mathbf{H} / \partial t$$

we see that a vertical component of  $\mathbf{H}$  is set up when  $\nabla \times \mathbf{E}$  has a vertical component, that is, when there is a lateral change in horizontal current density such that either  $\partial j_y / \partial x$  or  $\partial j_x / \partial y$  (but not  $\partial j_x / \partial x$ ) is nonzero. This  $H_z$  component must decay with distance from the contact. In the 2-D case this arises for  $j_y$  but not  $j_x$ ,

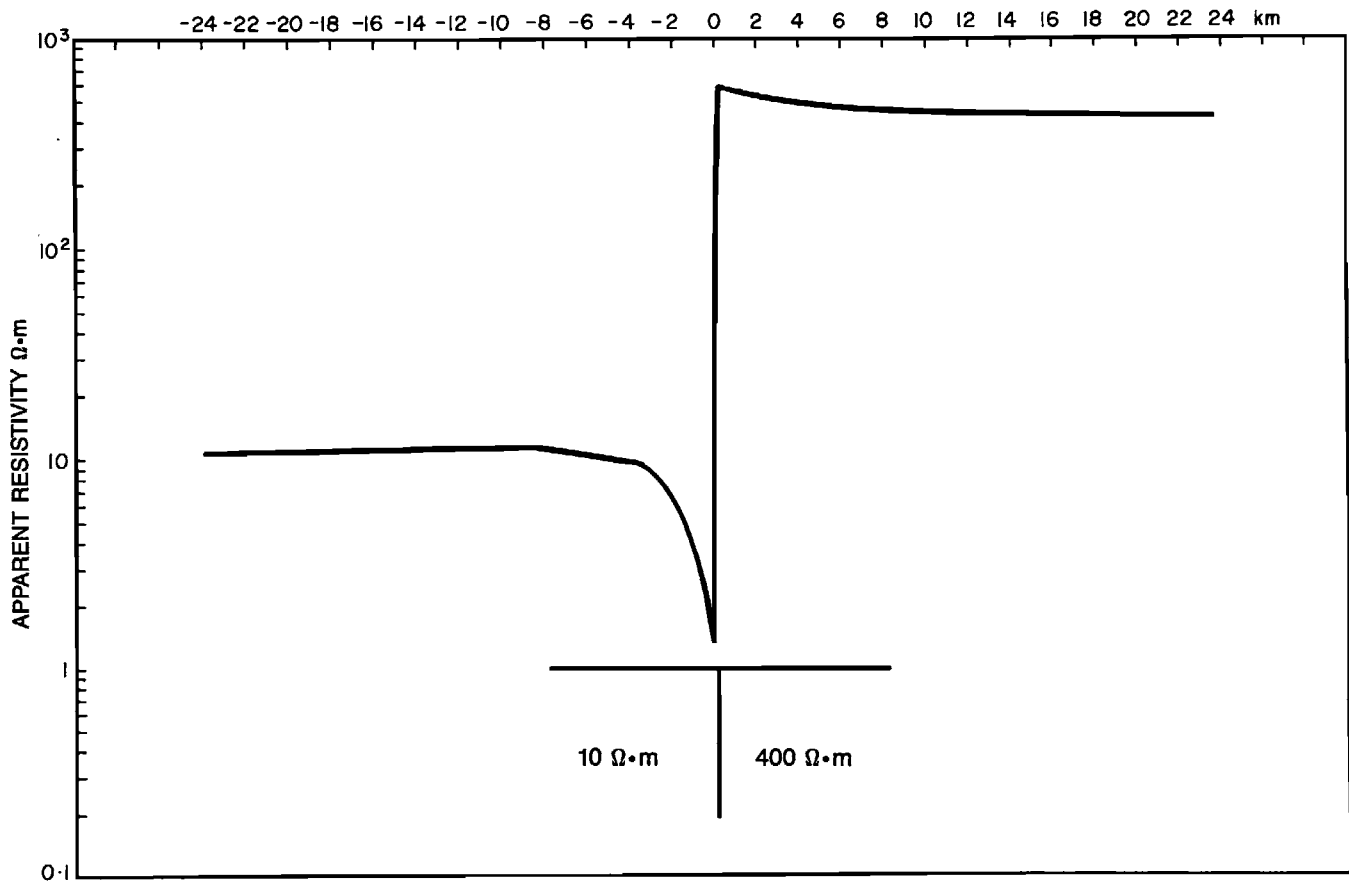


Fig. 29. Apparent resistivity versus position across the contact shown at bottom, at 0.1 Hz.

since  $\partial/\partial y = 0$ . When  $H_z$  is normalized by the horizontal magnetic field it is variously called the tipper, induction vector, Parkinson vector, or Weise vector. The magnitude of the tipper in our fault model (in percent) is also shown in pseudosection, since it varies with frequency and position (Figure 30c). This explains why care must be taken interpreting data from high and equatorial latitudes.

If the contact does not come to the surface, but is buried beneath an overburden (Figure 31), then its effects at the surface are smoothed to some extent. In this case the overburden consists of  $1.0 \Omega \cdot \text{m}$  material  $0.5 \text{ km}$  thick. The amount of smoothing depends on the thickness and conductivity of the overburden and on the frequency. At high enough frequencies, when the skin depth in the overburden is much less than its thickness, the contact will not be detected. We will see only the overburden, and the response will be independent of position and direction. In the low fre-

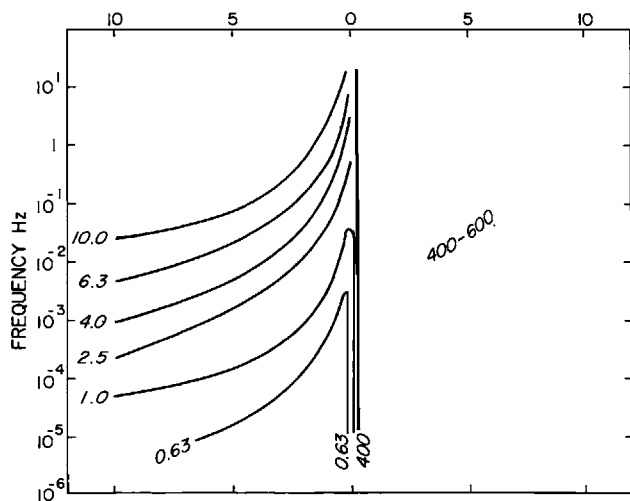
quency limit, when skin depth is much greater than overburden thickness, the overburden will be nearly invisible and the contact will appear to outcrop.

### Impedance Tensor and Tipper: 2-D and Almost 2-D Cases

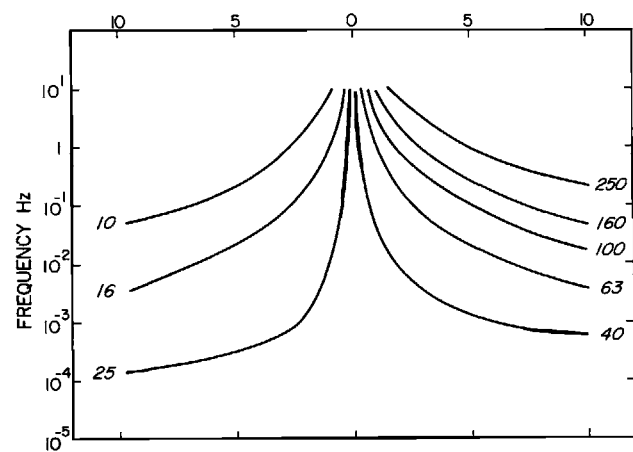
The relationships among the field components at a single site are systematically contained in the impedance and the tipper. They are the quantities from which conductivity structure is interpreted. In general,  $H_x$  has an associated  $E_y$  and some  $E_x$ , both of which are proportional to  $H_x$ . Likewise,  $H_y$  causes an  $E_x$  and some  $E_y$ , so that at each frequency we would expect a linear system to behave as

$$E_x = Z_{xy}H_y + Z_{xx}H_x$$

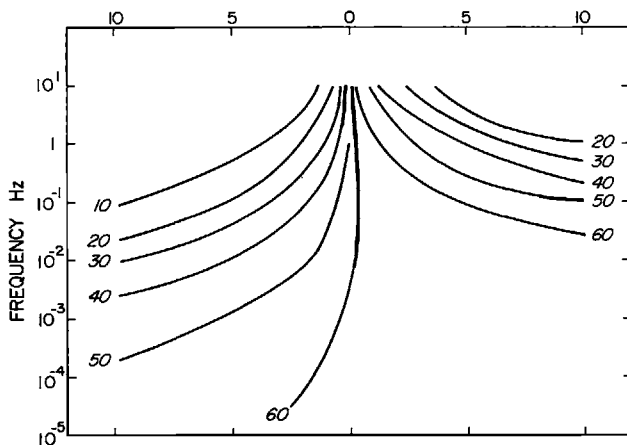
$$E_y = Z_{yx}H_x + Z_{yy}H_y,$$



30 (a)



30 (b)



30 (c)

Fig. 30. Pseudosections across the vertical contact. Horizontal axis is distance in kilometers from the contact. (a)  $\rho_a$  for  $E$  perpendicular to strike. (b)  $\rho_a$  for  $E$  parallel to strike. (c) tipper in percent.

where each term is frequency dependent. This is commonly written

$$\begin{pmatrix} E_x \\ E_y \end{pmatrix} = \begin{pmatrix} Z_{xx} & Z_{xy} \\ Z_{yx} & Z_{yy} \end{pmatrix} \begin{pmatrix} H_x \\ H_y \end{pmatrix} \quad (14)$$

or

$$\mathbf{E} = \mathbf{Z}\mathbf{H}.$$

In a uniform or horizontally layered earth,  $Z_{xx}$  and  $Z_{yy}$  are zero,  $Z_{yx} = -Z_{xy}$ , and the equations reduce to

$$E_x = Z_{xy} H_y$$

$$E_y = Z_{yx} H_x$$

$$= -Z_{xy} H_x.$$

In a 2-D case, if the  $x$  or  $y$  axis is along strike then

$$Z_{xx} = Z_{yy} = 0,$$

but

$$Z_{xy} \neq -Z_{yx}.$$

If neither axis is along strike then

$$Z_{xx} = -Z_{yy} \neq 0.$$

Equation (14), proposed in Cantwell (1960) and in Rokityanski (1961), assumed that the system is linear so that the electric fields are due only to magnetic fields, and the contributions of noise were ignored. The problems of unraveling these relationships in noisy data are discussed next.

As a practical matter we often wish to look at the fields or the tensor elements as if they had been measured in some other set of coordinate directions. For instance, the strike direction is seldom known very precisely at the time of a field survey. If we rotate

the vector  $\mathbf{E}$  field through angle  $+\theta$  (clockwise as seen from above) to be  $\mathbf{E}'$ , then

$$\begin{pmatrix} E'_x \\ E'_y \end{pmatrix} = \begin{pmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{pmatrix} \begin{pmatrix} E_x \\ E_y \end{pmatrix} \quad (15)$$

or

$$\mathbf{E}' = \mathbf{R}\mathbf{E}.$$

In the same way,

$$\mathbf{H}' = \mathbf{R}\mathbf{H} \quad (16)$$

and

$$\mathbf{Z}' = \mathbf{R}\mathbf{Z}\mathbf{R}^T, \quad (17)$$

where  $\mathbf{R}^T$ , the transpose of  $\mathbf{R}$ , is

$$\mathbf{R}^T = \begin{pmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{pmatrix}. \quad (18)$$

Starting from a tensor impedance  $\mathbf{Z}$  which is derived from measurements, and assuming 2-D conditions, several different ways have been used to find the rotation angle  $\theta_0$  between measurement direction and strike. One of these is to rotate the  $Z_{ij}$  in steps (say 5 degrees), plot them on a polar diagram, and pick an optimum angle from the plots. An optimum angle maximizes or minimizes some combination of the  $Z_{ij}$ . These interesting diagrams, called *polar figures* or *impedance polar diagrams*, are usually plotted at many frequencies, because in practice the strike direction often changes with depth. Figure 32 shows such plots for a 2-D model ( $Z_{xy}$  only), and Figure 33 shows a real example with both  $Z_{xy}$  and  $Z_{xx}$ .

Another way to find  $\theta_0$  is to use one of Swift's (1967) solutions, in which the expressions for  $Z_{xy}(\theta)$  and

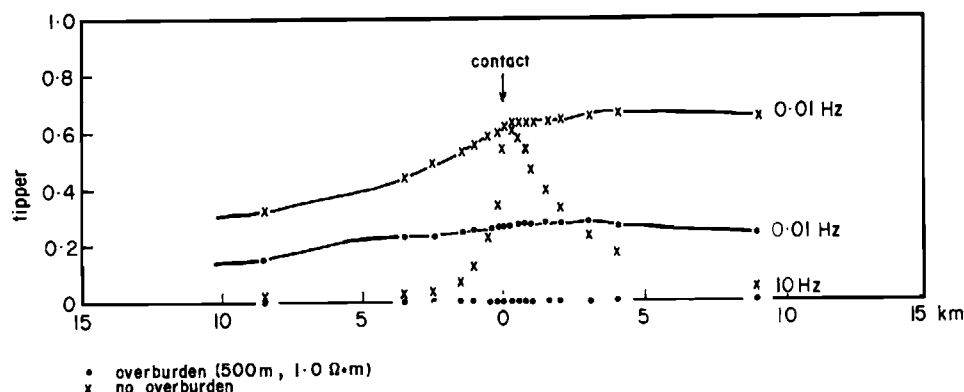


Fig. 31. Effect of overburden on tipper, at two frequencies. No overburden (x's) is for the same contact model as Figures 29 and 30. The overburden curves (dots) are for a layer 0.5 km thick of  $1 \Omega \cdot \text{m}$  material.

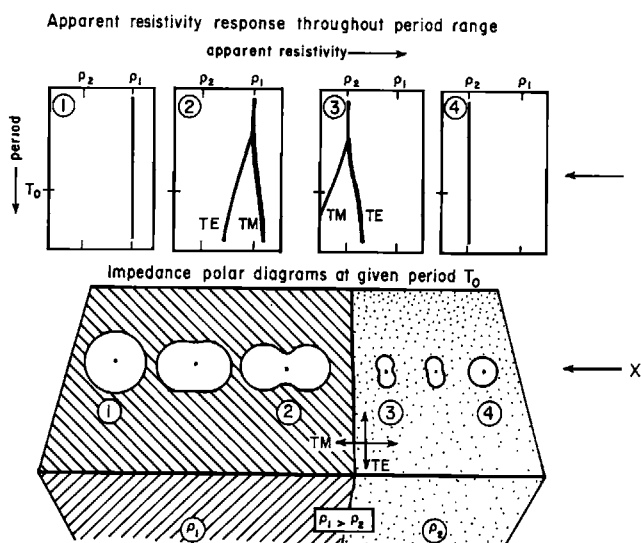


Fig. 32. Impedance polar diagrams (at one frequency), and apparent resistivities, at four sites on a simple 2-D contact model. TE and TM refer to Transverse Electric ( $E \parallel$  to strike) and Transverse Magnetic ( $H \parallel$  to strike) field polarizations, respectively. (After Phoenix Geophysics, Inc., pers. comm.)

$Z_{yx}(\theta)$  are differentiated with respect to  $\theta$  to give an angle  $\theta_0$  which optimizes

$$|Z'_{xy}(\theta_0)|^2 + |Z'_{yx}(\theta_0)|^2 \quad (19)$$

at each frequency. His solution

$$4\theta_0 = \tan^{-1} \frac{[(Z_{xx} - Z_{yy})(Z_{xy} + Z_{yx})^* + (Z_{xx} - Z_{yy})^*(Z_{xy} + Z_{yx})]}{|Z_{xx} - Z_{yy}|^2 - |Z_{xy} + Z_{yx}|^2} \quad (20)$$

also maximizes  $|Z_{xy}|$  and minimizes

$$|Z_{xx}|^2 + |Z_{yy}|^2.$$

There is no solution in the 1-D case, whereas in a clearly 2-D case it usually has a definite value. In the 3-D case its meaning is usually questionable and there is considerable research in progress into ways to present and interpret  $Z_{ij}$  in structural terms. The matter will be discussed further. Of the four values between 0 and 180 degrees, the "choice" of strike direction is started by evaluating equation (19) at two adjacent values, one a minimum and the other a maximum. This leaves four possible solutions at 90 degree intervals, or two possible strike directions. The choice between these solutions can only be made from

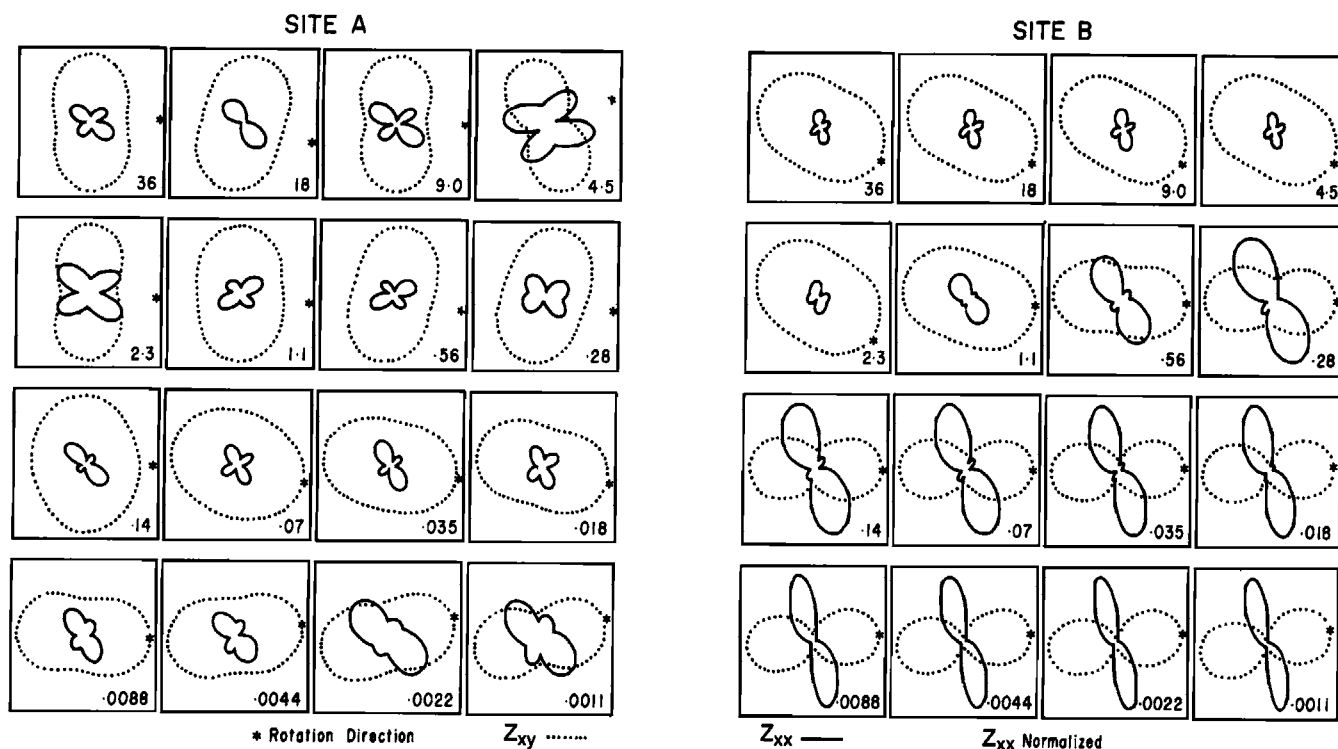


Fig. 33. Impedance polar diagrams at two sites (A and B) for 16 frequencies. Dotted curves are for off-diagonal ( $Z_{xy}$ ), solid curves are for ( $Z_{xx}$ ). Asterisk indicates Swift rotation direction. (After Phoenix Geophysics, Inc., pers. comm.)

independent information, usually the relations between vertical and horizontal  $\mathbf{H}$  components or from geological constraints.

Still another way of showing the angular variation in impedance elements is by means of *impedance rotation ellipses*, plotted in the complex plane, with properties that are explained in Word et al. (1971) and Eggers (1982). These ellipses should also be examined for many frequencies, and are used mostly to examine the general behavior of the tensor elements as rotation angle varies.

Several useful properties of  $\mathbf{Z}$  are mentioned here. When the coordinates are rotated, certain combinations of terms are constant even though the individual terms vary. These are

$$Z_{xx} + Z_{yy} = c_1, \quad (21)$$

$$Z_{xy} - Z_{yx} = c_2, \quad (22)$$

and

$$Z_{xx}Z_{yy} - Z_{xy}Z_{yx} = c_3, \quad (23)$$

where the absolute value is the determinant of the impedance tensor. The ratio  $c_1/c_2$  is the impedance skew,  $\alpha$ .  $c_1$  will be zero in (noise free) 1-D and 2-D models, so the skew is used as a measure of three dimensionality. It does not change with rotation of coordinates.

A quantity which *does* vary with setup direction is impedance *ellipticity*,

$$\beta(\theta) = \frac{Z_{xx}(\theta) - Z_{yy}(\theta)}{Z_{xy}(\theta) + Z_{yx}(\theta)}. \quad (24)$$

This is zero (for noise free data) in the 1-D case, and in the 2-D case when the  $x$  or  $y$  axis is along strike. Impedance ellipticity, like impedance skew, is used to indicate whether response at a site is 3-D.

It can usually be assumed that  $H_z \approx 0$  except near lateral conductivity changes, where  $\nabla \times \mathbf{E}$  has a vertical component (but see discussion in the Sources Section). There, the relationship between  $H_z$  and the horizontal magnetic field components at any frequency can be written

$$H_z = T_x H_x + T_y H_y \quad (25)$$

where the elements  $T_i$  are complex since they may include phase shifts. Given a 2-D structure with strike in the  $x'$  direction, in those coordinates equation (25) simplifies to

$$H_z = T_y' H_y'. \quad (26)$$

Here,  $T'$ , since it represents a tipping of the  $\mathbf{H}$  vector out of the horizontal plane, is called the *tipper*.  $T'$  is of course zero for the 1-D case. The modulus of  $T'$  is

rarely as great as 1, with .1 to .5 being the common range. The lower part of the range is often blurred by noise, since  $H_z$  is so weak. The required rotation angle  $\phi$  to  $x'$  can be estimated by the field data by finding the horizontal direction  $y'$  in which  $H(\phi)$  is most highly coherent with  $H_z$ . There is generally a definite solution in the "almost" 2-D situation. In that case the phases of  $T_x$  and  $T_y$  are the same, the ratio  $T_y/T_x$  is a real number, and

$$\phi = \arctan (T_y/T_x).$$

Another use of the tipper, besides helping resolve ambiguity in strike, is to show which side of a contact is more conductive. Near a conductor-resistor boundary, the near-surface current density parallel to strike is larger on the conductive side. If it is looked at as a simple problem in dc magnetic fields of the excess currents, the magnetic field in the vertical plane perpendicular to the contact will "curve around" the edge of the conductor, by Faraday's law. Thus  $H_z$  will be directed downward when the horizontal component is outward, and vice versa, depending on current direction. In a real situation, the phase of  $H_z$  depends on the conductivities, frequency, and distance from the contact. However in practice the relations can often be used to indicate the direction to a conductive region.

### The General 3-D Case

In practice 3-D problems commonly range from the effects of small isolated conductivity anomalies to those of large scale structures such as curved coastlines and complex mountain ranges. There is at the moment a particular interest in understanding the effects of superficial 3-D bodies, on account of the important static (vertical) shifts they can produce on  $\rho_a$  curves.

Our ability to interpret results from areas of 3-D structure is far less developed than for 2-D areas. The mode separation that helps so much to simplify the 2-D case does not occur in 3-D, where there are surfaces exposed to charge accumulation regardless of current direction. The vertical magnetic field, which is enhanced by long current paths in structures with a definite strike direction, is generally smaller and more difficult to interpret in the absence of this condition. Numerical model programs and results for 2-D have been in use since the middle 1960s, whereas those for 3-D have been available only since the early 1980s and are still considered costly and lacking in generality (Reddy et al., 1977; Jones and Vozoff, 1978; Ting and Hohmann, 1981; Ranganayaki and Madden, 1980; Wannamaker et al., 1984a; Pellerin and Hohmann, 1990). Berdichevski and Dmitriev (1976) and Larsen (1977) model special cases. In addition, a few analyt-

ical or semi-analytical solutions exist for special cases (e.g., Bailey, 1977; Fischer et al., 1978). Hohmann's (1988) paper is a very good review of the situation.

Examination of a few specific models is helpful. Figure 34, from Ting and Hohmann (1981), shows the distribution of apparent resistivity at a single frequency around a simple 3-D body, a conductive block in a more resistive half space. At the frequency used the skin depth in the half space is about 15 km, so the block is effectively quite shallow. The principal elements show the effects of currents being attracted into the conductive block. In  $\rho_{yx}$  the  $y$ -directed electric field and current density are enhanced where the current is "collected" by the block. They are reduced on the sides, where the current density has been reduced. This can also be interpreted as the effect of induced charges on the  $+y$  edge of the block augmenting the normal electric field. When the normal electric field is in the  $x$ -direction, in  $\rho_{xy}$  the same behavior is seen, rotated through 90 degrees. The two diagonal elements are seen to be near zero except around the corners of the block, where the induced charges distort current flow to set up "anomalous" magnetic fields.

In another specific case, Berdichevsky and Dmitriev (1976) modeled the low-frequency asymptotic (statics) effects on the principal axes of resistive and conductive (vertical) ellipsoidal cylinders in a thin surface layer. Both cylinder and layer were underlain by an infinitely resistive second layer, which in turn lay on an infinitely conductive basement, so their effects will be extreme. For a given contrast, they showed that resistive features create smaller disturbances than conductive ones, and in the opposite sense.

Groom and Bailey (1989) devised a composite 2-D and 3-D model, to study the behavior of various processing and interpretation techniques in such cases. In particular they used a numerical 2-D solution to represent a "regional" 2-D structure, and superimposed a small conductive hemisphere on the surface, to represent a body giving rise to statics effects. The hemisphere was assumed to affect the fields only in its immediate vicinity, and the effects of local  $E$  and  $H$  fields were computed numerically. Lines through the center of the hemisphere in the principal directions of the 2-D model are the principal axes of the fields around the hemisphere. The authors studied impedances and directions just outside the hemisphere, finding marked differences with position relative to the principal axes. Along those lines the impedances contained only off-diagonal terms, and those were static-shifted versions of the 2-D impedances at those positions. At sites off the principal axes of the hemisphere, all four impedance elements were mixtures of the two 2-D terms.

West and Edwards (1985) set out general rules of response to an EM dipolar source of a small, simple conductive 3-D object in a conducting layered half-space. They show the factors affecting the induced source strength within the scatterer, current gathering from the host, and induction from the field. Inductive interaction with the host is shown to be relatively weak. The arguments are useful in the MT case when the "target" is compact, although they apply strictly only to systems with dipolar source and receiver. The discussion is primarily a list of caveats.

Similarly, Park et al. (1983) and Park (1985) classified the first order effects of 3-D bodies on MT fields into three categories: vertical current distortion, horizontal current distortion, and local induction (Figure 35). ("Current distortion" is also called "current gathering" or "current channeling".)

Vertical current distortion also occurs in the TM mode for 2-D models (Madden, 1971), and a look in passing is instructive. Figure 36 shows the effect of width in a 2-D body at the surface of an otherwise uniform half-space. Apparent resistivity versus frequency is plotted for various body widths and for the 1-D asymptote when width reaches infinity. The observation point is in the center of the body. The decrease in low-frequency apparent resistivity as the body becomes narrower is remarkable and indicates how small superficial conductors can produce such important static shifts. Figure 37 shows pseudosections for a very similar model.

Horizontal current gathering can only occur in 3-D bodies. The effect in 3-D slabs was studied in Ting and Hohmann (1981). They showed that the  $E$  field of the charge built up on the edges, and therefore the statics effect, persists to surprisingly large distances.

When the model is complicated by adding layers beneath the inlier, those layers will also affect its ability to attract current. For example, a more conductive half-space beneath the body makes it easier to gather current from below. This has been found to be an important factor in regional studies. In the case of the resistive deep crust-conductive upper mantle system, the combination forms an effectively anisotropic layer (Ranganayaki and Madden, 1980). Park et al. (1983) show that, in such a situation, a surface conductor must be very wide in order to gather current from the conductive mantle beneath. The length involved, called the "adjustment distance," is  $t = \sqrt{SR}$ , where  $S = \sigma\Delta z$  and  $R = \rho\Delta z$  are the integrated conductivity and integrated resistivity, respectively, over the zone of thickness  $\Delta z$  and typically is hundreds of kilometers. The effect introduces a strong downward bias on  $E$  at low frequencies in measurements near continental margins and on islands but is less important in many problems of exploration scale.



Downloaded 08/26/15 to 130.194.20.173. Redistribution subject to SEG license or copyright; see Terms of Use at <http://library.seg.org/>

As shown later, at low enough frequencies, current gathering effects become frequency independent, but local induction effects remain proportional to frequency, and are thus fundamentally different from current gathering (Park, 1985; West and Edwards, 1985). Adequate data can provide a way of distinguishing the two phenomena in particular cases.

These conclusions regarding the domains of induction and distortion are borne out by other studies of

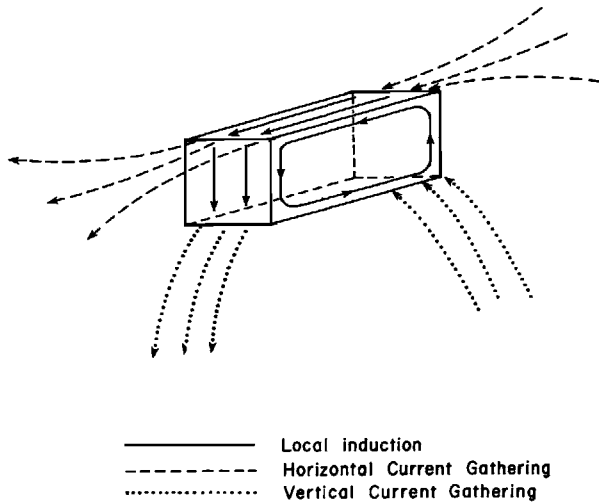


Fig. 35. Patterns of current flow due to an isolated 3-D conductor in a more resistive host. (After Park, 1985.)

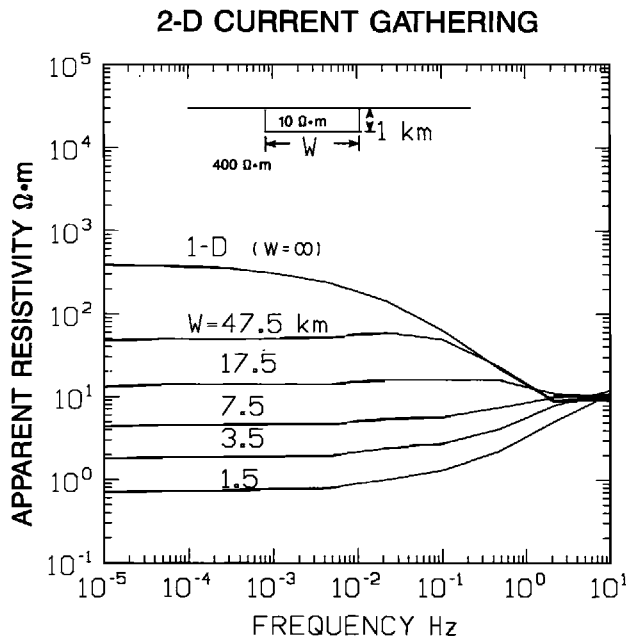


Fig. 36. Effect of vertical current gathering on apparent resistivity for  $E$  across strike in a 2-D model.  $W$  is the width of the rectangular conductive cylinder. The observation point is in the center of the body on the surface.

low-frequency asymptotic behavior of 3-D models. The studies demonstrated that the secondary electric field becomes a fixed percentage of the primary, so that the surface charge effect on  $E$  and thus on  $\rho_a$  is present down to dc. However as a consequence of Equation (10) the effect on the phase vanishes as was shown for specific models in Ranganayaki (1984, their Figures 8 and 9), and analytically in Wannamaker et al. (1984b). The latter, in a fundamental contribution, show that at the surface of a layered medium containing a 3-D body

$$E_h = [\underline{I} + \underline{P}]E_i \quad (27)$$

and

$$H_h = [\underline{I} + \underline{Q}\underline{Z}_\ell]H_i, \quad (28)$$

where  $\underline{Z}_\ell$  is the impedance of the layered medium without the body,  $\underline{P}$  and  $\underline{Q}$  are scattering tensors,  $\underline{I}$  is a  $2 \times 2$  identity matrix, and the subscripts  $h$  and  $i$  indicate horizontal components and incident fields, respectively. In this case,

$$\underline{Z} = [\underline{I} + \underline{P}]\underline{Z}_\ell[\underline{I} + \underline{Q}\underline{Z}_\ell]^{-1}, \quad (29)$$

where  $\underline{Z}$  is the complete impedance tensor. When the 3-D body is relatively conductive and frequencies are low enough that skin depth *within* it is larger than the body, then they show that  $\underline{P}$  and  $\underline{Q}$  become real constants, independent of frequency, and  $\underline{Q}\underline{Z}_\ell$  vanishes (as  $\underline{Z}_\ell$ ) as frequency goes to zero. In that case,

$$\underline{Z} \approx [\underline{I} + \underline{P}]\underline{Z}_\ell \quad (30)$$

so that the four (unrotated) apparent resistivities become

$$\begin{aligned} \rho_{xx} &= \rho_\ell P_{xy}^2 \\ \rho_{xy} &= \rho_\ell (1 + P_{xx})^2 \\ \rho_{yx} &= \rho_\ell (1 + P_{yy})^2 \end{aligned}$$

and

$$\rho_{yy} = \rho_\ell P_{yx}^2 \quad (31)$$

where  $\rho_\ell$  is the apparent resistivity at that frequency for the unperturbed layered medium. The coefficients of  $\rho_\ell$  are always positive so that the apparent resistivities are always positive.  $\rho_{xy}$  and  $\rho_{yx}$  can be greater than, less than, or equal to  $\rho_\ell$ , depending on the signs of  $P_{xx}$  and  $P_{yy}$ . They will depend on the resistivity and position of the inlier relative to the measurement point. The phases are those of the *layered* medium in this approximation. Hence, simply using 1-D inversion gives model resistivities too large by  $\rho_{ij}/\rho_\ell$  and depths

too great by  $\sqrt{\rho_{ij}/\rho_e}$ . When the superficial feature is 2-D with strike in the  $x$  direction, then no surface charge distribution is created and  $P_{xx}$  is zero. Then a 1-D inversion of  $\rho_{xy}$  ( $\mathbf{E} \parallel$  to strike) will give results less distorted than the  $\rho_{yx}$  ( $\mathbf{E} \perp$ ) component, at least at low frequencies.

Wannamaker et al.'s (1984b) conclusions specifically extend and support the earlier semi-analytical results of Berdichevsky and Dmitriev (1976) and are confirmed in the model computation of Groom and Bailey (1990).

Two other characteristics of 3-D responses which are observed in practice are their enhancement at low frequency when resistivity increases with depth, and systematically reduced tippers as compared with those of 2-D features. This behavior, which is anticipated intuitively, is also explained by the results of Wannamaker et al. (1984b) and Ting and Hohmann (1981).

The number and variety of response functions which are used for interpretation pose a problem in data presentation. In a combined case history and model study, Ranganayaki (1984) experiments with ways of presenting these and related data. In particular, she

considers an apparent resistivity and phase defined by the determinant of  $\rho_a$

$$\rho_d = \frac{1}{\mu\omega} |Z_{xx}Z_{yy} - Z_{xy}Z_{yx}|$$

and its phase

$$\phi_d = \text{phase of } (Z_{xx}Z_{yy} - Z_{xy}Z_{yx}).$$

The recovery from the effects of superficial features with decreasing frequency of the individual phases and of the phase of the determinant, and the persistence of those effects in  $\rho_a$  and  $\rho_d$ , are illustrated graphically. Clearly, phase is easier to interpret than apparent resistivity in these situations, as shown by Figure 37 from Ranganayaki (1984).

While the ultimate goal of 3-D research might be to produce inexpensive, accurate, inversion tools, at the present stage it would be highly desirable just to have simplified, if less accurate, ways to do 3-D modeling. Flores and Edwards (1985) use the simplified behavioral guidelines of West and Edwards (1985), applied to plates, to interpret the results from a complex geothermal area. Their model could be used as input to

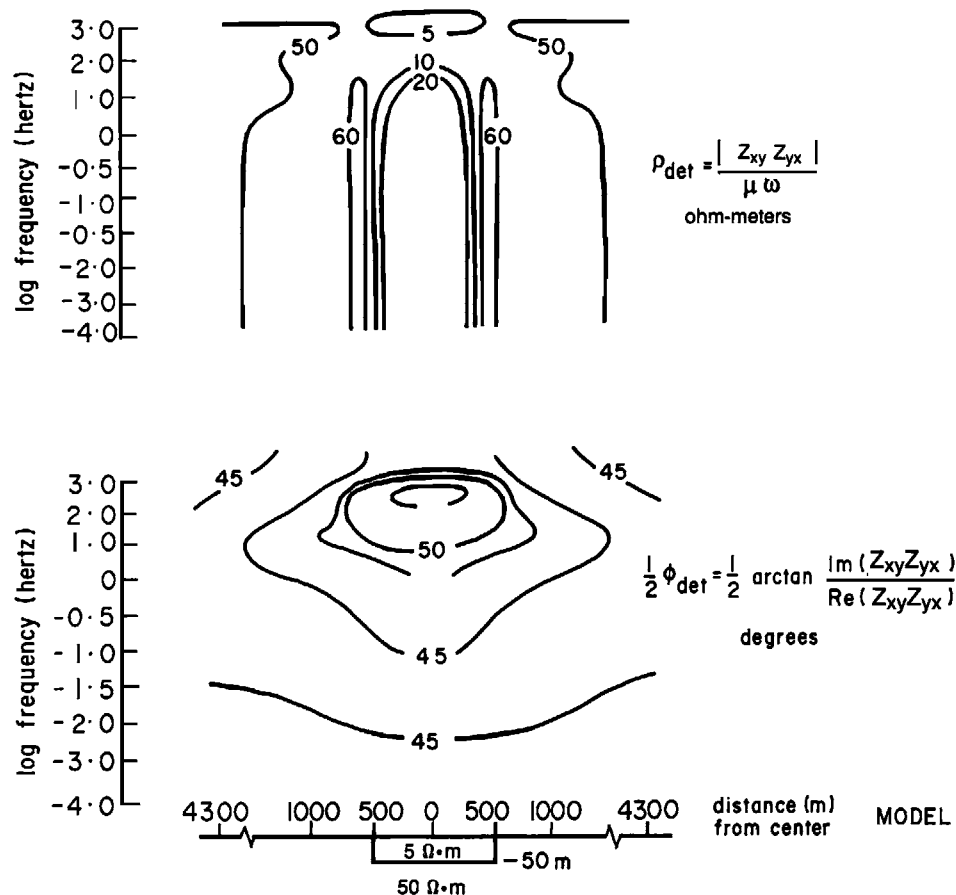


Fig. 37. The effects of a shallow body on apparent resistivity and phase. Model, shown at bottom, is very similar to that of Figure 36. (After Ranganayaki, 1984.)

a more accurate forward program if desired, but it could equally well be used to start a 3-D inversion.

### Statics, Topographic, and Regional Effects

The results of the previous section are, of course, fundamental to an understanding of statics effects, which became increasingly noticeable as station spacing in surveys decreased. In this section I summarize the definition and appearance of statics effects and their relationship to the 3-D theory of Wannamaker et al. (1984b). I also point out the existence and cause of regional effects, the importance of which Madden and Mackie (1989) emphasized. Topographic effects are also important in some locations and have some of the characteristics of statics effects. Certain aspects are discussed at the end of this section and are also described qualitatively in the Zonge and Hughes CSAMT chapter (this volume), where references are given to more quantitative work in the literature.

Statics effects are defined as vertical displacements of apparent resistivity curves, between adjacent sites or between the two curves at one site, without other differences in either the shapes of the curves or of the phases. An illustration is shown in Figure 38, from Sternberg et al. (1985). The implication is that the offset is a result of a conductivity feature at surface that is so thin that its effect on phase has disappeared above our highest frequency. Hence by definition, frequency is so low that induction effects in the causative body have vanished. If we went to still higher frequencies, then differences would appear first in the shapes of the  $\rho_a$  curves, and then in the phases. If we also recorded at more closely spaced sites we would expect to be able to map and interpret the responsible feature. In a sense, statics are due to undersampling in both time and space. The effects are most evident in CSAMT data because of the close station spacings and short dipoles normally used: if the sites were further apart the differences might be mistakenly attributed to structural complications. Likewise, longer dipoles will hide the effects of small statics features by averaging the field across the entire length of the dipole.

Static shifts can have either 2-D or 3-D source regions. In the 2-D case, the charge buildup in the TM mode will shift that apparent resistivity by a constant factor when the thickness of the anomalous body is much less than the skin depth in either the host or the body. Then, the TM apparent resistivity will depend on the width of the body and on position, but a shift would be predicted even for very narrow bodies. Much lesser shifts are expected for the TE mode, and if the body is narrow enough then no shift should be seen. Berdichevsky and Dmitriev (1976) include type curves of the statics effects of 2-D bodies in a conduct-

ing layer overlying an insulating layer on a perfectly conducting basement.

In the 3-D case the observed effects are anticipated from Equation (30). By definition, the frequency is low enough that  $\mathbf{P}$  is a real constant. Intuitively the magnitude of the statics effect of a 3-D feature would be smaller than that of a 2-D feature with the same cross section, because current can bypass the former. However, there will be a TM-like response in the 3-D case regardless of field polarization.

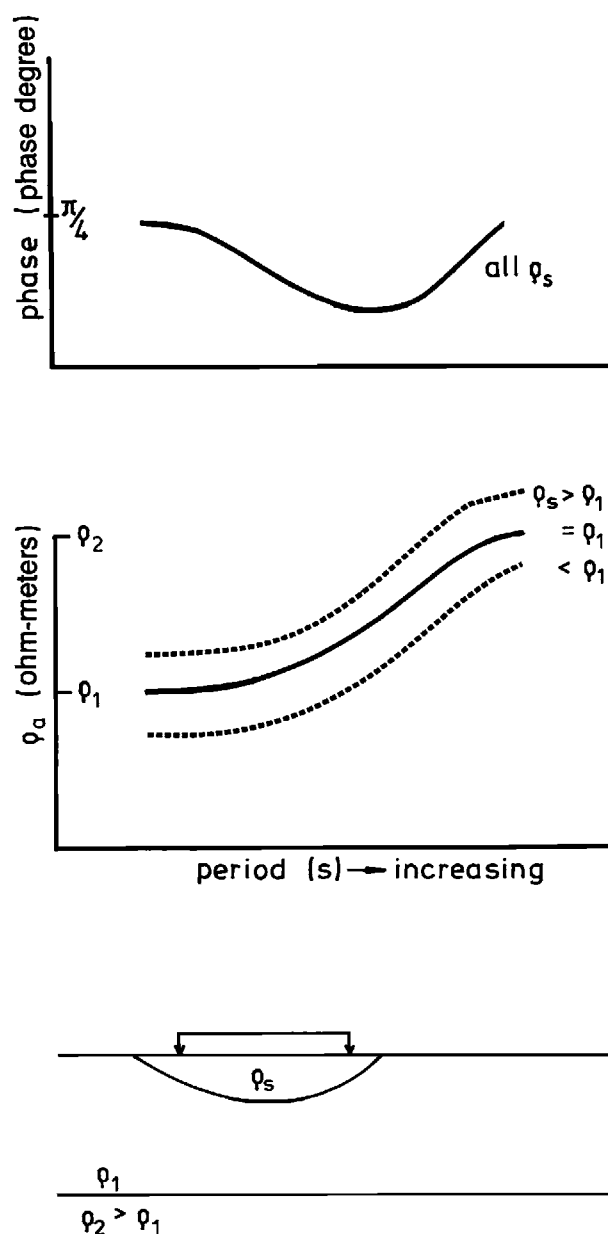


Fig. 38. Sketch of model used to explain static effects. The middle apparent resistivity curve is that which would be observed if the inlier had the same resistivity as the layer containing it. (After Sternberg et al., 1985)

Two special 3-D cases have been described in the literature. Zhang et al. (1987) are concerned with a Shield area having a 2-D regional "grain" in one direction overlain by a very thin cover with a 2-D grain in a different direction. They show that  $\underline{Z}$  takes on particularly simple forms if the measurements are made along either of the two directions. In particular, if a measurement axis lies along the superficial strike direction, then the two diagonal terms are linearly related. If one of the measurement axes is along regional strike, then  $Z_{xx}$  is a real multiple of  $Z_{yx}$  and  $Z_{yy}$  is a (different) real multiple of  $Z_{xy}$ .

The other special case is that of a small 3-D body in a regional 1-D or 2-D environment (Berdichevsky and Dmitriev, 1976; Groom and Bailey, 1989, 1990). That problem is treated using specific models. Berdichevsky and Dmitriev (1976) use a vertical elliptic cylinder in a thin conductive layer over an insulating middle layer underlain by a perfectly conducting half-space. At points along the axes of the ellipse they examine its distortion effect on the impedance, compared with that of the unperturbed layered model. Results are described in terms of distortion coefficients, which are different from 1.0 according to whether the inclusion causes a "flow around" or "concentration" of the current at the observation point. The distortion coefficients are real scalars.

The Groom-Bailey model consists of a very small conductive hemisphere near a vertical fault in a thin layer. The layer is underlain by a more conductive half-space. The purpose of the fault is to provide a 2-D "regional" environment for the hemisphere, and the authors examine the impedance variation with azimuth from the center of the hemisphere. They find that, on azimuths along the regional principal axes, impedances are shifted by real scalars, in accordance with the results in Berdichevsky and Dmitriev (1976). However, on azimuths intermediate to the two principal directions, the effect of the hemisphere must be described by a real tensor rather than a scalar. The analysis and decomposition of these tensors are described in two later sections.

In some applications, as in the study of sedimentary basins, the field within the hemisphere is of primary importance. The regional fields impose a systematic bias on both  $E$  and  $H$  at low frequencies, which means that measurements within it, as well as interpretive models, must account for this condition (Madden and Mackie, 1989) as was discussed previously.

Topographic features force currents to flow in patterns different than what they would be if the surface were flat, and hence they affect both magnetic and (especially) electric fields at the surface. Impedances are, therefore, affected as well. This influence is predicted theoretically and is often observed in prac-

tice. Topographic effects have been modeled with the same computer programs used to model 2-D and 3-D effects of features buried in a flat earth, e.g., Wannamaker et al. (1984b), Wannamaker et al. (1986), etc. In addition, Jiracek and his students (Jiracek, 1973; Reddig, 1984; Kojima, 1985) have used their general Rayleigh-FFT technique to model topographic effects, and Holcombe (1982) developed special purpose 3-D finite element code for the purpose, in response to the problems of interpreting MT surveys in mountainous geothermal areas of the Western U.S. (Mozley, 1982). I calculated the 2-D effects that might be anticipated in a proposed traverse into the Himalayas (Vozoff, 1984), and Jiracek and I modeled the (more likely) 3-D situations of measurements along the roads, which follow river valleys transecting the linear ranges at right angles. In the 2-D case there is a marked difference between the TE and TM cases, as is usual in other 2-D models. TM effects can be described as primarily galvanic, slope dependent, and weakly frequency dependent as compared with TE effects, which behave inductively and are generally smaller and smoother. In the general 3-D situation, it is recognized that every polarization gives galvanic effects, although their magnitude will be smaller than in the 2-D case. Inductive effects will be small because of the finite dimensions of the conductors. Considering that the thrust zones of interest in the Himalayas are locally expected to be nearly 2-D conducting planes, they should be detectable in the tipper at least. Two good recent reviews (Jiracek, 1990; Singer, 1990) deal with topographic effects and statics effects.

### Alternative Definitions of Resistivity and Impedance

As previously mentioned other definitions of impedance are sometimes used. Although this is usually done for theoretical reasons, there are some practical reasons also. With one of these alternative definitions, impedance can be measured without using electrodes. Schmucker and Weidelt (1975) show that

$$\underline{Z} = -\rho_0 H'_x / H_x$$

where

$$H'_x = \partial H_x / \partial z$$

and  $\rho_0$  is the resistivity over the depth range where the derivative is measured. This makes it possible to do MT in locations where  $\underline{E}$  cannot be measured on account of surface conditions, or in an oil well, by differencing the outputs of a pair of vertically offset magnetometers or by using a gradiometer (Vozoff, 1982). The local resistivity must then be determined by other means, such as a well log or a surface electrical measurement. Edwards et al. (1988) used this ap-

proach to make measurements in shallow portions of the Arctic Ocean. They obtained the gradient by placing induction coils just beneath the ice and on the seafloor. Seawater resistivity was estimated from its temperature and salinity, and the results provided the first MT soundings in the Arctic Ocean.

Spies and Eggers (1986) point out that many definitions of apparent resistivity, including that of Cagniard, are somewhat arbitrary, and that others can be devised that have more intuitively appealing behavior. In the MT case, instead of  $|Z_{ij}|^2$  it is equally legitimate to use  $\text{Re} [(Z_{ij})^2]$ , or  $\text{Im} [(Z_{ij})^2]$ , as examples. Figure 39 compares three-layer  $\rho_a$  curves  $\rho_a$ ,  $|z|$  obtained in the conventional way, using the real part of  $Z$   $\rho_a$ ,  $\text{Re}(z)$ , and in the time domain  $\rho_a$ ,  $T.D.$  The second and third are better behaved than the first in that they do not show the well known overshoot and undershoot which are characteristic of the Cagniard definition, and their variations are confined to a smaller range of frequency.

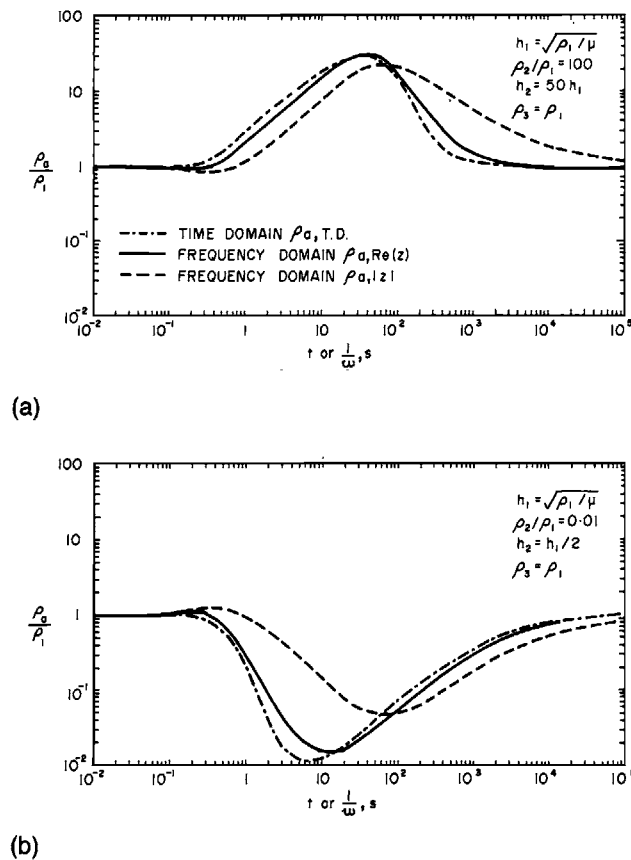


Fig. 39. Alternative apparent resistivities for three-layer models. Dashed curve is for usual definition, solid curve is from real part of impedance. Dot-dash curve is from time domain. (a) Resistive second layer. (b) Conductive second layer. (After Spies and Eggers, 1986.)

Since the introduction of the impedance (Cantwell, 1960), mathematical difficulties have been noted in dealing with the  $\underline{Z}$  tensor in which the diagonal elements are commonly near zero and the major elements are off-diagonal (Swift, 1967; Eggers, 1982). Cevallos (1986) has overcome this difficulty by showing that it is due to an improper physical formulation of the problem. Using classical EM theory, he shows it arises from the fact that the magnetic field  $\mathbf{H}$  is a *pseudovector* or *axial vector*, rather than a true vector. This is not an obvious property, nor a question that arises very frequently in geophysics. It is only apparent in the behavior of  $\mathbf{H}$  under reflection, and is related to the fact that  $\mathbf{H}$  arises from the *curl* operation. On the other hand,  $\mathbf{E}$  is a true vector. The result is that, as defined,  $\underline{Z}$  is a *pseudotensor* which does not transform as a true tensor (Jackson, 1975). The proper definition of the MT impedance tensor  $\underline{Z}$  is

$$\mathbf{E} = \underline{\mathbf{Z}}(\mathbf{H} \times \mathbf{n}). \quad (32)$$

Similarly, a proper admittance tensor can be defined by

$$\mathbf{H} = \underline{\mathbf{Y}}(\mathbf{E} \times \mathbf{n}).$$

There is a simple relationship between the  $Z_{ij}$ , the elements of  $\underline{Z}$  and  $\zeta_{kl}$ , the elements of  $\underline{\mathbf{Z}}$ , namely  $\zeta_{11} = Z_{xy}$ ,  $\zeta_{12} = -Z_{xx}$ ,  $\zeta_{21} = Z_{yy}$  and  $\zeta_{22} = -Z_{yx}$ . Thus

$$\underline{\mathbf{Z}} = \begin{pmatrix} Z_{xy} & -Z_{xx} \\ Z_{yy} & -Z_{yx} \end{pmatrix}. \quad (33)$$

The major terms are now on the diagonal and combinations of  $\underline{\mathbf{Z}}$  and its complex conjugate transpose are hermitian, overcoming the objections pointed out in Swift (1967) and Eggers (1982). In particular the hermitian forms  $(\tilde{\mathbf{Z}}\underline{\mathbf{Z}})$  and  $(\tilde{\mathbf{Z}} + \underline{\mathbf{Z}})$  are shown to have fundamental physical significance, their principal directions are real, and  $(\tilde{\mathbf{Z}}\underline{\mathbf{Z}}/\omega\mu)$  is an apparent resistivity.

### Source Effects and CSAMT

We started by assuming that the source fields are effectively plane waves. It is observed in both practice and theory that the impedances taken from fields due to nearby sources are different from those obtained or expected with plane waves. The farther away the (natural or artificial) source, the closer the impedance approaches that for a plane wave. The justification for the plane wave assumption in MT was debated in early literature but was finally accepted on the basis of a report by Madden and Nelson written in 1964 but unpublished until recently (Madden and Nelson, 1986).

At audio frequencies the use of an artificial source is

now well established as the Controlled Source Audio Frequency MT (CSAMT) method, and thus appears in this volume. The  $E/H$  ratios from a CSAMT transmitter or from a visible thunderstorm are often very different from those obtained from remote storms. In such cases the impedance depends on the kind of source (loop, grounded wire, horizontal or vertical lightning stroke), its orientation, and its distance in skin depths. Among other complications there is no single obvious plotting position for an interpretive model since both transmitter and receiver positions are involved in the result. Because skin depth decreases with increasing conductivity and frequency, the impedance may well depend on transmitter location at low frequencies but approach the plane wave value at high frequencies. In CSAMT surveys an effort is made to place the transmitter as far from the survey positions as possible, consistent with adequate field strength, so as to take advantage of the simplicity of interpretation for plane wave source fields. Figure 40 shows apparent resistivities at a site from AMT and CSAMT measurements at the same dipole position. The CSAMT transmitter was parallel to the receiver dipole and about 10 km away along a line at right angles to the dipoles. Agreement is very good down to a transition frequency of 10 Hz ( $\rho_a$ ) or 30 Hz (phase).

How far does the receiver have to be from the source? Literature estimates range from seven to three skin depths (Bannister and Williams, 1974; Zonge and Hughes, this volume). Plots in the CSAMT chapter showing the fields from a grounded wire source indicate that the answer depends on azimuth relative to the source, as well as on distance. Dmitriev and Berdichevsky (1979) show that the fields can vary linearly across a region without affecting impedance, but this applies only to the 1-D (horizontal layers) case. There are several papers dealing with the TM approximation and discrete sources, including Law and Fannin (1961) and Quon et al. (1979). Yamashita et al. (1985) describe one approximate technique of correcting CSAMT data for source effects at low frequencies. This and a few other approximate methods are discussed in the Zonge and Hughes CSAMT chapter, this volume.

No methods proposed to date are very satisfactory: the only solution with CSAMT is to use a more powerful source at greater distances. The real problem arises when conductive near-surface rocks are underlain by even a thin resistive zone. Without other, independent information it is then difficult to tell whether the increasing apparent resistivity at low frequencies is a source effect or is due to increasing resistivity at depth.

## SENSORS

Proper choice of sensors is critical to the collection of good field data. We will confine this discussion to onshore measurements, although marine MT is done both for deep crustal studies (e.g., Filloux et al., 1985) and for petroleum exploration (Hoehn and Warner, 1983).

The electric field is measured by recording the voltage variation  $V$  versus time between a pair of grounded electrodes, distance  $\ell$  apart, and setting  $E = V/\ell$  in  $V/m$ . This assumes that  $E$  is constant over the line and gives the average value. To obtain  $E_x$  the line joining the electrodes must be in the  $x$  direction, and  $\ell$  must be big enough that  $V$  is much greater than the noise generated in the electrodes themselves. The major problems associated with the  $E$  measurement are electrode-generated noise and noise induced in the wire by wind motion. An electric field  $E_m = v_m \times B_0$  is induced in a wire moving with velocity  $v_m$  in the earth's main field  $B_0$ . Motion-induced noise is also a major problem when large loops are used to measure  $H$  components, as is often done with  $H_z$ . Wires and electrodes must be fixed so that they do not move during measurement. At high frequencies the wire joining the electrodes must be straight or an additional voltage will be induced in the wire by time variations in  $H_z$  (Swift, 1967).

An electrode in contact with the earth is an electrochemical cell. A voltage exists at the interface which depends on the chemical natures of the materials involved, on their interaction, and on the temperature. This voltage is of no interest to us, but it can be a major noise source if our electrodes are not chosen properly and treated carefully. Several types of electrodes are now in wide use. In addition to calomel standard cells, the common electrodes are nonpolarizing  $Cu - CuSO_4$ ,  $Cd - CdCl_2$ , and  $Ag - AgCl$  cells. (Cadmium cells are not recommended because of their toxicity.) Petiau and Dupis (1980) describe a  $Pb - PbCl_2$  system in Plaster of Paris or clay which can be very stable if properly made (Pelton, pers. comm.). Figure 41 shows the noise spectra of a variety of electrodes. However, different electrodes made of the same material can have vastly different noise, and each should be tested periodically. In the field it is common practice to install electrodes a day before use if possible, to give the system a chance to come to equilibrium. The effect of stabilization is also shown in Figure 41. No such data are available for frequencies above 10 Hz. Petiau and Dupis suggest that electrode noise is near signal at 1 Hz for a 100 m electrode spacing, and that signal/noise improves at both higher and lower frequencies. This improvement would obviously depend on resistivity and signal level. Karmann

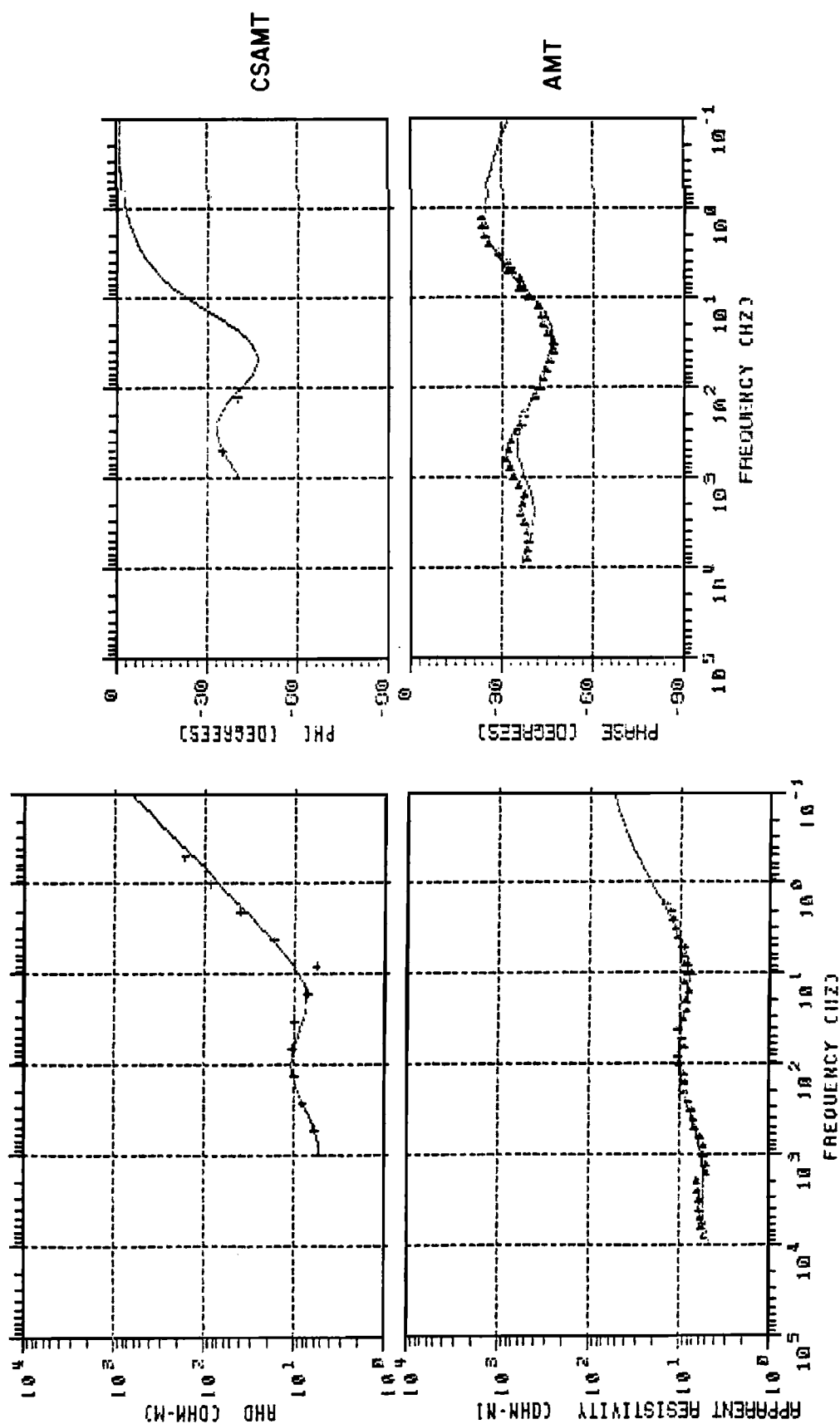


Fig. 40. Observed AMT and CSAMT apparent resistivities on the same dipole.

(1986, pers. comm.) attempted to measure broad band noise in a commercial electrode but found that it was less than the noise of the amplifiers he used.

The electric field measurements are often affected by very rapid, large changes in earth currents due to power systems and lightning. These changes can appear as spurious output signal unless the preamplifiers have very high common mode noise rejection and rapid response.

In exploration practice it is usually possible to increase the electric signal just by increasing  $\ell$ , the electrode spacing. This luxury doesn't exist for the  $H$  measurement. If the magnetic field strength is less than the equivalent magnetic noise in the sensors then the situation is serious and better technique, better conditions, or better sensors are needed for satisfactory results. A completely satisfactory magnetometer has yet to be devised. Two varieties of magnetometer are in general use, induction coils and SQUIDS, and both are commercially available.

At its simplest the induction coil is a loop of wire which produces a voltage proportional to its area

multiplied by the time derivative of  $B$  across the area. Given a sinusoidal  $B$ , the output voltage increases in proportion to frequency. The phenomenon is described in first year university physics textbooks. Since the fields of interest are very small, various steps must be taken to increase the loop voltage even before it can be electronically amplified. Its area can be increased substantially if it is to be laid out flat on the surface, to sense  $B_z$ . In addition use of many turns in series results in output proportional to the number of turns. Large, multiturn loops are, in fact, used for many surveys. However, such a large area air-core loop is obviously impractical for sensing the horizontal components of the field. Instead, effective loop area is increased by winding turns around a core of high magnetic permeability. Materials such as mumetal or moly-permalloy, alloys of molybdenum and nickel in steel, are used for lower frequencies. Nonconductive ceramics such as MN-60 (Ceramic Magnetics Inc.) are used for higher frequencies. As can be seen in Figure 42, the effective magnetic permeability depends on the material and its length/diameter ratio (Bozorth, 1964). These cores and the coils wound on them are long and narrow, so as to obtain the smallest demagnetization factor and thus the highest effective permeability of the core.

Real coils made up of many turns of wire on a permeable rod thus constitute circuits including distributed resistance, inductance, and capacitance, as well as core losses. A simplified equivalent circuit is

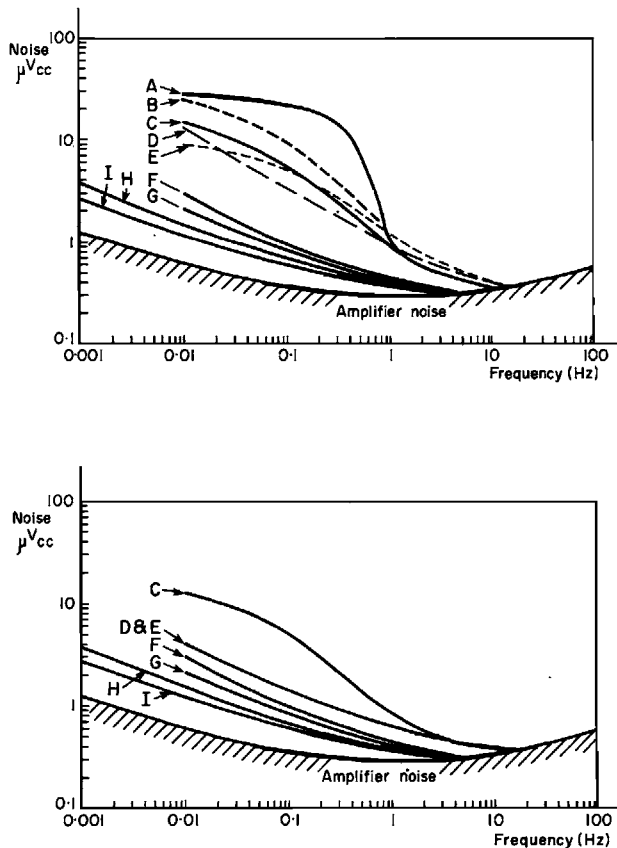


Fig. 41. Electrode noise spectra for several kinds of electrodes. Upper curves—1/2–1 hour after setup. Lower curves—after 24 hours. The two quietest are  $Pb - PbCl_2$  and  $Ag - AgCl$ . (After Petiau and Dupis, 1980.)

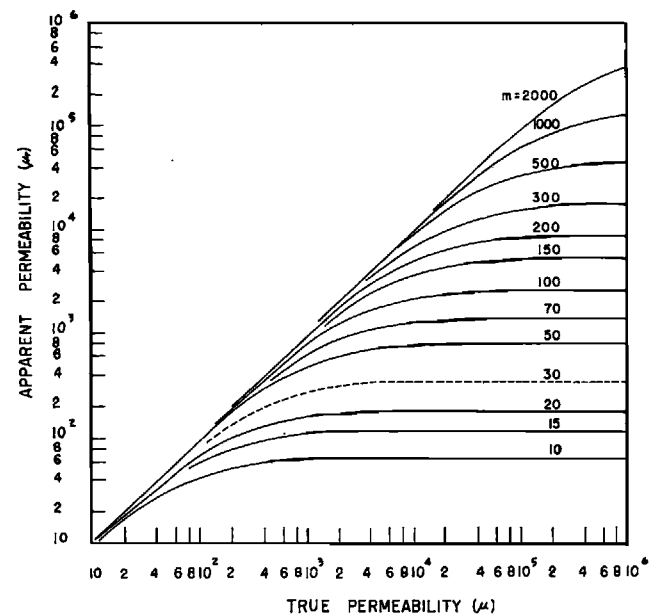


Fig. 42. Apparent magnetic permeability versus true permeability and length, for permeable rods.  $m$  is length/diameter ratio. (After Bozorth, 1964.)

shown in Figure 43. This circuit will have a resonant frequency, above which the voltage output decreases with increasing frequency. Because of the distributed nature of the circuit components there will be other resonances at higher frequencies as well. It is desirable to avoid operating at frequencies much above the first resonance because the response becomes difficult to compensate and is often temperature sensitive. Good coil design is an engineering problem requiring matching to a particular amplifier circuit and optimization for a particular frequency range and application. The basic references on coil design are the theses by Clerc (1971) and Karmann (1975), but Karmann (1977) is more easily available. In addition, useful literature is available from time to time from universities and manufacturers. Stanley and Tinkler (1982) give complete construction details and performance curves for a magnetometer designed for subaudio frequencies.

The frequency dependence of induction coil response below resonance is ideally suited to the measurement of natural fields below 10–100 Hz, where it very nearly compensates for the  $1/\text{frequency}$  falloff of the average natural spectrum (Figure 19). This prewhitens the sensor output, effectively extending the dynamic range of the entire system. For higher frequencies, where the spectrum flattens and then increases, a frequency-independent magnetometer response is desirable. This frequency independence can be achieved by feeding back into the coil a magnetic field proportional to the current. The scheme is described in Clerc and Gilbert (1964), MacIntyre (1980), and Labson et al. (1985). The technique can be carried a stage further, as shown in the response curve of the Metronix KIM802 (Figure 44). However this approach may not be worth the effort because no two data sets are alike and spectral behavior is only regular on average. On the other hand, if the interest is in applied fields, then it would seem reasonable to tailor system response to a highly predictable spectrum.

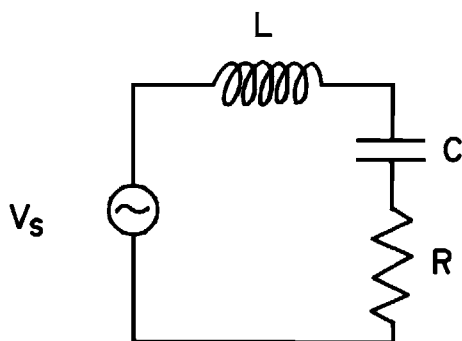


Fig. 43. Simplified equivalent of induction coil magnetometers. In practice the circuit elements are frequency dependent.

The intrinsic dynamic range of an induction coil is very great. At the large signal end the range is limited by the range over which the incremental permeability of the core material is linear, whereas at the small signal end it is limited by quantum effects in the core material and thermal effects in core and windings. Thus as long as the signal is much greater than thermal noise, and less than the Tesla level required to saturate the core material, coil output at a given frequency will be proportional to the input field. In a well designed system the limitations will lie in the amplifier. Typical noise in a good instrumentation preamplifier is  $1/2 \mu\text{V}$  rms. With a typical supply voltage of 10 V this gives a range of  $2 \times 10^6$  or 126 dB.

The other magnetometer commonly used in MT is the SQUID, based on the Josephson junction effect in superconductors. These should combine a high dynamic range with the lowest noise theoretically possible. The operating principle of the SQUID magnetometer depends on a quantum effect discovered in 1964 (Jaklevic et al., 1964). The principle is not as easily described as the induction coil, nor does it yet appear in first year physics textbooks. Clarke (1976), Zimmerman and Campbell (1975), and Falco and Schuller (1981) discuss the principle. A special Society of Exploration Geophysicists publication is devoted to the SQUID and its applications (Weinstock and Overton, 1981). Other descriptions of the SQUID are found in literature obtainable from manufacturers.

Two varieties of SQUID magnetometers, dc and RF, have been built, but only the RF SQUID is available in geophysical magnetometers. The central element is a very thin ring of material which becomes superconducting (resistance = 0) at liquid helium temperatures, around 4 degrees Kelvin. The material most commonly used is the metal niobium, although many others are available. The niobium is deposited on an insulating rod (Figure 45), and the circuit is nearly broken at one point by a notch, the Josephson

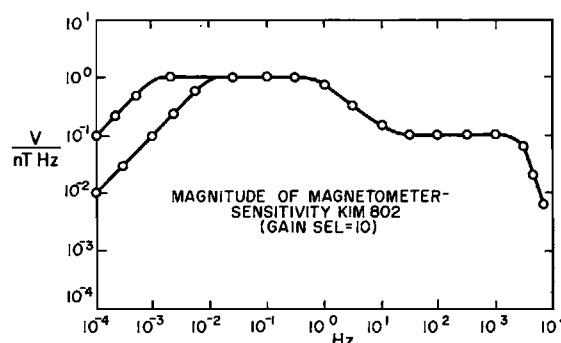


Fig. 44. Response of one commercial magnetometer system designed to prewhiten the natural spectrum.

junction, forcing the superconducting current through a tiny area. When it is superconducting, and in the absence of the notch, whatever magnetic flux threads the loop cannot be changed, since additional supercurrents will be induced which exactly cancel any new applied field. With the notch, both the current and the flux will change with the applied field, but not in a simple linear way. During part of the change the flux through the loop ( $\phi = B_n \times \text{loop area}$ ) increases more slowly than the applied field  $\phi_e$ . As  $\phi_e$  is increased further (Figure 46)  $\phi$  jumps discontinuously by amount  $\phi_0(1 - \phi_c/\phi_{ec})$ , where  $\phi_c$  and  $\phi_{ec}$  depend on the

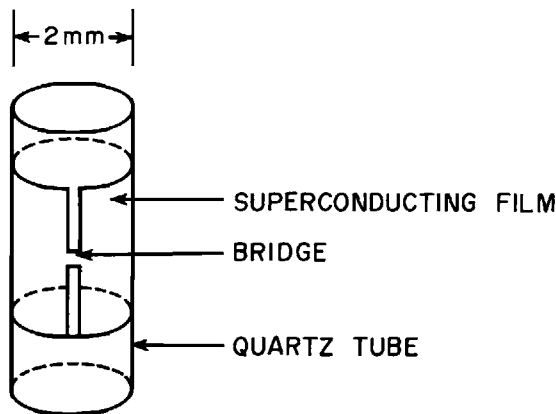


Fig. 45. Sketch of an RF SQUID element consisting of a thin film of superconducting alloy on an insulating cylinder. The weak link (bridge) in the alloy circuit enables the Josephson effect to occur. (After Clarke, 1976.)

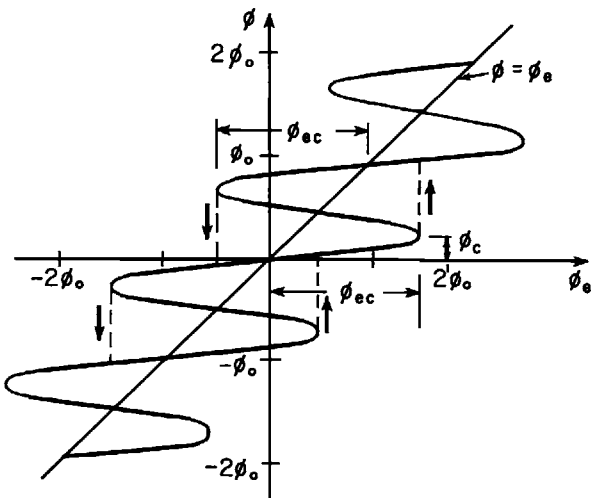


Fig. 46. Flux  $\phi$  passing through the superconducting loop in an RF SQUID, versus applied flux  $\phi_e$ .  $\phi$  changes discontinuously (arrows) as  $\phi_e$  increases (decreases) beyond a local maximum (minimum) in the smooth curve, thus changing the effective inductance of the circuit. (After Clarke, 1976.)

geometry of the loop and notch and  $\phi_0$  is the flux quantum,

$$\phi_0 = h/2e \approx 2 \times 10^{-15} \text{ Wb}.$$

If  $\phi_e$  is further increased the same happens again, but if  $\phi_e$  decreases then  $\phi$  initially decreases more slowly and then drops discontinuously. Thus during parts of this sequence the ring appears diamagnetic and in the other parts it appears paramagnetic. In the RF SQUID this response is sensed by an inductance bridge consisting of a tuned circuit whose inductor is magnetically coupled to the ring (Figure 47). The voltage in the circuit then depends on the state of the ring.

There have been numerous implementations of magnetometers using this effect. In most, a detector is used to look at the RF voltage (as in Figure 47), and the current is fed back into the bridge inductor to keep  $\phi_e$  at a minimum, the distance between minima corresponding to  $\pm\phi_0$ . The feedback signal is then proportional to the magnetic field change. If the signal becomes large enough to indicate that  $\phi_e$  has changed by  $\phi_0$  then the feedback current is reduced by an amount equivalent to  $\phi_0$  and the system shifts to the next minimum. A counter keeps track of the number of such transitions to add to the output. In the recent generation of SQUIDS the counting and output are digital. Then the A/D conversion is arranged so the least significant bit corresponds approximately to the rms noise. This arrangement varies from  $10^{-4}$  to  $10^{-5}$  nT in different equipment. [Very recent developments appear to have achieved noise levels of  $10^{-7}$  nT (Henderson, R.J., pers. comm., 1987)].

To keep the sensing element in liquid helium, special containers (dewar flasks) are necessary. Unlike other SQUID applications, the container for a magnetometer must be nonmagnetic. They are made of fiberglass, with double walls separated by a vacuum. Dewars lose helium through evaporation as heat slowly leaks through the walls. In addition, vibration in transport is

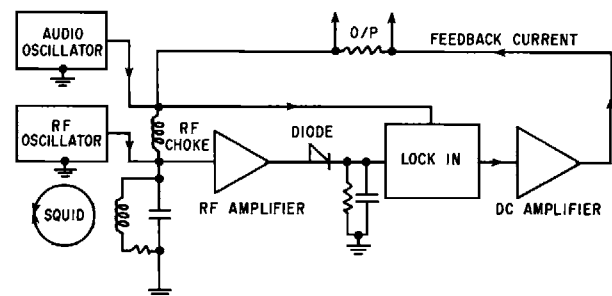


Fig. 47. Simplified circuit of an RF SQUID system. The tuned circuit (lower left) is driven by the RF oscillator and senses changes in the effective inductance of the SQUID element. (After Clarke, 1976.)

converted to heat in the helium, causing further evaporation. Hence containers must be refilled periodically. To avoid too frequent refilling the containers must be large enough to hold at least enough helium for a day, and preferably for several days. Capacities of 5–10 liters are made, but 30–50 liter sizes are more common (Figure 48). Larger containers of 100–300 liter capacity are used for bulk storage. Liquid helium is sometimes difficult to obtain and awkward to transport, which has tended to discourage the use of SQUID magnetometers.

A persistent problem with SQUID magnetometers has been the effect of spherics. Their high frequency content shifts  $\phi_e$  faster than the response time of the electronic circuitry. The system shifts to a new minimum, but the amount of shift is uncertain and the system is said to have lost synchronization. To overcome this behavior it is now common to low-pass filter the field by building a conductive shield into the dewar or by wrapping a shield around the element. The frequency response can be selected very precisely. In addition, the faster response of newer circuitry also helps overcome this situation.

The dynamic range of SQUIDS must be larger than that of induction coils because SQUID response does not prewhiten the natural pink spectrum. One manufacturer quotes a range of 193 dB (32 bits) with a resolution to  $10^{-5}$  nT over the entire range.



Fig. 48. Commercial 3-component SQUID magnetometer. (Courtesy Biomagnetic Technologies, Inc.)

Finally, it is important to look at the noise levels of various sensors, with respect to signal levels and to one another. Some but not all of this data is provided by the manufacturer. Figure 49 shows many of the available noise data in terms of equivalent magnetic field spectral density. It is particularly interesting to look at the SQUIDS data. On the basis of laboratory measurements and theoretical predictions SQUIDS should have a noise level of about  $10^{-5}$  nT/(Hz)<sup>1/2</sup>, yet in the field the level is consistently about 10 times larger, based on remote reference analysis (see the Data Processing and Analysis Section). Numerous explanations have been offered, especially that the noise level is a result of microseismic motion. A series of experiments begun in 1983 has shown that much of the anomalous SQUID noise is seismic in origin (Goubau et al. 1984; Morrison et al. 1984; Nichols et al. 1988). By using one SQUID to cancel the field at another several kilometers away, they showed that much of the residue is highly coherent with a tiltmeter. They then used the tiltmeter output to cancel the residue. In all, the cancellation amounted to over 70 dB in parts of the spectrum.

At present, without such cancellation, induction coils have lower noise than SQUIDS above 100 Hz, but it remains to be determined which is superior in the critical 0.1–10. Hz band.

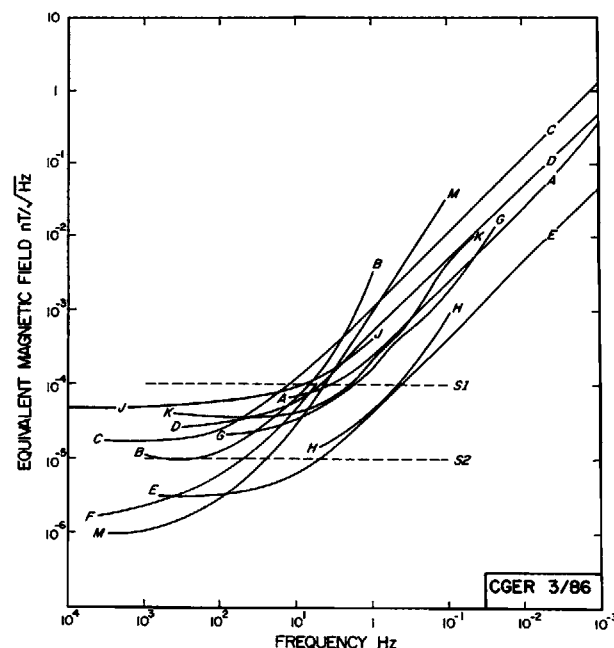


Fig. 49. Noise figures reported for various magnetometers. S1 and S2 are, respectively, typical observed and theoretical figures for SQUIDS.

## FIELD PROCEDURES

Careful field procedure is probably more crucial to successful results in MT and AMT than in any other EM technique, simply because the fields are so small. Site selection and sensor installation are the two major factors governing data quality. The cost of MT surveys depends upon the length of time which must be spent at each site. From experience (Vozoff et al., 1985), given good recording conditions and considering the time to install the TEM transmitter, MT measurements at a single site should take less time than transient EM measurements for the same transfer function bandwidth and the same error bars. MT has an even greater advantage if tensor measurements are desired. On the other hand, once the TEM transmitter has been installed, subsequent TEM measurements can be done more quickly. Hence TEM has a distinct advantage over MT for intensive scalar surveys over an area which is accessible from a single transmitter. Conversely, MT has an advantage where tensor information is required over a large area.

As a general statement, the better the installation, the better the data and the less time required at the site. However there are many variables over which the crew has no control. Many crew-years of experience have taught us to avoid powerlines, pipelines, electric railways, electric fences, welding activities, smelters (especially aluminum smelters), irrigation pumps, and radio and radar transmitters. Towns, vehicular traffic, and even pedestrian traffic can act as local field sources for which the impedance differs from the plane wave value. Metal structures such as billboards, flagpoles, (passive) metal fences, and metal strands on wooden fenceposts can locally distort the magnetic field, or cause sensor motion when the wind blows. Large grounded conductors such as culverts, rails, and telephone and power grounding circuits will distort the E field locally.

The amount of flexibility in locating sites differs in every case, and the range of interference differs for every source. If possible, avoid large, active sources such as powerlines, railways, and distributed, well-grounded conductors such as pipelines and metal fences by at least 1 km; poorly grounded passive conductors and vehicular traffic by 100 m; and small, passive, static objects by 50 m. With experience in a given area some of these rules can be relaxed but others may be inadequate. Wherever practicable avoid natural features which might affect the fields, especially the electric field. These objects include abrupt topographic features such as sinkholes and pinnacles, small salt pools, outcrop in areas that are otherwise soil covered, etc.

In common with other geophysical field methods,

certain gnawing animals pose more or less severe problems to cables and other equipment. It is prudent to pack up and remove everything possible when leaving an area overnight, the exception being electrodes intended for use the following day.

Even in the most remote regions, care must be taken to obtain good data. Magnetometers, which must be shielded from wind and sun, are commonly buried in trenches or holes covered by dirt or a wooden sheet. Depending on soil conditions, this may or may not be easy for horizontal, permeable core induction coils which require only shallow trenches. It is more difficult for SQUIDS because of the size and shape of the dewars, so a combination of partial burial and a streamlined shield are commonly used. Induction coils for  $H_z$  require drilling a vertical hole. Given an adequate thickness of soil the hole can usually be dug quickly by power auger. Where soil conditions are more difficult, a small drill can put down enough holes for many days' operation in a few hours. In some situations it is more convenient to lay an open loop on the surface, anchored against wind motion by clumps of dirt. As noted earlier, coils should be placed as far as possible from the roots of trees, because the trees will move when the wind blows.

Electrodes should be installed, at least a few hours and preferably a day before they are to be used, in shallow holes which have been water saturated, and covered to retard evaporation and temperature change. Resistance of the electrode circuit must be checked before recording begins, as is done automatically by some systems. If the resistance is too high for the input impedance of the preamplifier, breaking up the surrounding soil and adding more brine can overcome the problem. This procedure is not always effective enough. Where resistance is unacceptably high, an alternative is to install several electrodes in parallel. These electrodes must be as far apart as possible to be effective, but their maximum spacing has to be a small fraction of the end-to-end spacing. The consequences of high contact resistance can be severe, in terms of noise and lost signal. The consequences are illustrated and discussed at length in the Zonge and Hughes CSAMT chapter this volume, (Zonge and Hughes, 1989), where it is suggested that impedance-matching amplifiers be used at the electrodes. The problem is most important at high frequencies and large electrode spacings. The wire between electrodes should be as near a straight line as possible. Otherwise the wire forms part of a horizontal loop which responds to changes in  $H_z$ , the effect increasing with frequency (Swift, 1967).

When remote reference acquisition is carried out [Clarke et al. (1978)] then consideration must be given to the location of the reference site and to means of

collating the data. Data collection is done by radio telemetry, by hard wire link, or by synchronized recording. The first is preferred if the reference is used for a number of recording sites and when the distances are more than 1–2 km. Hard wire is simplest in principle, but synchronized recording is a practical option with the crystal clocks which are now available. The spacing between the reference and the data sites depends on the nature of the major noise source anticipated: if it is instrumental then they can be quite close together, as discussed in the next section.

Major determinants of survey cost, aside from accessibility of the area and the benevolence of the environment, are the spacing between sites and the recording time required. Whereas recording time is increased by adverse noise conditions, as noted above, the minimum is determined by the depth of interest and the resistivities of the section. The greater the depths and the lower the resistivities, the lower are the frequencies required and, therefore the greater the recording time. Nominal safe recording times for various geological environments in the absence of serious noise contamination are shown in Vozoff (1972, Figure 36). The best indication of minimum time is obtained from inversion on *assumed* models. This inversion can show directly, for particular noise assumptions, the frequency range and record length necessary to resolve a chosen target. For petroleum exploration the recording times range from 1/2 day to 2 days. In geothermal problems, 1/2 to 1 day is typical and in AMT surveys we have spent 2–6 hours at a site, depending on signal strength.

Minimum site spacing also depends on the conductivity structure and the degree of lateral resolution needed. In (conventional) tensor surveys, spacings of 5–10 km might be used in reconnaissance, and 1/2–1 km or less where greater detail is wanted.

As experience with the method increased and closer spacings were used for better lateral resolution, it became clear that MT, like the reflection seismic method, sometimes suffered from what are now called “statics” problems (Sternberg et al., 1982; Warner et al., 1983; Andrieux and Wightman, 1984; Sternberg et al., 1985). That is, entire apparent resistivity curves can be vertically shifted on account of large but superficial lateral conductivity contrasts. The problem is discussed later in this chapter. Special procedures have been devised to cope with this situation. The most elaborate of these is the EMAP technique, which requires contiguous *E* field measurements along a profile (Bostick, 1985, 1986a and b). This is obtained by using a multiconductor seismic cable with takeouts to electrodes spaced the order of 500 m apart. Lateral changes in the electric field are smoothed, over distances equivalent to skin depth at that frequency and

position, with a spatial filter applied to the array of *E* measurements. For reasons of practicability, only one local and one remote magnetometer are used in EMAP. Hence only partial tensor information is obtained from the EMAP measurement, although full tensor data could be collected at the same time if enough equipment and data channels were available. Processing and presentation of EMAP data are discussed in a later section.

Finally, there are now (late 1990) appearing general multichannel receivers permitting many tens of channels to be digitally recorded simultaneously. These advances are bound to affect field procedures, making it feasible to collect data at more closely spaced stations at reasonable cost. The same devices can in principle be used for measurements with applied fields, reducing the cost of statics compensation procedures.

## DATA PROCESSING AND ANALYSIS

### General Considerations

The object of data processing is to extract from the noiselike signals a set of smooth, repeatable functions representing the earth's response, which can be used to interpret conductivity structure. In this area and in interpretation our capabilities are changing most rapidly as a result of new developments in robust estimation, 3-D modeling, and impedance tensor analysis. The response functions which we seek correspond to those which can be computed for models. Basically they consist of only two functions, impedances and tippers, and their attributes phases, special directions, skew, and ellipticity. Apparent resistivities are calculated from the impedances. Various estimates of noise can also be computed from repeated field measurements, for use in estimating confidence limits on the parameters of the models which are fitted to the response functions. These estimates include the coherencies among field components, sample variances, and signal/noise ratios. With the incursion of ever greater computing capability into the field, much work is now done in real time or almost real time. Figure 50 is a summary plot of conventional response functions from a single site, as delivered by one contractor. It illustrates the need in MT for rapid and flexible graphics output even in the field. The emphasis in the section is on the processes rather than on specific implementations.

The bulk of data processing is done in the frequency domain. There have been proposals for time domain processing (Wait, 1954; Kunetz, 1972), but these proposals have never been implemented for routine surveys, as far as I know. The following discussion deals

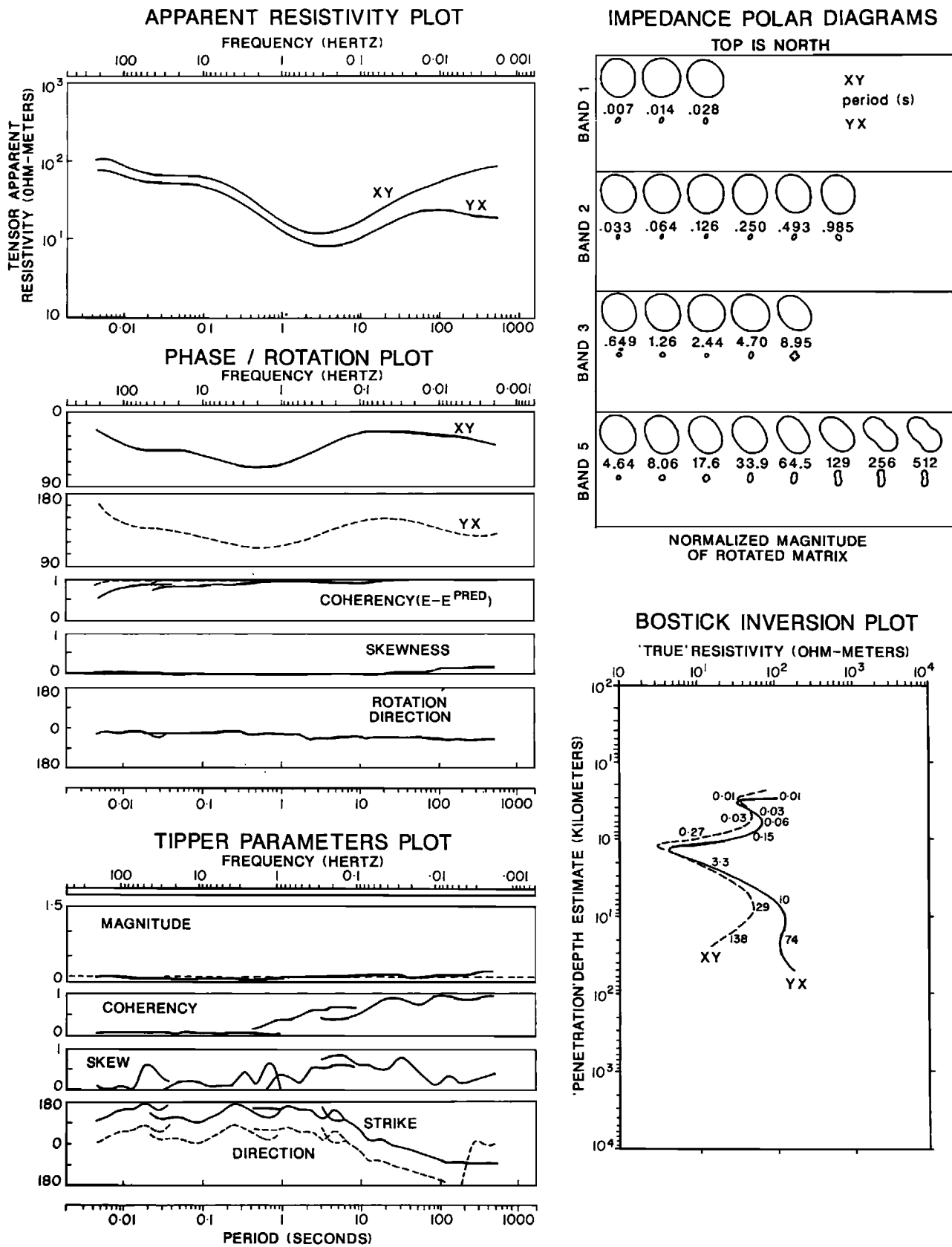


Fig. 50. Summary of the processing results at one station. (After Z-Axis Exploration, Inc.)

with frequency domain processing, but the time domain ideas are described at the end of this section.

In a homogeneous or horizontally layered earth, and in the absence of noise, apparent resistivity could be determined simply by simultaneously measuring the amplitudes of  $E_x$  and  $H_y$  at a specific time and frequency (using band-pass filters, say), squaring the ratio, and multiplying by the appropriate constant. This could be done for a sequence of frequencies to define as much of the curve as desired. Much MT was done this way in the early years, and it is even the basis of one new system. However, in practice there is always noise in the data and there are usually lateral resistivity changes occurring close enough to a site that apparent resistivity depends on measurement direction. Hence almost universal practice is to measure both horizontal components of  $\mathbf{E}$  and  $\mathbf{H}$  in order to find  $\mathbf{Z}$ , instead of trying to use just one perpendicular  $\mathbf{E}$ - $\mathbf{H}$  pair.

I first describe the conventional methods used to extract smooth estimates of the  $Z_{ij}(f)$ , etc., from the noisy signals  $E_i(t)$  and  $H_j(t)$ , where  $i$  and  $j$  can be  $x$  or  $y$ . The equations that must be solved are [Equations (14)]

$$E_x(f) = Z_{xx}(f)H_x(f) + Z_{xy}(f)H_y(f)$$

$$E_y(f) = Z_{yx}(f)H_x(f) + Z_{yy}(f)H_y(f),$$

i.e., two complex equations in four complex unknowns, the  $Z_{ij}(f)$ . The equations are complex because all quantities have both magnitude and phase. They can be solved because we can obtain many independent estimates of the  $E_i(f)$  and  $H_j(f)$ . Solutions can be obtained in either of two ways: by averaging repeated time measurements since the  $Z_{ij}$  will not change with time, or by averaging over a number of closely spaced frequencies since they change only very gradually with frequency. If the data have a large signal/noise ratio the results are the same either way.

Assume that the signals are in digital form: virtually all MT and AMT data now collected are amplified, filtered, multiplexed, and digitized, typically by (10 to 16 bit) analog-digital converters (ADC). Frequency ranges and conversion rates are chosen to suit the geophysical problems and available hardware. Some signal conditioning must be done before digitization and further signal preparation may be desirable or essential before spectral analysis. Signals must be low-pass filtered before the ADC. The low-pass filter output must conform to the requirement that there be no significant energy above the Nyquist frequency entering the ADC (Bendat and Piersol, 1971). In addition, high-pass filtering is sometimes done to overcome instrumental drift and reduce the demands on

dynamic range of the low frequency part of the signal spectrum. Repetitive signal such as powerline noise must be notch filtered. These are most important for the 10–12 bit ADC's and when long data sets are desired. (One of the SQUIDS described in the previous section has a dynamic range of 32 bits in order to deal with its dc response to the  $1/f$  spectrum.) Removal of the mean, detrending (subtraction of a linear trend) or low-pass filtering, and windowing (tapering the ends by a cosine or similar function) are most conveniently done after digital conversion (Young, 1981).

Conversion from time to frequency domain is usually done by Fast Fourier Transform (FFT) because of its speed. After Fourier transforming, a correction is made to the resulting spectrum of each channel for the absolute complex frequency response of that channel's instrumentation, resulting in true spectra of the  $E$  fields in  $V/m \sqrt{Hz}$ , and in  $nT/\sqrt{Hz}$  for the  $H$  fields. Most algorithms require that  $M$ , the number of data points analyzed, be  $M = 2^n$ , where  $n$  is an integer. Typical values of  $n$  are 5 to 10 ( $M = 32 - 1024$ ) and rarely exceed 12. Values of the complex spectrum are produced at  $N = M/2$  frequencies, linearly spaced from  $f_1$  to  $f_N$ , where  $f_N = 1/(2\Delta t)$  is the Nyquist frequency and  $\Delta t$  is the sampling period of the ADC.

Each spectrum can be expressed either in terms of its real and imaginary parts, or in the equivalent magnitude and phase. From the physics involved, this phase and (the logarithm of) impedance must be smooth functions when plotted on a logarithmic frequency scale. The curves can easily be defined by a half dozen points per frequency decade. Hence the FFT generally yields many more frequencies than can be used for interpretation, especially at higher frequencies. In one approach, impedance values are averaged over adjoining groups of frequencies, or windows (Figure 51) to solve the underdetermined equations, in order to provide smoother results and to reduce the number of data to be dealt with. To obtain  $n$  windows per decade of frequency, the ratio of center frequencies of adjacent windows is  $\log(f_{j+1}/f_j) = 1/n$ .

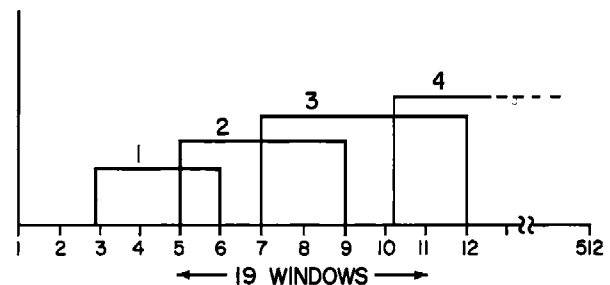


Fig. 51. Windows (1–19) used for logarithmic averaging of linearly spaced harmonics (1–512).

Another approach is to digitally filter many short data sets at only the desired frequencies, thus obtaining multiple solutions of the impedance equations at a few frequencies. These solutions can then be stacked or otherwise averaged, as discussed in the Errors and Noise Section. Avoiding the full FFT computation can save computing time.

An ingenious scheme for continuous MT acquisition and processing, called cascade decimation, is given in Wight and Bostick (1981). In their implementation, short data sets (32 points) are collected in dual buffers, one of which is being filled while the other is being processed. The latter is sine- and cosine-transformed at a few mid-band frequencies and those values are saved. The data are then low-pass filtered to half the original Nyquist frequency and alternate points are discarded. The same is then done to the data in the other buffer while the first is being refilled. The two decimated data sets are then joined to form a new 32 point data set at half the original sampling frequency and the process is repeated. This could be continued indefinitely. The few transform values which are kept from each data set can be used to obtain autopower and crosspower in the same way as the averages described next, but they must be averaged over several data sets rather than over a band of several adjacent frequencies from a single data set.

In our AMT work we found that useful results were generally obtainable only when the data set included one or more obvious spherics, at least for frequencies from 8 Hz to 10 kHz. Presumably this was because the signal/noise ratio was too small otherwise. For that reason the data acquisition system was designed to accept data in a flow-through buffer until a selected "trigger" level was detected. The buffer memory retained 100–200 points before the onset of the trigger signal and continued so as to make up a 1024-point data set for each channel. The 5 (or 7) channels of 1024 points were then automatically transferred to another buffer and simultaneously displayed, via a digital/analog converter, on the screen of an oscilloscope. The operator had the option of saving and processing that data set for possible stacking, or of discarding it. The trigger was reset as soon as a data set had been transferred, so very little signal was lost while processing. The results were available on a monitor screen within a minute or so of capture, and it soon became clear which kinds of spherics gave useful results. This procedure also allowed selection of spherics on the basis of their source direction, by switching the trigger circuit to any desired component. The approach worked very well for frequencies above 5–8 Hz, the waiting time for triggers varying from nil to a few minutes. It was not successful for frequencies between 0.3 Hz and 8 Hz, partly because of poor signal/noise

ratio at those frequencies in our induction coils. Interestingly, in a simple sedimentary basin, we were able to obtain smooth and repeatable apparent resistivities directly from the Fourier transforms of signals from an orthogonal sensor pair, without averaging (Holliday, 1981).

### Auto- and Cross-Spectra

Once we have a data set in the frequency domain, the basic tools of the band averaging procedures are the auto-spectra and cross-spectra. Say the corrected spectral value of channel  $A$  at frequency  $f_j$  is  $A_j = a_j + ib_j$ , ( $i = \sqrt{-1}$ ). Then an estimate of the average auto-spectral density, or spectrum, in the band  $(f_{j-m}, f_{j+m})$  around  $f_j$  is

$$\begin{aligned}\langle A(f_j) \rangle &= \sqrt{\frac{1}{2m+1} \sum_{k=j-m}^{j+m} A_k A_k^*} \\ &= \langle A_j A_j^* \rangle^{1/2},\end{aligned}\quad (34)$$

which is real in this case. The square of this is the autopower spectral density, or autopower, at  $f_j$ . In the same way we can calculate the crosspower density at  $f_j$  of two channels, say  $A$  and  $B$ , by taking

$$\begin{aligned}\langle A(f_j), B(f_j) \rangle &= \frac{1}{2m+1} \sum_{k=j-m}^{j+m} A_k B_k^* \\ &= \langle A_j B_j^* \rangle,\end{aligned}\quad (35)$$

which will generally be complex. The square root is called the cross spectral density, and is also complex. The *coherence* or *coherency* of the two channels at frequency  $f_j$  is defined as

$$\begin{aligned}\text{coh}(A_j, B_j) &= \frac{1}{2m+1} \sum_{k=j-m}^{j+m} \frac{A_k B_k^*}{\sqrt{(A_k A_k^*)(B_k B_k^*)}} \\ &= \text{coh}(A, B)_j\end{aligned}\quad (36)$$

which is complex with modulus between 0 and 1. It will have a phase given by  $\arctan[\text{Im}(\text{coh})/\text{Re}(\text{coh})]$ . The coherency and crosspower are very powerful tools in the extraction of information from the pulsation signals. A simple calculation will show that coherency is large when the ratios of imaginary to real parts of the  $A_j$  and  $B_j$  are about the same over the summation band. That is, high coherency implies that any variations in the moduli or phases of the two functions over the band occur together.

### Solutions to the Impedance and Tipper Equations

The conventional way of solving the impedance equations assumes that the  $Z_{ij}$  are constant over an averaging band (window), which is physically reasonable if the bands are narrow enough. In each band, each equation has crosspower taken with  $H_x$  and  $H_y$  in turn, giving pairs of equations such as

$$\begin{aligned}\langle E_x H_x^* \rangle &= Z_{xx} \langle H_x H_x^* \rangle + Z_{xy} \langle H_y H_x^* \rangle \\ \langle E_x H_y^* \rangle &= Z_{xx} \langle H_x H_y^* \rangle + Z_{xy} \langle H_y H_y^* \rangle\end{aligned}\quad (37)$$

or two complex equations in the two complex unknowns  $Z_{xx}$  and  $Z_{xy}$ . The real and imaginary parts of each  $Z$  are found by solving the pair of simultaneous equations, viz

$$Z_{xx} = \frac{\begin{vmatrix} \langle E_x H_x^* \rangle & \langle H_y H_x^* \rangle \\ \langle E_x H_y^* \rangle & \langle H_y H_y^* \rangle \end{vmatrix}}{\begin{vmatrix} \langle H_x H_x^* \rangle & \langle H_y H_x^* \rangle \\ \langle H_x H_y^* \rangle & \langle H_y H_y^* \rangle \end{vmatrix}}\quad (38)$$

giving a least-squares estimate of  $Z_{xx}$  over the band. The same is done for the other three tensor elements. From additional data records one can obviously compute additional estimates of each  $Z_{ij}$  which will be different because of noise variation between records.

Having estimated values for the  $Z_{ij}$  there is now the interesting possibility of using Equation (14) to predict what  $E_x$  and  $E_y$  should be if they were due entirely to the observed  $H$  components, that is, if they were noise-free. This is done by writing

$$\begin{aligned}E_x^p &= Z_{xx} H_x + Z_{xy} H_y \\ E_y^p &= Z_{yx} H_x + Z_{yy} H_y\end{aligned}\quad (39)$$

where the superscript indicates a predicted value. Differences between like components of  $E$  and  $E^p$  arise only on account of the part of the noise which is not common to  $E$  and  $H$ . It is then a straightforward matter to compute the coherencies between the observed and predicted components,  $\text{coh}(E_i, E_i^p)$ . These have become a very useful measure of data quality. They are called *multiple coherencies* or *predictabilities*, and are real numbers in the range (0, 1).

In estimating the  $Z_{ij}$  as in Equations (37) and (38), crosspowers could of course be taken with  $E_x$  and  $E_y$  instead of  $H_x$  and  $H_y$ . In fact six separate estimates are available. Sims et al. (1971) examined the possibility of using some combination of the six possible kinds of estimates to get smoother impedances. They also studied the *admittances*,  $\underline{Y}$ ,

$$\underline{Y} = \underline{Z}^{-1},$$

calculated from

$$\underline{H} = \underline{Y} \underline{E}.$$

From this, an  $H^p$  and a coh ( $\underline{H}\underline{H}^p$ ) could also be calculated. Sims found that results were best when channels with the least noise were used to predict the other channels.

The tipper is found by solving Equation (25) in the same way as used to find the impedance. That is, the crosspowers of  $H_z$  are taken with  $H_x$  and  $H_y$  to give two equations in  $T_x$  and  $T_y$ ,

$$\begin{aligned}\langle H_z H_x^* \rangle &= T_x \langle H_x H_x^* \rangle + T_y \langle H_y H_x^* \rangle \\ \langle H_z H_y^* \rangle &= T_x \langle H_x H_y^* \rangle + T_y \langle H_y H_y^* \rangle.\end{aligned}\quad (40)$$

As with the horizontal components, an  $H_z$  predictability can be calculated from coh ( $H_z^p, H_z$ ) using  $H_z^p = T_x H_x + T_y H_y$  and the  $T_i$  from equation (40).  $H_z^p$  will also be real.

Tipper direction and tipper strike were explained in a previous section. Their derivation in the 2-D case is straightforward. In the more general case, the ratio  $T_y/T_x$  is complex and there are several possible ways to define the rotation, none of them completely satisfactory (Vozoff, 1972; Jupp and Vozoff 1976; Gamble et al., 1982). This is to be expected because the concept is basically derived from 2-D models. One commonly used rotation, which maximizes the cross power between horizontal and vertical components, is

$$\phi_1 = \frac{(a^2 + c^2) \arctan(c/a) + (b^2 + d^2) \arctan(d/b)}{T^2}\quad (41)$$

where  $T_x = (a + ib)$ ,  $T_y = (c + id)$ , and  $T^2 = (a^2 + b^2 + c^2 + d^2)$ . Another definition, from Sims and Bostick (1969), maximizes  $T_y'$  in Equation (26). That gives

$$\phi_2 = (1/2) \arctan \frac{2(ac + bd)}{(a^2 + b^2) - (c^2 + d^2)}.$$

In each case, 90 degrees should be added or subtracted for strike direction. When these points are badly scattered, plots of several of the varieties of  $\phi$  can sometimes be helpful.

Each tipper definition gives a different phase. Phase  $\phi_T$  corresponding to definition (41) is

$$\Phi_T = \frac{(a^2 + b^2) \arctan(b/a) + (c^2 + d^2) \arctan(d/c)}{T^2},$$

and is the one usually plotted in routine surveys with  $(-90^\circ \leq \Phi_T \leq +90^\circ)$ . Like  $|T|$ ,  $\Phi_T$  is invariant under

rotation. In the 2-D case  $\Phi_T$  is the phase of  $T_y'$  in equation (26).

A third attribute of the tipper is its skew, defined by  $S_T = 2(ad - bc)/T^2$ .  $S_T$  is zero in the (noise-free) 1-D and 2-D cases and is invariant under rotation. Jupp and Vozoff (1976) give a more detailed description of tipper analysis. (Note that the first equation in that paper is incorrect. Equation (20) here is correct.)

### Remote Reference

The well known problem with the above procedure is the biasing effect of noise through the autopowers. That is, if  $A = S_a + N_a$ , and  $B = S_b + N_b$ , where  $S$  and  $N$  are signals and noises, respectively, then

$$AA^* = S_a S_a^* + 2S_a N_a^* + N_a N_a^*$$

and

$$AB^* = S_a S_b^* + S_a N_b^* + S_b N_a^* + N_a N_b^*.$$

If it is assumed that signals are coherent but that signals and noise are incoherent (coherency  $\approx 0$ ), then  $AA^* > S_a S_a^*$  but  $AB^* \approx S_a S_b^*$ . Thus the autopower estimates will be biased upward whereas the cross-powers will be unbiased. Every equation (such as equation 38) which contains autopower will suffer a bias error. If we let  $H_x$  be  $A$  and  $H_y$  be  $B$ , the solutions to the four simultaneous equations can be written

$$\begin{aligned} Z_{xx} &= \frac{\langle E_x A^* \rangle \langle H_y B^* \rangle - \langle E_x B^* \rangle \langle H_y A^* \rangle}{\langle H_x A^* \rangle \langle H_y B^* \rangle - \langle H_x B^* \rangle \langle H_y A^* \rangle} \\ Z_{xy} &= \frac{\langle E_x A^* \rangle \langle H_x B^* \rangle - \langle E_x B^* \rangle \langle H_x A^* \rangle}{\langle H_y A^* \rangle \langle H_x B^* \rangle - \langle H_y B^* \rangle \langle H_x A^* \rangle} \\ Z_{yx} &= \frac{\langle E_y A^* \rangle \langle H_y B^* \rangle - \langle E_y B^* \rangle \langle H_y A^* \rangle}{\langle H_x A^* \rangle \langle H_y B^* \rangle - \langle H_x B^* \rangle \langle H_y A^* \rangle} \\ Z_{yy} &= \frac{\langle E_y A^* \rangle \langle H_x B^* \rangle - \langle E_y B^* \rangle \langle H_x A^* \rangle}{\langle H_y A^* \rangle \langle H_x B^* \rangle - \langle H_y B^* \rangle \langle H_x A^* \rangle} \end{aligned} \quad (42)$$

Each of these equations is subject to bias error since each contains autopower terms.

Much of the scatter observed in many MT results can be explained this way. Each data set has different noise and hence each  $Z_{ij}$  estimate is different. A development having major impact on the effectiveness of MT has been the remote reference (RR) method (Clarke et al., 1978; Gamble et al., 1979a), in which two independent signal channels are recorded for use as  $A$  and  $B$ , in place of  $H_x$  and  $H_y$  in Equation (42). By providing a kind of synchronous detection, the method helps compensate for noise, both internal and external to the equipment. Figure 52 compares apparent resistivities calculated in both ways from the same data set.

The remote channels, usually designated  $R_x$  and  $R_y$ ,

are most commonly signals from  $H_x$  and  $H_y$  magnetometers a few hundred meters to a few kilometers away. Sometimes it is more convenient to use a pair of  $E$  field receivers. Recent field experiments showed that the reference sensors need only be a few tens of meters from the central array to be effective (Goubau et al., 1984), but this must depend on the origins and spatial distribution of the noises. Situations do arise when it is impractical to separate the two sites sufficiently, and the reference site is subject to the same noise as the main site. No benefit can be expected from RR measurements in such cases.

One fascinating aspect of the RR approach is that the noise spectrum itself can also be calculated for each channel (Goubau et al., 1978; Gamble et al., 1979b). It is not unusual for noise to exceed signal in some parts of the spectrum of some data sets, particularly near 1 Hz (Figure 53). The RR method has made it possible to routinely extract usable results even under those conditions (Figure 54). As a byproduct, the ability to estimate a noise spectrum for each signal channel provides a useful means of identifying noise sources in equipment and in an environment.

The present assessment of RR is that it is definitely cost-effective under "average" conditions, but may be unnecessary in very quiet locations and is unable to help in extremely noisy locations. RR is widely used in routine surveys.

### Errors and Noise

Despite the RR method, data acquired in noisy environments does show scatter. For both RR and conventional data, a variety of processes have been adopted to deal with scatter in the estimates of  $Z_{ij}$ , and results by and large have been good. To begin, it

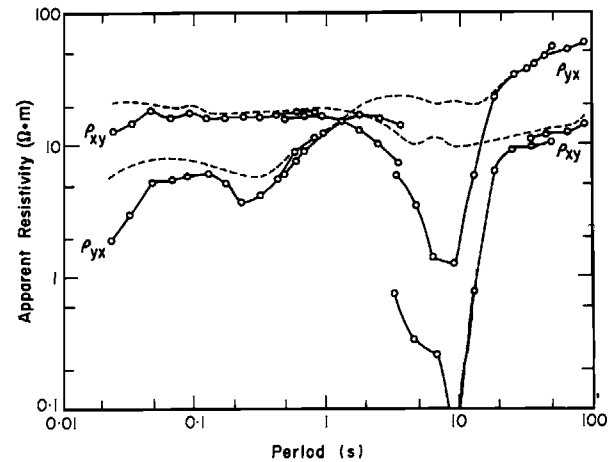


Fig. 52. Comparison of apparent resistivities obtained by conventional processing (open circles) and by Remote Reference (dashed lines). (After Gamble et al., 1979a.)

should be pointed out that the kinds of summation stacking of signals which are carried out on repetitive signals (radar, reflection seismic, transient EM) will not work with MT: random signals will merely stack to their long term mean value of zero. The same is true for their Fourier transforms. Instead, stacking must be done with the autopowers and crosspowers, which are nonnegative and will not sum to zero. In-field comput-

ers have made this a practical part of commercial surveys.

Simply stacking powers produces arithmetic averages, and the contribution of each data set is, therefore, weighted according to its signal strength. There are many situations in which this is not appropriate, e.g. when the largest signals are due to occasional bursts of noise (Egbert and Booker, 1986). Hence it is desirable to provide for some other weighting using a measure such as predictability, or the product ( $E_x^p E_y^p$ ). The same weight must be applied to every autopower and crosspower from that data set.

When in-field computing power is not adequate, the approach has sometimes been to obtain many estimates of apparent resistivity in each band and take the mean, or better, the geometric mean of those values, and the mean of the phases. The median is still better than the geometric mean and gives more stable results (Moore, 1977) although it requires that the estimates be sorted. This is sometimes refined by discarding outliers which are more than one or two standard deviations from the median, and computing a new median.

Still more improvements are emerging from the statistical research specialty called robust estimation, which deals with such problems in the presence of large outliers and nonGaussian distributions (e.g., Huber, 1981). The median filter is one type of robust estimator. From repeated measurements many estimates are obtained of each impedance element at each frequency, and a 'best' estimate is determined from them. Conventional least squares processing assumes that the estimates have a Gaussian distribution. Each point is weighted according to the square of its difference from the best estimate, which has the well known result of giving greatest importance to extreme values. This is optimal when values have a Gaussian distribution, but it was shown in Egbert and Booker (1986) and Sutarno and Vozoff (1989, 1991) that this is often far from true.

The absolute value ( $L_1$  norm) method weights each estimate according to its absolute difference from the best estimate, minimizing the sum of the differences. This method still gives undue influence to outliers. In the Robust-M (maximum entropy) estimation being applied to geomagnetic problems, estimates beyond a specific distance from the current best estimate are given weights which become less as the difference increases. Different kinds of weighting are still possible, and are a topic of ongoing research. Robust methods require much more computation than least squares. However, the methods now seem capable of reducing the time which must be spent in the field and possibly doing away with the need for a remote reference, easily offsetting increased computing costs.

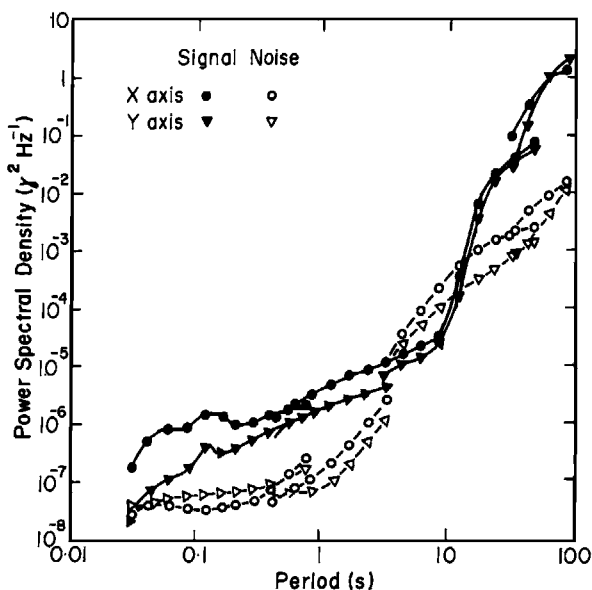


Fig. 53. Spectral densities of magnetic signal (solid lines) and noise (dashed lines) obtained by Remote Reference processing. Noise exceeds signal in the period range 3–20 s in these data sets. (After Gamble et al., 1979b.)

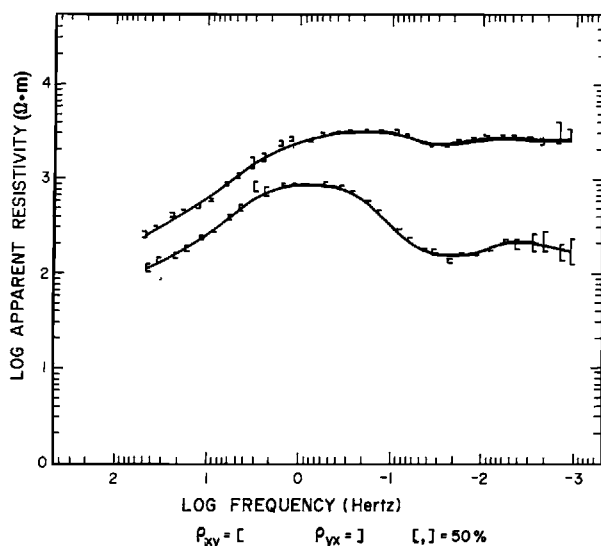


Fig. 54. Apparent resistivities with bars showing 50 percent sample confidence limits. Solid curves are the result of smoothing. (After Phoenix Geophysics, Inc.)

Use of robust- $M$  approaches was stimulated by Egbert and Booker (1986) and Chave et al. (1987). More recent comparative papers are Jones et al. (1989) and Sutarno and Vozoff (1989). The latter work was done with an RR data set in which both survey and reference sites were highly contaminated with noise from a large smelter. Robust estimation improved the apparent resistivities remarkably. However, it appeared to improve apparent resistivities much more than phases. We also found that analyses without the RR were superior in parts of the spectrum where the reference signal was highly contaminated. Thus there are benefits to be had by analyzing data from noisy environments both with and without the reference data. However, one characteristic of these results is that they generally do not give smooth curves when the data are noisy.

Most recently Sutarno and Vozoff (1991) showed how to get still greater improvement by using the physical constraints which apply. At each step in the robust estimation process Hilbert transforms are applied to the real and imaginary parts of the impedances, forcing them to vary smoothly with frequency. One strong measure of the effectiveness of this process is the agreement between RR data processed both with and without RR processing. The results obtained in the examples done to date appear very satisfactory (Figure 55). The strict mathematical validity of the procedure remains to be demonstrated.

In presenting apparent resistivities and other results, sample variances

$$s^2 = \frac{\sum (x_i - x_{\text{avg}})^2}{(n - 1)}$$

(for  $n$  samples), sample standard deviations  $s$ , or the range of some proportion of the estimates, (say, 50 percent) are often computed in real time and plotted at the same time (see error bars in Figure 54). Recording can be discontinued when these statistics have become satisfactorily small.

To compute population variance, standard deviation, or confidence limits (usually 95 percent), it is usually assumed that the scatter has a normal (Gaussian) distribution. Among its other uses, the variance is often used in inversion, to define numerical confidence limits in the model parameters. Gamble et al. (1979, a and b) derived the commonly used expressions for population variance and confidence limits of the impedance elements, apparent resistivities, rotation angle, skew, etc., in single site and RR data, assuming Gaussian distributions. However, confidence limits based on an assumed population should be used with caution because distributions are often nonGaussian.

### Tensor Analyses for 3-D Sites

It is now accepted, on the basis of both field results and 3-D modeling, that the impedance does not always degenerate to an off-diagonal (or diagonal) tensor in some "optimum" coordinate system, and that the failure to do so is not simply due to noise. Instead, in the general 3-D case it is known that there are no special directions along which modes decouple, in the sense they do in 2-D structures. A linearly polarized  $H$  field will generally induce an elliptically polarized  $E$  in this case. The special directions are replaced by polarization states. Each state consists of a pair of ellipses, one each for  $E$  and  $H$ , which are related in a relatively simple way. The ellipses are characterized by spatial directions of their principal axes, ellipticities, phases, and a ratio of the lengths of the major axes. The  $H$  ellipses induce the  $E$  ellipses, so that the ratio of the major axis of an  $E$  ellipse to that of the corresponding  $H$  ellipse is an impedance. The polarization states are special in that those two pair of ellipses have the largest and the smallest ratios possible at the site, for that frequency.

Much of the work so far on 3-D interpretation has focussed on the statics effects of shallow bodies. These effects are simpler than the general case because they are frequency independent, but are still very important.

In areas of slightly disturbed 2-D structure, rotation angles derived from the conventional Swift 2-D tensor analysis can still provide useful structural information (Jones and Vozoff, 1978; Ting and Hohmann 1981). While the resulting arctangents are complex, their imaginary parts may be small enough to be ignored. From the spatial patterns of impedance and tipper rotation directions it is possible in this "almost 2-D" situation to locate and map simple anomalous regions. Accurate determination of depth, thickness, and conductivity of these bodies can be done by modeling and/or inversion. In a sense this very simple 3-D case is like the 2-D case except that more data are required.

However the more usual 3-D case involves regions with complicated boundary shapes, and it is still very difficult to obtain more than a descriptive interpretation, using techniques like that used in Berdichevsky et al. (1980) where a survey area is divided into regions in which the apparent resistivity curves have similar appearance. The directions obtained from the Swift analysis in such cases can be highly variable with frequency and position, and simply reflect local current directions.

Two tensor analysis methods more general than that of Swift are being developed. They are eigenanalysis and explicit tensor decomposition. The first includes the work of Eggers (1982), LaTorraca et al. (1986),

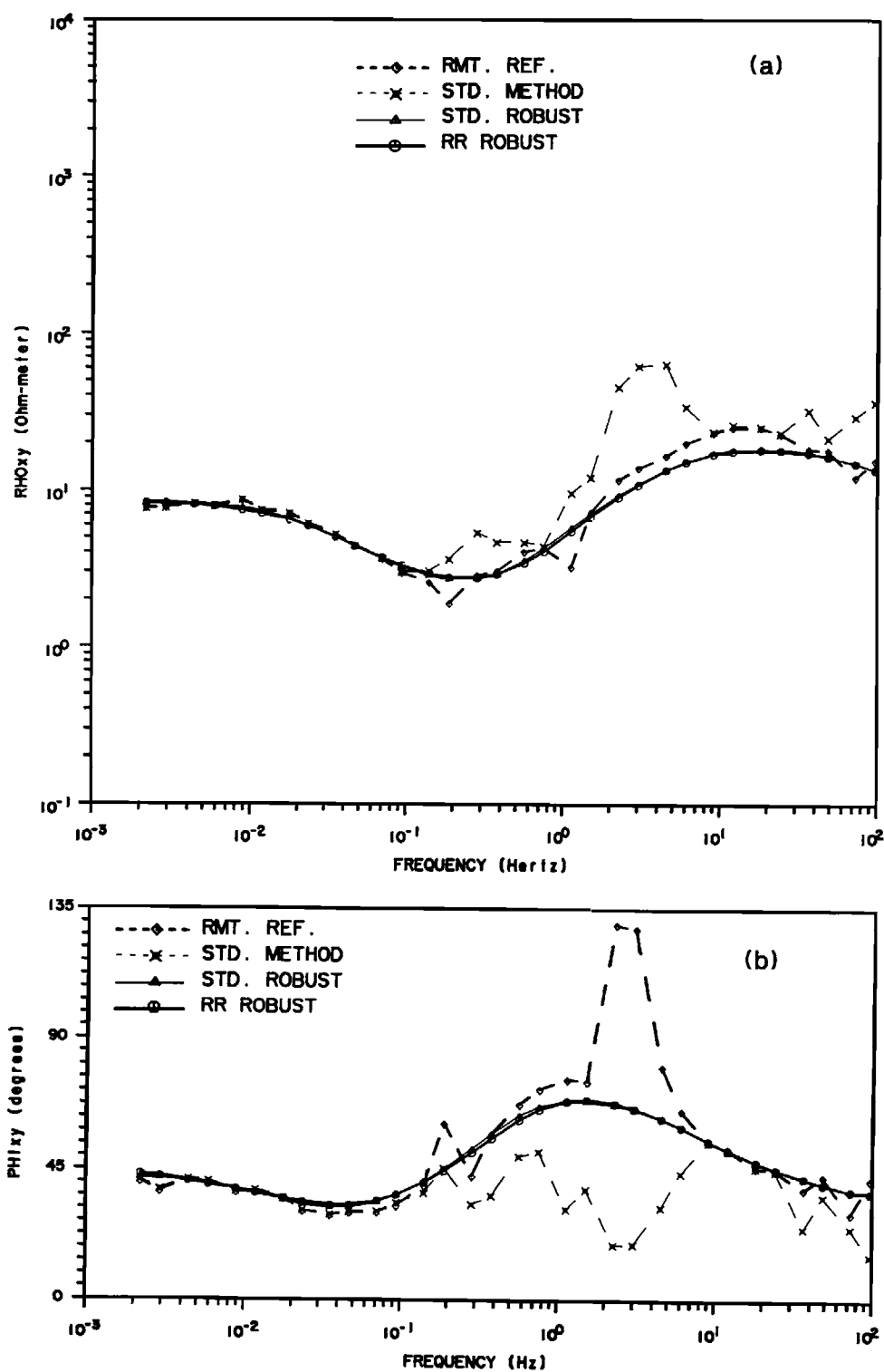


Fig. 55. A comparison of conventional and robust estimation of the impedance where both the survey site and the reference site were contaminated by large scale industrial noise. The robust estimates using Hilbert transforms are very nearly the same with and without the RR. Both are substantially better than the standard 4-channel estimates and the normal RR estimates. (From Sutarno and Vozoff, 1991)

Counil et al. (1986) and Cevallos (1986). The second is represented by Spitz (1985), Yee and Paulson (1987), and Groom and Bailey, (1989, 1990). It should be emphasized that these general tensor analyses are a difficult topic of active research and debate, and the usefulness of many of the results is only now (late 1990) being demonstrated on real data (Groom, personal comm.).

The eigenanalyses are based on the fact that the tensor equations (14) are simply a pair of complex linear equations, that the elements of  $E$ ,  $H$ , and  $Z$  change systematically as the coordinate system rotates, and that the elements of  $Z$  are simplest in some sense for specific directions. The object is to find those directions, since they should be related to structural features. Eigenanalysis of the impedance was discussed in Swift (1967), but the first implementation was in Eggers (1982). Eggers looks for a tensor  $\underline{Z}$  and vectors  $\mathbf{E}^i$  and  $\mathbf{H}^i$  such that

$$\mathbf{E}^i = \underline{Z}^i \mathbf{H}^i \quad (43)$$

assuming that

$$\mathbf{E}^i \cdot \mathbf{H}^i = 0, \quad (44)$$

with  $i = 1, 2$ . The solutions are pairs of eigenstates, with

$$\underline{Z}^i = \begin{pmatrix} 0 & \lambda^i \\ -\lambda^i & 0 \end{pmatrix}, \quad (45)$$

and the two eigenvalues  $\lambda^i$  are found from

$$\lambda^i = c_2/2 \pm \sqrt{c_2^2/4 - \det \underline{Z}}, \quad (46)$$

where  $c_2 = (Z_{xy} - Z_{yx})$ . The corresponding pair of eigenvectors are

$$\mathbf{H}^i = \begin{pmatrix} -Z_{yy} \\ Z_{yx} + \lambda^i \end{pmatrix} \quad (47)$$

and

$$\mathbf{E}^i = \begin{pmatrix} Z_{xx} \\ Z_{yx} + \lambda^i \end{pmatrix}. \quad (48)$$

All the parameters are complex. The eigenvectors  $\mathbf{E}^i$  and  $\mathbf{H}^i$  are related to each other by a 90 degree space rotation, scaling by  $|\lambda^i|$ , and a phase shift of  $\arctan [\text{Im}(\lambda^i)/\text{Re}(\lambda^i)]$ . Then the  $\rho^i = |\lambda^i|^2/\omega\mu$ . The solutions represent two situations ("states") in which an elliptically polarized  $H$  field will induce an  $E$  field which has the same ellipticity and a major axis perpendicular to that of  $H$ , as required by Equation (44). Thus they are related by a real multiplication (by  $\lambda^i$ ) and a 90 degree rotation. Eggers shows that the results reduce to the familiar ones in the 1-D and 2-D situations.

Figure 56 shows the two pairs of polarization ellipses for the tensor

$$\underline{Z} = \begin{pmatrix} (0.097, 0.208) & (1.140, 0.957) \\ (-.274, -.457) & (0.297, -.138) \end{pmatrix},$$

which has been rotated into its principal coordinates. The two eigenvalues are  $\lambda^1 = (1.060, 0.945)$  and  $\lambda^2 = (0.354, 0.469)$ . The advantages given for this approach are that it is not intrinsically 2-D, and that it uses all of the information in the impedance.

LaTorraca et al. (1986) use a different formulation of the eigenvalue problem, comparing it with that of Eggers for the same example impedance, and then showing the results for a simple model. They use the singular value decomposition (SVD) or "shifted eigenvalue analysis" of Golub and Van Loan (1983) and Lanczos (1961), respectively. Instead of the two pairs of ellipses and two real eigenvalues obtained in Eggers, LaTorraca et al. obtain the  $E$  ellipses for a pair of orthogonal, linearly polarized  $H$  fields, and two complex eigenvalues. The major axes of their  $E$  ellipses are not perpendicular to their respective  $H$  vectors, the deviation being a redefined skew angle. These new eigenstates correspond to the direction of maximum and minimum  $|\mathbf{E}|/|\mathbf{H}|$ . Figure 57 shows the two sets of eigenvectors for the example used in Eggers; clearly the directions are very similar, as are the magnitudes of the eigenvalues. Figure 58 shows the pattern of the two sets of  $E$  vectors at four frequencies around the model. Superficially at least these look very much like the patterns obtained in Jones and Vozoff (1978).

Eigenanalysis of Cevallos' (1986) new impedance gives precisely the same results as Eggers, without the orthogonality assumption (44). However Cevallos then avoids the problem of complex eigenvectors by using the Hermitian forms  $\tilde{\mathbf{z}}\mathbf{z}$  and  $\mathbf{z} + \tilde{\mathbf{z}}$ . The directions

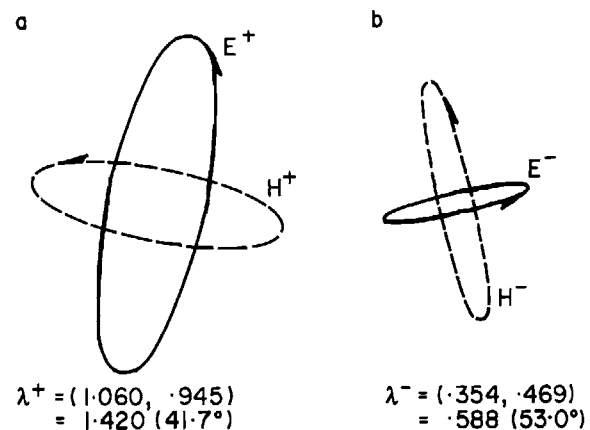


Fig. 56. Eggers' (1982) eigenstates for his impedance example. The  $\lambda$  are his two eigenvalues.

for the first correspond to one of LaTorraca's principal directions while the latter are different from any of Eggers' or LaTorraca. In the absence of noise, Cevallos' two directions are virtually identical.

The most general, and certainly the most elegant analysis to date is that of Yee and Paulson (1987). Two different canonical tensor decompositions are derived, one in polarization parameters and the other in more familiar elliptic parameters, and the results are related to those of Eggers (1982), Spitz (1985), Counil et al. (1986) and LaTorraca et al. (1986). They show that their polarization parameter decomposition represents a physical system in which a magnetic field having the polarization characteristics of one of the  $H$  eigenstates induces an electric field with the corresponding  $E$  eigenstate by simple (scalar) multiplication and phase shift. In that case the scalars are just the singular values of  $Z$  (or of  $Z^\dagger Z$ , which are the same. The  $\dagger$  indicates hermitian conjugation). The analysis deals with relations at a discrete frequency, and concludes that there are 8 degrees of freedom (independent parameters) at each frequency. While some results of

the decomposition applied to real data were provided by Paulson (personal comm., 1988), none have thus far been published.

The least general but most successful analysis to date is that of Groom and Bailey (1989, 1990). These authors treat the problems of strike determination, statics, and information content of the tensor impedance in a model consisting of a 2-D structure perturbed by a small hemisphere (Figure 59). The 2-D structure provides a regional impedance background in which local 3-D channeling takes place. In analyzing this model they also answer some of the questions of information content which were previously noted. Writing  $E = \mathbf{C}E_r$ , where  $E$  is the "measured" electric field,  $E_r$  is the "regional" field and  $\mathbf{C}$  is a channeling tensor,  $E$  can be written  $E = \mathbf{Z}_m H$ , where  $\mathbf{Z}_m$  is the measured impedance. Then if  $\mathbf{Z}_2$  is the assumed regional impedance,  $\mathbf{Z}_m$  can be decomposed as  $\mathbf{Z}_m = \mathbf{R}\mathbf{C}\mathbf{Z}_2\mathbf{R}^T$ , where  $\mathbf{R}$  is the rotation matrix from the measurement axes to regional coordinates. Groom and Bailey then show that  $\mathbf{C}$  can be written as the product of three tensors and a scalar, i.e.,  $\mathbf{C} = g\mathbf{T}\mathbf{S}\mathbf{A}$ . The tensors are responsible for "Twist", "Shear", and "Anisotropy", effects, respectively, and  $g$  is necessary for normalization. The three operations are illustrated with a simple model (Figure 60). Then with  $\mathbf{Z}_m = g\mathbf{R}\mathbf{T}\mathbf{S}\mathbf{A}\mathbf{Z}_2\mathbf{R}^T$ , they demonstrate that it is not possible to tell whether  $g\mathbf{A}$  is different from the identity matrix  $\mathbf{I}$  on the basis of a single local measurement, so the terms  $g\mathbf{A}\mathbf{Z}_2$  are lumped together. Then  $\mathbf{Z}_m = \mathbf{R}\mathbf{T}\mathbf{S}\mathbf{Z}'_2\mathbf{R}^T$ , where now  $\mathbf{Z}'_2 = g\mathbf{A}\mathbf{Z}_2$ . This representation of  $\mathbf{Z}_m$  contains seven real parameters, three angles, and two complex principal impedances. They are found by least squares or other fitting procedure to the data, giving a misfit as the eighth parameter. This eighth parameter could be reinterpreted as an imaginary component of the Twist. In very good, noise-free data it would measure the suitability of the channeling assumptions. Otherwise it would simply indicate signal/noise levels.

This decomposition was tested on real data from a complex geological area, and dramatically improved the smoothness and credibility of the apparent resistivities compared with those obtained with conventional processing (Cavaliere et al, 1986). Although smoothness is desirable simply on the basis that model responses are smooth, it does not yet guarantee interpretability.

Groom and Bailey (1990) and Bahr and Groom (1990) show that perturbations in both electric and magnetic fields could be included by means of distortion tensors, the magnetic tensor being complex, frequency dependent and going to unity at low frequency. Comparing the computed results with the parameters of their 2-D regional model, they found that impedance

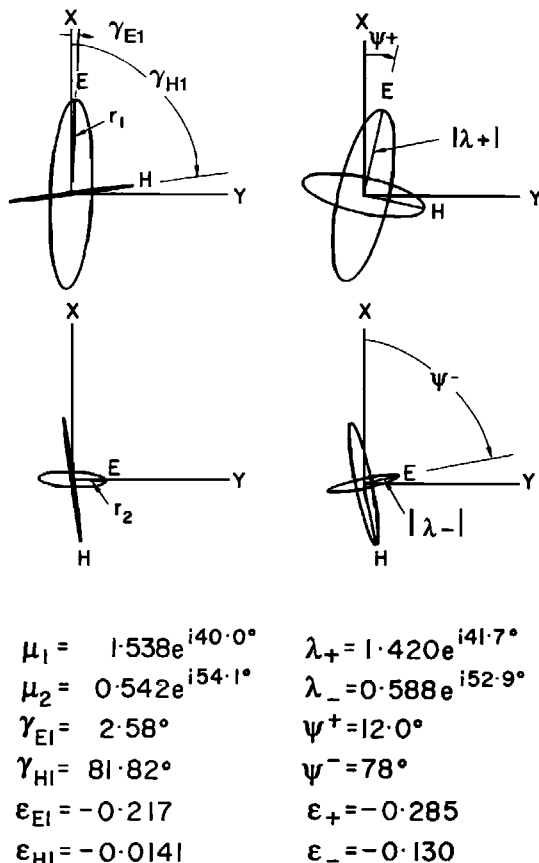


Fig. 57. Comparison of the eigenstates of LaTorraca, et al, 1986, (left) and Eggers, 1982, (right). (After LaTorraca et al., 1986.)

was in general a mixture of the two regional impedances, except on lines along one of the regional directions through the center of the hemisphere. In some positions the minor principal impedance was due entirely to contamination of that electric field by the major regional electric field, so the two apparent resistivity curves were parallel. The magnetic field perturbation resulted in small phase changes at high frequencies.

Those authors went on to do a thorough comparison of the analyses of Eggers (1982), Spitz (1985), LaTorraca et al. (1986) and Counil et al. (1986) for points immediately outside the small hemispherical perturbation. They concluded that, except for stability in the presence of noise, the methods of Eggers, LaTorraca et al., and Spitz (Q parameter) give the same results as those of Swift. These are controlled by the local current direction which, in turn, is dominated by the hemisphere. As regards ability to extract the regional directions and impedances, only theirs and the other Spitz' orientation ( $U$ ) were successful.

Groom (pers. comm., 1989) extended his comparisons to include the Yee-Paulson analysis. He found some difficulties with numerical stability, particularly in the 1-D and 2-D cases. When stable results were obtained, they were very similar to those of the other

four methods. This does not necessarily detract from the validity of the Yee-Paulson results, but indicates additional effort will be required before they can be used.

The related question of the number of degrees of freedom (independent parameters) given by an impedance is still not fully resolved. An impedance measures the amount of new information which a site provides. There are limits to the effort expended in acquiring data which tells nothing new, so if there were no degrees of freedom then the data would have no value. An impedance  $Z$  contains eight parameters at each frequency, corresponding to the real and imaginary parts of the four  $Z_{ij}$ , but these degenerate to one parameter in any horizontally layered structure through the dispersion relations and azimuthal independence (Weidelt, 1972). In the principal coordinates of the 2-D case these parameters give us two complex impedance elements (four parameters), two directions—strike and perpendicular to strike (two parameters)—and two combinations of tensor elements—usually skew and ellipticity—which vanish. However, of these, only one of the two directions can be independent, and the phases are again related to the apparent resistivities through the Hilbert transform (Fischer and Schnegg, 1980). Hence there are only two

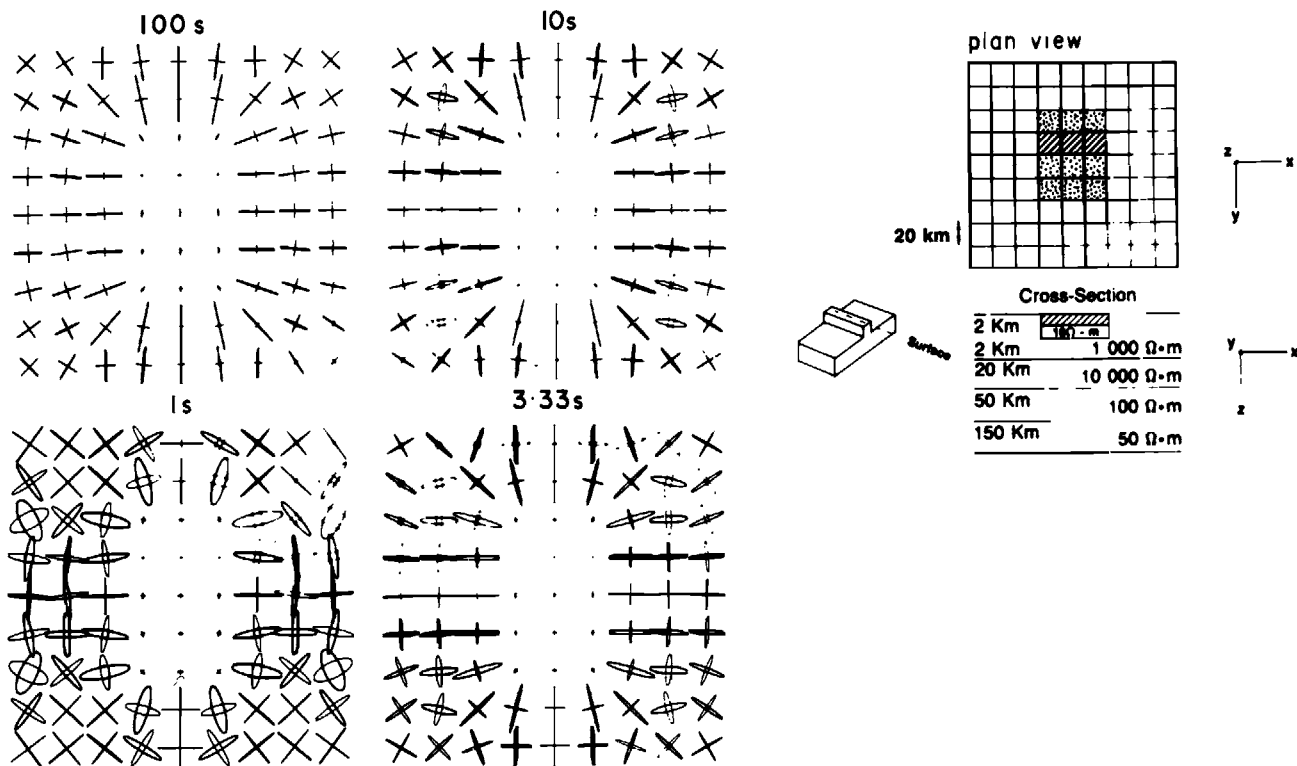


Fig. 58. The distribution of LaTorraca E field eigenstates at four frequencies around the 3-D model shown. (After LaTorraca et al., 1986).

degrees of freedom in 2-D. Yee and Paulson (1988) proved from fundamental principles that the same dispersion relations hold in the general 3-D case, at least for the principal terms, so that there are at most six (and perhaps fewer) independent frequency functions in any (four channel) MT data set. (There is still debate about this result.) Also possible relationships between the real and imaginary parts of the diagonal terms of  $Z$  in the general 3-D case remain to be discovered, so further work has to be done in this area (Yee, pers. comm., 1989).

The major challenge that remains is to devise practical procedures for quantitative interpretation without the expense and inconvenience of computing eight (or even only four) parameters for 3-D models at many sites, over a broad spectrum for comparison with field data.

### Imaging and Time Domain Processing

We saw in a previous section that the solution to Maxwell's equations in a layered earth consists of downgoing and upgoing waves in each layer, the upgoing waves having been reflected from the bottom of the layer. The picture holds equally well in time domain if we consider source fields that are impulsive. The waveforms will change rapidly on account of dispersion, but this can be modeled with little difficulty.

Kunetz (1972) explored some characteristics of modeling and processing in time domain, using con-

cepts from reflection seismology. He considered only the 1-D case, showed how an impedance could be derived without Fourier transformation, computed model curves (Kunetz's Figure 4-14) and outlined several alternative inversion procedures. Spies and Eggers (1986) discussed the behavior of time domain responses in comparison with alternative frequency domain definitions of apparent resistivity. They conclude that the time domain  $\rho_a$  has much more intuitively appealing character in its lack of oscillation and rapid convergence to its asymptotic values. The method has not received much use with real data, as far as I know, but appears to even be applicable to tensor measurements. The question raised in both papers about the transient approach is the practical difficulty anticipated in accurately measuring transients of significant length. Recalling that MT data acquisition now rarely includes less than four frequency decades, and that TEM systems obtain three time decades only with difficulty, it is unlikely that a natural field transient system would reliably record waveforms more than one decade in duration. Thus time domain seems an unrewarding approach to MT data acquisition.

On the other hand, time domain could be a useful way of presenting and interpreting data and might also have some benefits in processing. Whittall and Oldenburg (1986) actually invert on impulse responses, but they obtain them from the frequency-domain impedance, assuming several different models of noise.

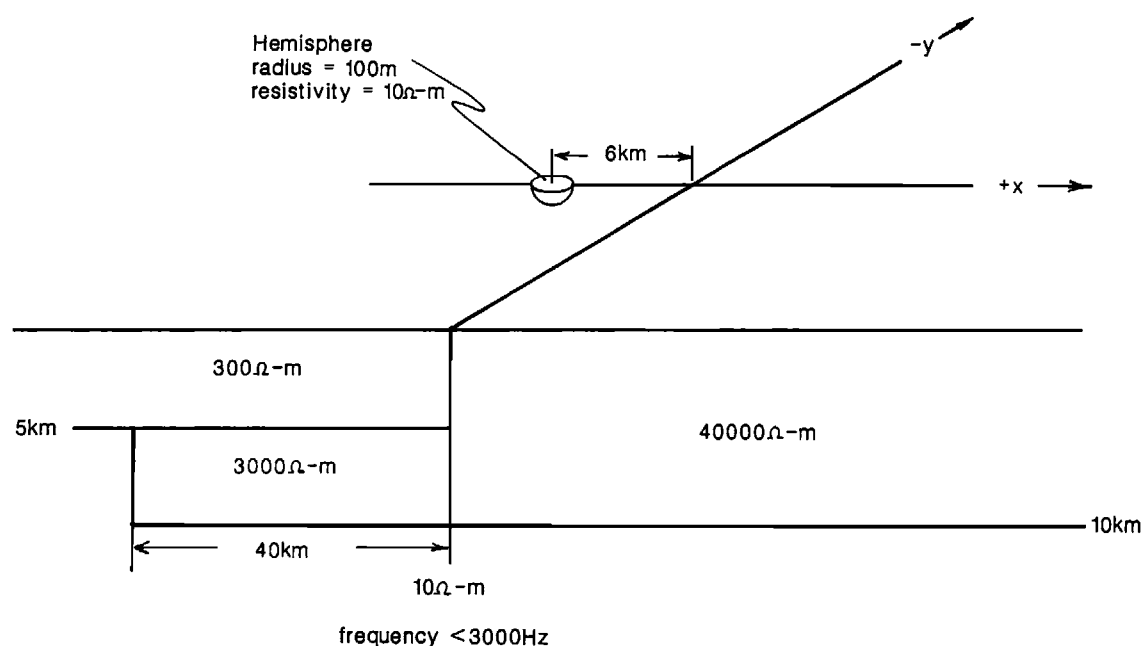


Fig. 59. The perturbed 2-D model for which impedances were computed at points  $w$  around the hemisphere. (Groom and Bailey, 1991)

Their impulse responses are plotted in the same way as deconvolved seismic reflectivity time sections, which has a certain psychological appeal. Levy et al. (1986) discuss the possibility of using either the TE or TM modes, or both, to produce such sections from 2-D data and show a synthetic section for a 2-D model, and Stinson et al. (1986) model near-surface effects on imaging and apply the method to Kurtz's Vancouver Island field data. Wang et al. (1985, 1986) and Liu et al. (1987) also apply time-domain techniques to 2-D models and demonstrate their use on real data.

Lee et al. (1987) extended Claerbout's (1976) 2-D numerical modeling technique to the MT case, for both the forward and inverse problems. The latter is the same as Claerbout's migration, and produces good images of EM reflectivity. Vertical resolution is found to depend on the range of frequencies and station spacing used, and the imaging must be done separately for each frequency. Conductivity determination appears to be semiquantitative. Ten to twenty frequencies were used for the results shown. Lateral resolution is said to be on the order of half the station spacing. The method, referred to as "phase-field imaging", is evidently fairly computer-intensive. It would be of considerable interest to evaluate this method and the impedance series inversion on com-

mon data sets, particularly with regard to their suitability for 3-D interpretation.

## INTERPRETATION

### An Example

Interpretation, the process of drawing geological conclusions from the geophysical data, is most effectively done with reference to conceptual geological models. In addition, interpretation requires a thorough understanding of the physics of the method and the technical details of the measurement, so that data artifacts are not attributed to geological causes. As with most geophysical methods, MT surveys raise difficult questions which the practicing interpreter must do his best to answer every day. This section describes the tools available and the way they have been brought to bear on these problems.

To illustrate the nature of MT interpretation problems it is easiest to begin with a real example, since they rarely conform to the simple theoretical pictures. I selected an AMT survey across the Torrens Hinge Zone of South Australia (Figure 61). Ten sites (Figures 62 and 63) were surveyed across the feature, thought

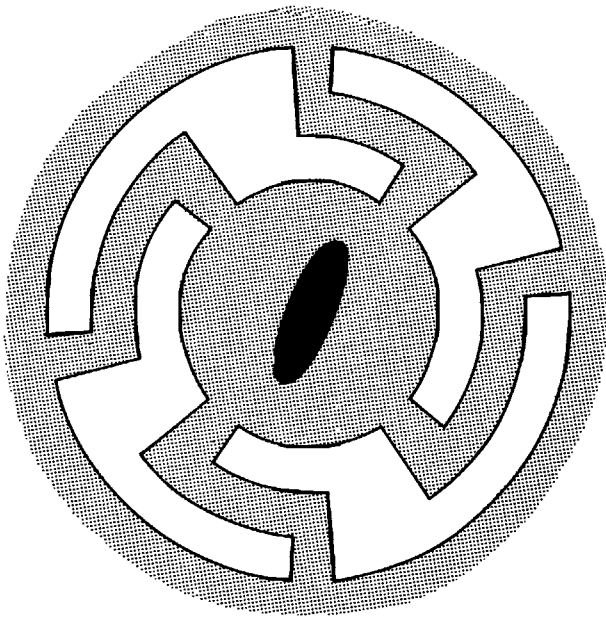


Fig. 60. Model incorporating the three types of distortion underlying the Groom-Bailey decomposition. Measurement location is at the center of the very conductive (black) region. White region is resistive and the spotted region has intermediate conductivity. The regional  $E$  field is at angle  $\theta_r$  from the vertical, and currents reach the center by twisting  $T$  through  $\theta_t$ . The central ellipsoidal conductor is responsible for anisotropy  $A$ . (From Groom and Bailey, 1989.)

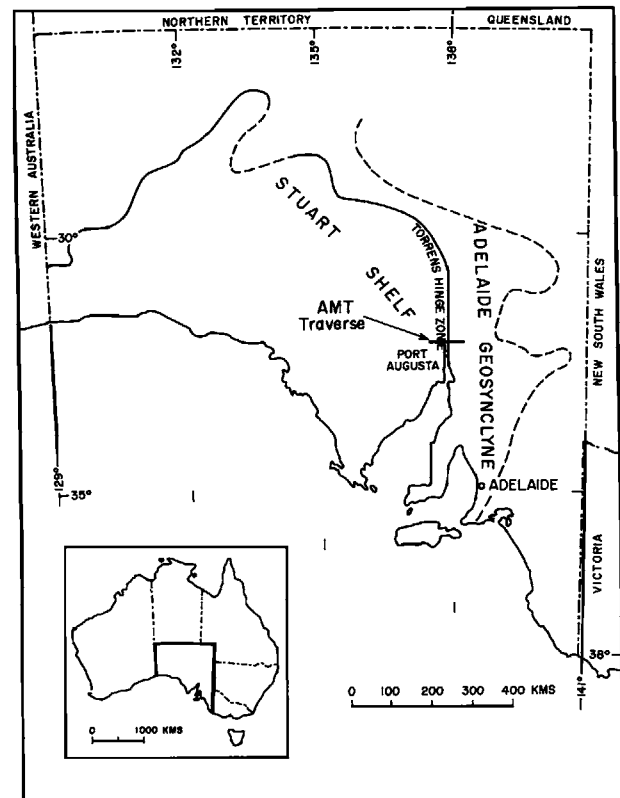


Fig. 61. Location map, Torrens Hinge Zone AMT traverse, South Australia.

Fig. 62. Site locations, Torrens Hinge Zone traverse.

to be a major geosuture (Figure 64), during March 1983.

A brief geological description of Figure 64 from an unpublished report by Nelson and Greenhalgh for the South Australian Department of Mines and Energy follows.

The "Northern Eyre Peninsula, which lies in the western part of the transect is structurally part of the Gawler Craton, a basement complex of rocks, highly folded and intruded by granites, and ranging in age from Archean to Middle Proterozoic. The oldest known rocks are highly metamorphosed basic rocks and metasediments of probable late Archean age. These are overlain by granitoid rocks of the Hutchison Group. Mildly metamorphosed rocks of Carpenterian age overlie Hutchison Group rocks (e.g. metaquartzites of the Moonabie Formation). The final phase of consolidation of crystalline basement was the extrusion of a vast sheet of porphyritic rhyolitic lavas—the Gawler Range Volcanics—extending from the Gawler Craton eastward, possibly across the future site of the Adelaide Geosyncline.

The thick sedimentary sequence that forms the Adelaide Geosyncline accumulated during Late Proterozoic in a subsiding trough east of the Gawler Craton. It has been recognized as an aulocogene, or

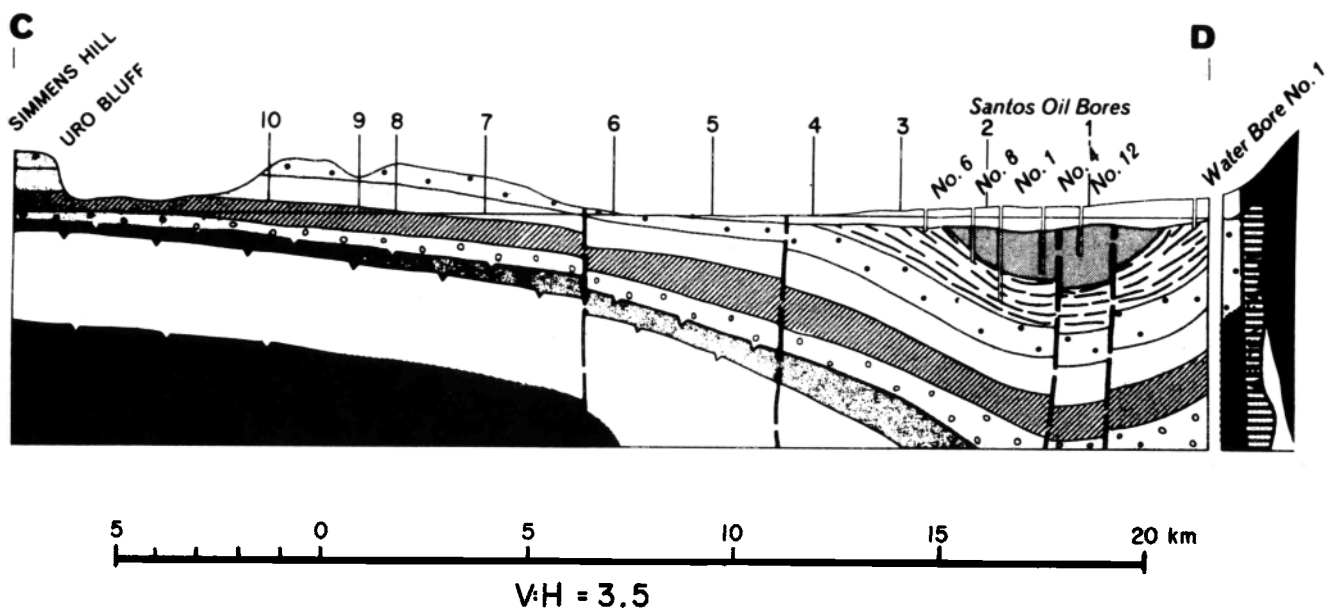
fault-bounded basin within the continental crust. On the western edge of this geosyncline, a contemporaneous, gently dipping shelf facies was deposited over the irregular, undulating basement surface. This is known as the Stuart Shelf, and the zone of intensive faulting and deep fracturing that separates it from the geosyncline is known as the Torrens Hinge Zone, part of a northerly trending, ancient lineament of continental proportions.

Northwest of Port Augusta, the Adelaidean sediments are known to overlie unconformably the Gawler Range Volcanics, but little is known about the preAdelaidean basement on the Stuart Shelf in the area immediately north of Port Augusta, and virtually nothing about basement beneath the Adelaide Geosyncline."

In fact, neither is much known about the superficial unconsolidated sediments of the region.

The survey area is sandy and dry most of the time, relatively flat, with low scrub, and isolated homesteads tens of kilometers apart. Lakes and streams shown on the maps are often empty for years at a time. Torrens, Gairdner, Frome and Eyre are salt lakes whose levels are also widely variable. The major noise source in the region is an (50 Hz) earth return power grid. Fortunately, current densities are fairly low.

### SECTION C-D



GEOLOGICAL CROSS SECTION SHOWING AMT SITES  
TORRENS HINGE ZONE AMT

Fig. 63. Geological cross section, Torrens Hinge Zone. Numbers across the top show AMT sites.

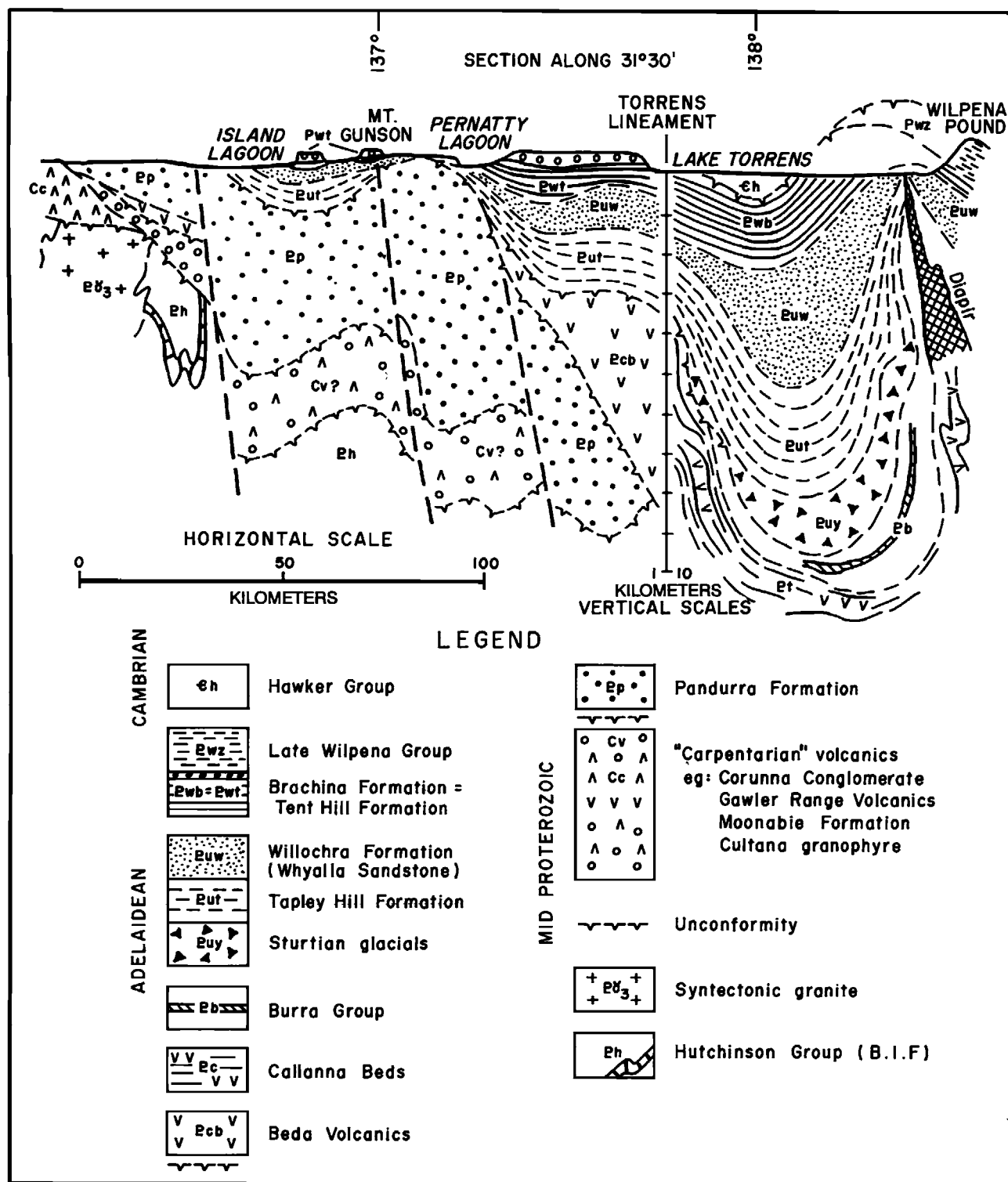


Fig. 64. Larger scale cross section showing interpreted geology, Torrens Hinge Zone AMT. Note depth scale change at the Lineament at the west edge of the Adelaide Geosyncline. (After Nelson et al., pers. comm.)

Between sites 5 and 7 the traverse crosses the Torrens lineament, extending north from Port Augusta, which is thought to be the trace of the geosuture.

Pairs of scalar (Cagniard) apparent resistivities with their phases are shown in Figure 65, while the coherencies, rotation angles, tippers, impedance skews and  $H$  predictabilities appear in Figure 66. Coherencies for  $(E_x, H_y)$  and  $(E_y, H_x)$  in the range (0.7–1.0) are plotted at the top. Predictabilities  $H_x^p$  and  $H_y^p$  for the same range are immediately below. The third box shows tensor rotation angles from equation (20) (theta 1) and from Eggers (1982) E polarization (theta 2). Impedance skew is plotted in the range (0–0.5). Beneath that is the tipper modulus  $|T|$  (range 0–1.0). The last pair of plots are tipper rotation angles, from equation (41) (angle A) and another (angle B) suggested by D. Kerr (pers. comm.). We did not plot tipper skew, tipper phase, or  $H_z^p$ . Data quality is poor to fair by comparison with much of the current generation of commercial MT output. The data were acquired with an AMT system designed and built at Macquarie University in 1980 and described in summary in Zile et al. (1983).

Scatter in these plots can arise for any of several reasons. If the site is approximately 1-D over part of its frequency range, then the two apparent resistivities will of course have the same value. In that case the computed rotation angles are undefined and the points will scatter. If the situation is truly 1-D we should also find that the  $\rho_a$  and  $\phi$  plots are smooth because they will be unaffected by rotation. In those conditions, large scatter is also expected in tipper rotation angles because  $H_z$  will be small and the tipper rotation angles will be poorly defined. Of course, when data are noisy then coherencies and predictabilities will be low, in which case all of the response functions including  $\rho_a$  and phase are expected to scatter.

The band of low signal level due to poor propagation near 2 kHz is seen in the apparently enhanced tipper and in the scatter in  $\rho_a$ , phase, skew, and rotation angles. The decreasing S/N ratio below 5 Hz in these data causes similar artifacts. (A rapid change in phase of some of these data at frequencies above 3 kHz was caused by the electrodes. This undesirable characteristic was not evident until late in the field program.)

## Discussion

The first step in interpretation, after discarding the worst data points, was to assess the results for internal and external consistency. This assessment was done in order to establish the range of resistivities and depths represented. Layered models were fitted to each of the apparent resistivity-phase curve pairs. With the rotation angles and the tippers, these were then compared with the geological models to see whether they were

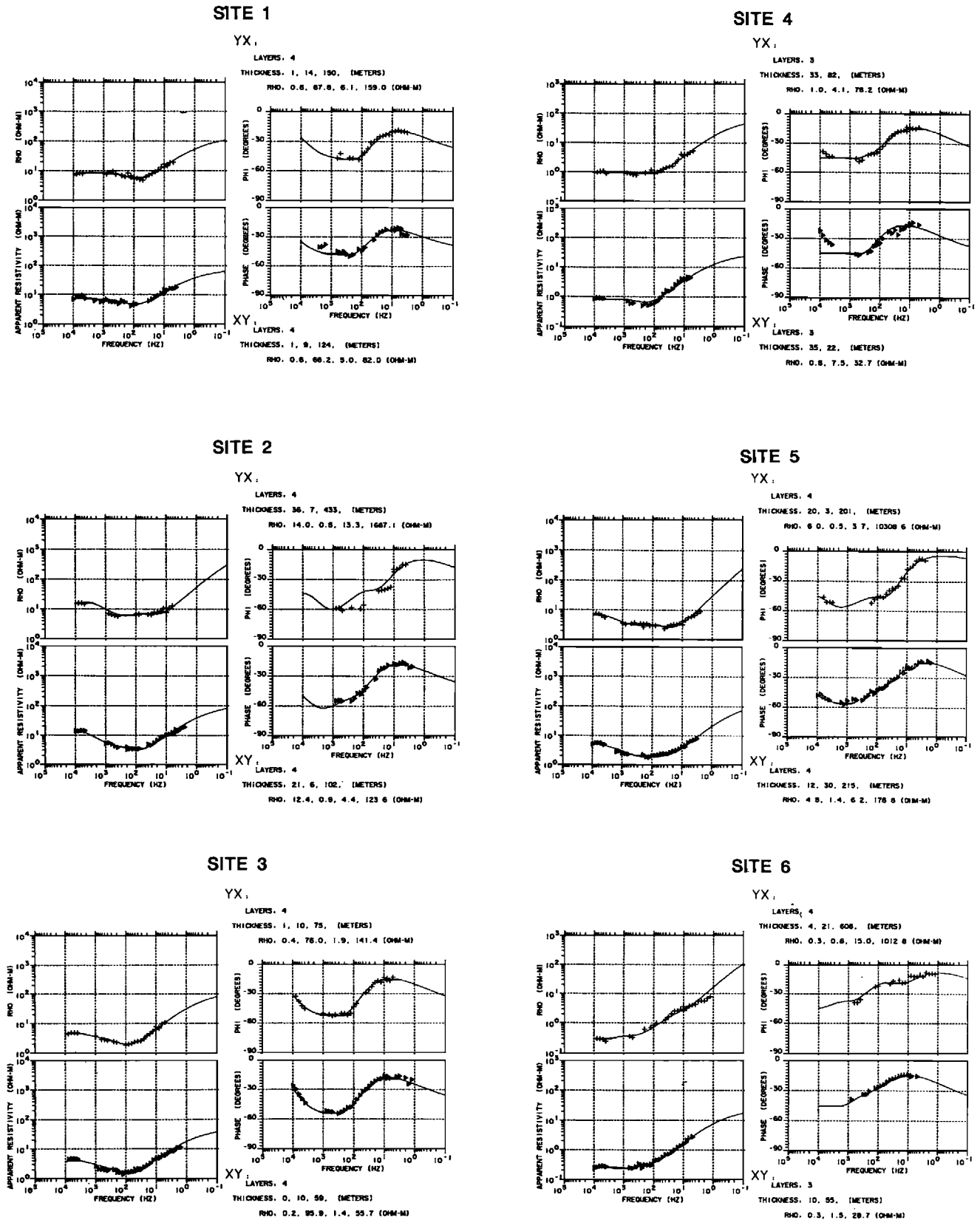
compatible. In particular, there should be a general agreement between strike directions, at least for shallow depths, and the resistivity values should be in the broad range expected for the geological environment, the prevailing hydrological regime and any independent information.

In the present survey, dc resistivity depth soundings were carried out at each site by the South Australian Department of Minerals and Energy (Figure 67). These soundings were used to confirm the near-surface resistivities and for joint inversion with the MT (Jupp and Vozoff, 1977).

Both MT and dc results showed near-surface resistivities of less than  $10 \Omega \cdot m$  and often less than  $1 \Omega \cdot m$ , which was consistent with the known high salinity of the region. The values at each site were in excellent agreement. Tippers were generally 0.1 or less above 10 Hz, increasing toward lower frequencies, and largest at site 6. This fact, that tippers are largest at suture locations, as predicted by theory, agrees with the expected suture location. Impedance skews were scattered, on account of noise. Where they form smooth plots the values are small ( $\approx 0.1$ ), increasing toward lower frequency. Skews at sites 4 and 9 were uniformly larger (0.1–0.2), and at site 5 they were off-scale ( $>0.5$ ). This indicated that local structure is most complex at site 5, although the apparent resistivity and phase curves were well behaved and the impedance rotation angles were the best defined of the survey.

Close comparison of the  $\rho_a$  curves showed that sites 7, 8, 9, and 10 were almost 1-D, from highest frequencies to between 10 and 100 Hz, with a resistive basement becoming shallower to the west. Sites 1, 3, and possibly 2 were also nearly 1-D from 10 kHz to 10–100 Hz. Site 3 appeared to have a static shift (discussed below) of about 0.5 relative to sites 1 and 2. Sites 4, 5, and especially 6 showed the greatest departures from 1-D, but even these are not very marked. At sites 4 and 5 the departure is apparent only in the mid-band. Thus the tensor rotation angles are fairly indefinite and the tippers are small, as was expected. Under these circumstances, and in light of the large spacing between sites relative to the skin depths achieved, there seemed little point in going beyond a 1-D interpretation, possibly after some static shift compensation at site 3.

There was a question regarding the tipper rotation angles (they were not tipper strike angles), which we would have expected to be east or west, for geological reasons. Instead, where they did cluster they were consistently northward, except at the lowest frequencies at sites 1–3 where they were easterly. No explanation is available, but the evidence suggests an east-west cross structure, such as a fault very near the traverse, superposed on the major geosyncline. An-



65 (a)

Fig. 65. Summary of apparent resistivities and phases. Solid curves are responses of inversion models whose parameters are shown. (a) Sites 1-6.

other alternative was suggested by the results of Groom and Bailey (1989), where poorly conditioned matrices led to numerical instability and physically meaningless results.

Figure 68 is a composite of 1-D inversions on the TE (xy) curves. This simple presentation confirms the view from Figure 65 that statics were not a serious problem on this line. The zone of shallow high resistivities in the center suggests a low porosity zone in mid-line, between sites 5 and 7. If the traverse line grazed one end of such a block it would help explain the tipper rotations. The TE inversions were usually used for this purpose because that mode changes more gradually between sites, which makes correlation easier.

In this case we had a conceptual model of the structure, including a definite idea of strike direction. This model was used to lay out the traverse, and site

spacing was taken to be some convenient value (2 km) with the idea that additional sites might be added later. The situation was fairly typical, as were the results. Interpretation started with the classification of curves as TE or TM at each site. It is quite usual to find that, with a single line traverse, the choice will sometimes be difficult or arbitrary. This difficulty arises because of 3-D conditions at one or more of the sites. For that reason it is strongly advisable to get areal coverage instead of single traverse lines.

In practice, there are many situations in which there is no clear strike direction, because an area is virtually layered, or 3-D, or because of noise. A great deal of effort has gone into devising ways of extracting values of effective strike and impedances from the data in some of these cases (Jones and Vozoff, 1978; Gamble et al. 1982; Wannamaker et al. 1984b; Zhang et al.

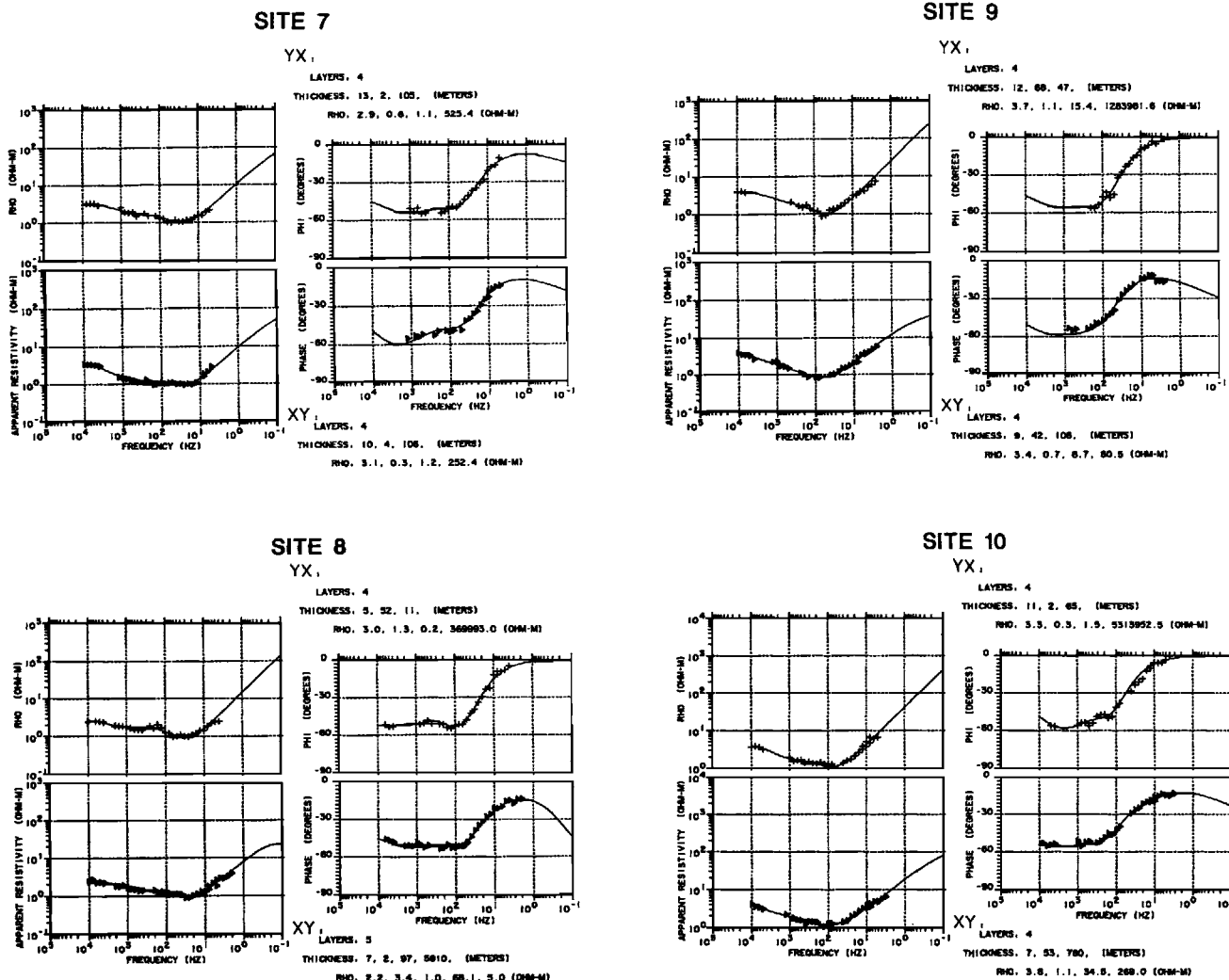


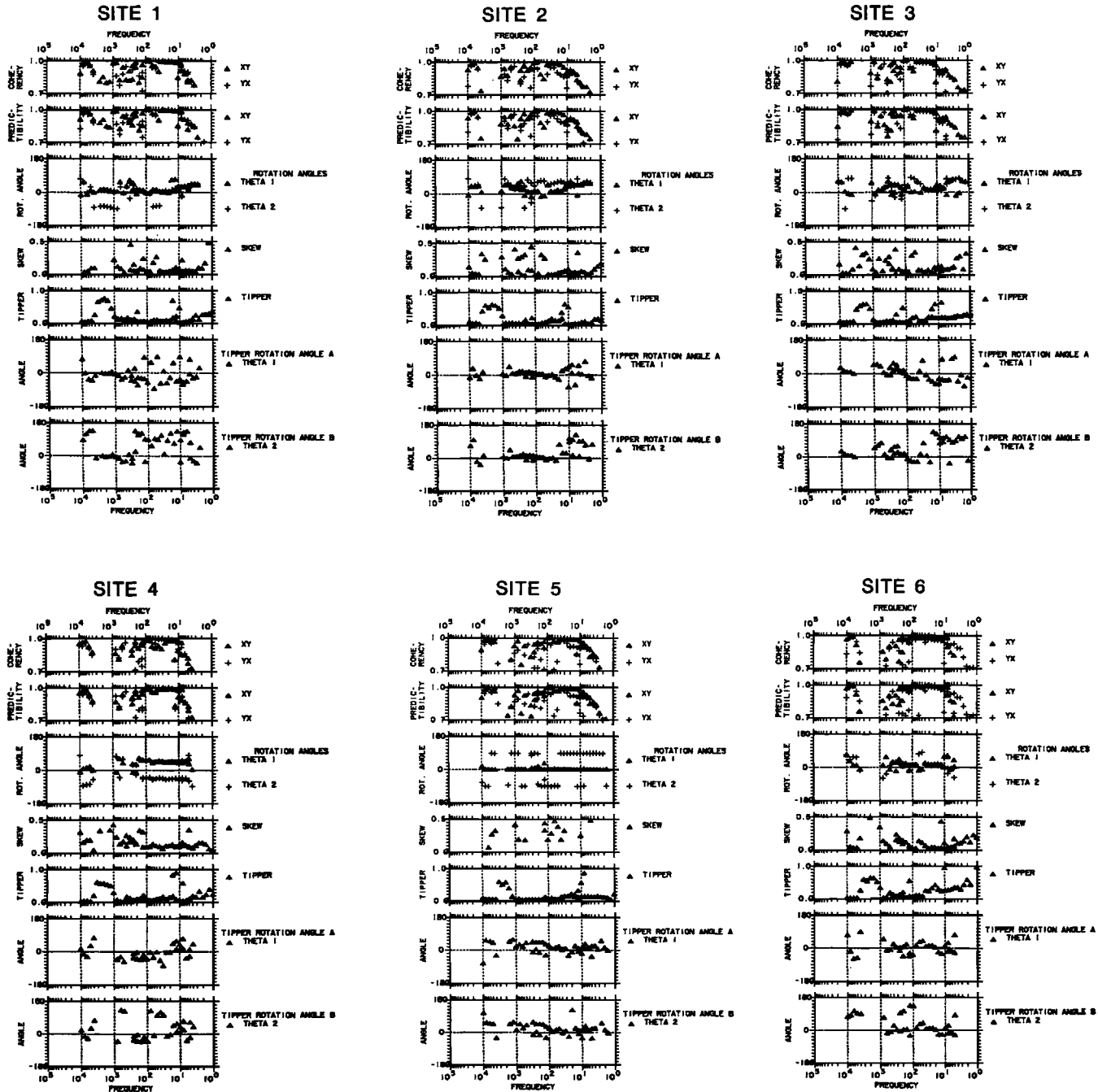
Fig. 65, continued, (b) Sites 7-10.

1987; Groom and Bailey, 1989). These situations are discussed later in this section.

The sources of statics effects were described previously, and their treatment is discussed in the Statics Compensation Section. Ultimately the problems of statics and strike determination are closely intermingled, since both usually result from 3-D phenomena.

The paper of Zhang et al. (1987a) and especially the Groom and Bailey papers (1989, 1991) deal with both problems.

Another departure from the simple picture of MT field behavior, discussed briefly in a previous section, arises in the presence of topographic relief, which is a major problem where slopes and resistivities are large.



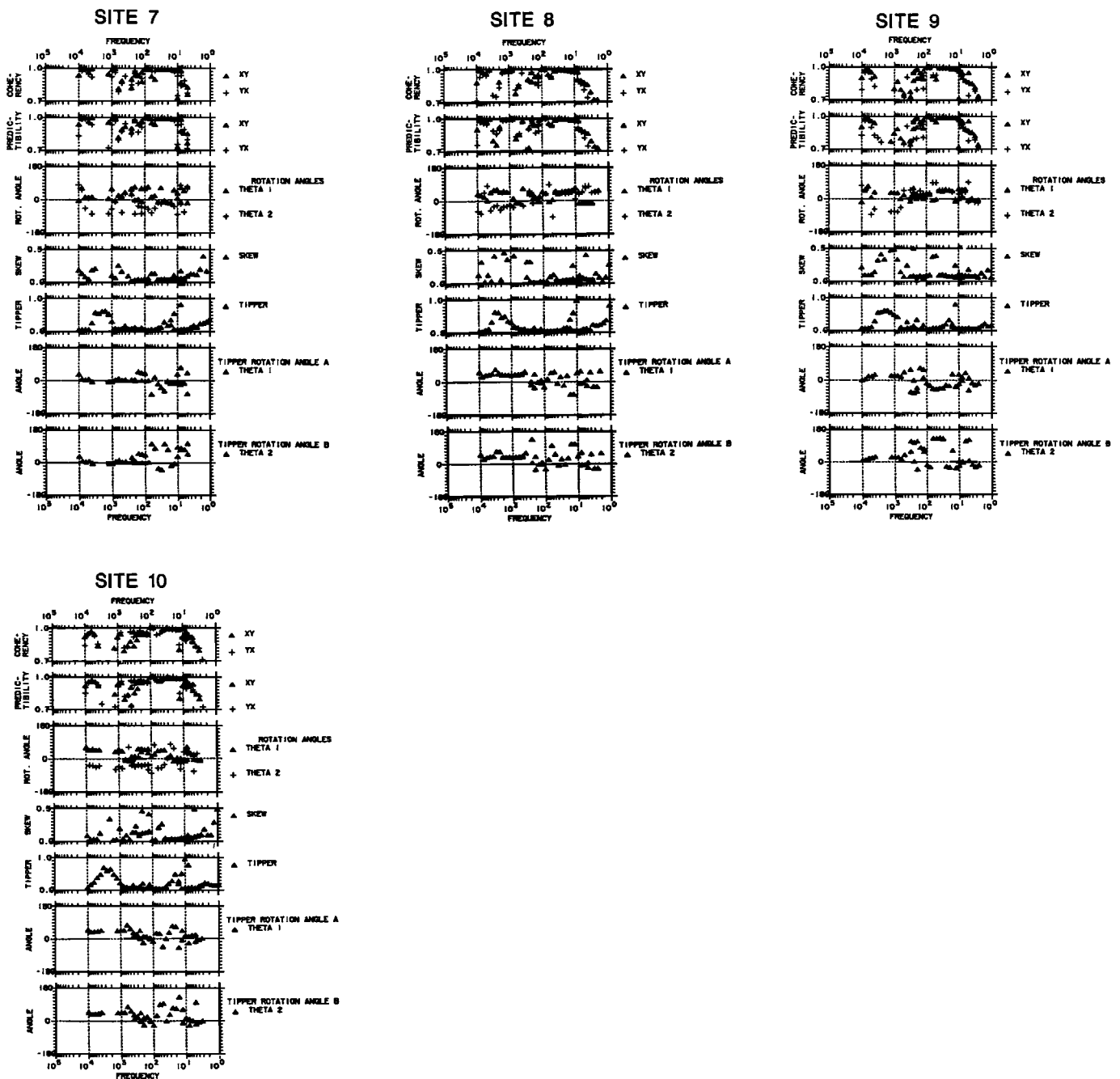
66 (a)

Fig. 66. Summary of coherencies, rotation angles, tippers and skews. (a) Sites 1-6.

## Inversion

MT responses are usually interpreted through models. Occasionally it may be enough to find the location of some anomalous feature such as a minimum in apparent resistivities or a direction change, but usually the result sought is in terms of the distribution of resistivities. In the early years, 1-D models were fitted

to the observations at each site by trial and error, adjusting layer thicknesses and resistivities until the model response showed a reasonably good visual fit to the field data. However because this procedure was time consuming for the geophysicist, it soon became apparent that computer inversion could do a better, more economical job. Always keep in mind that inver-



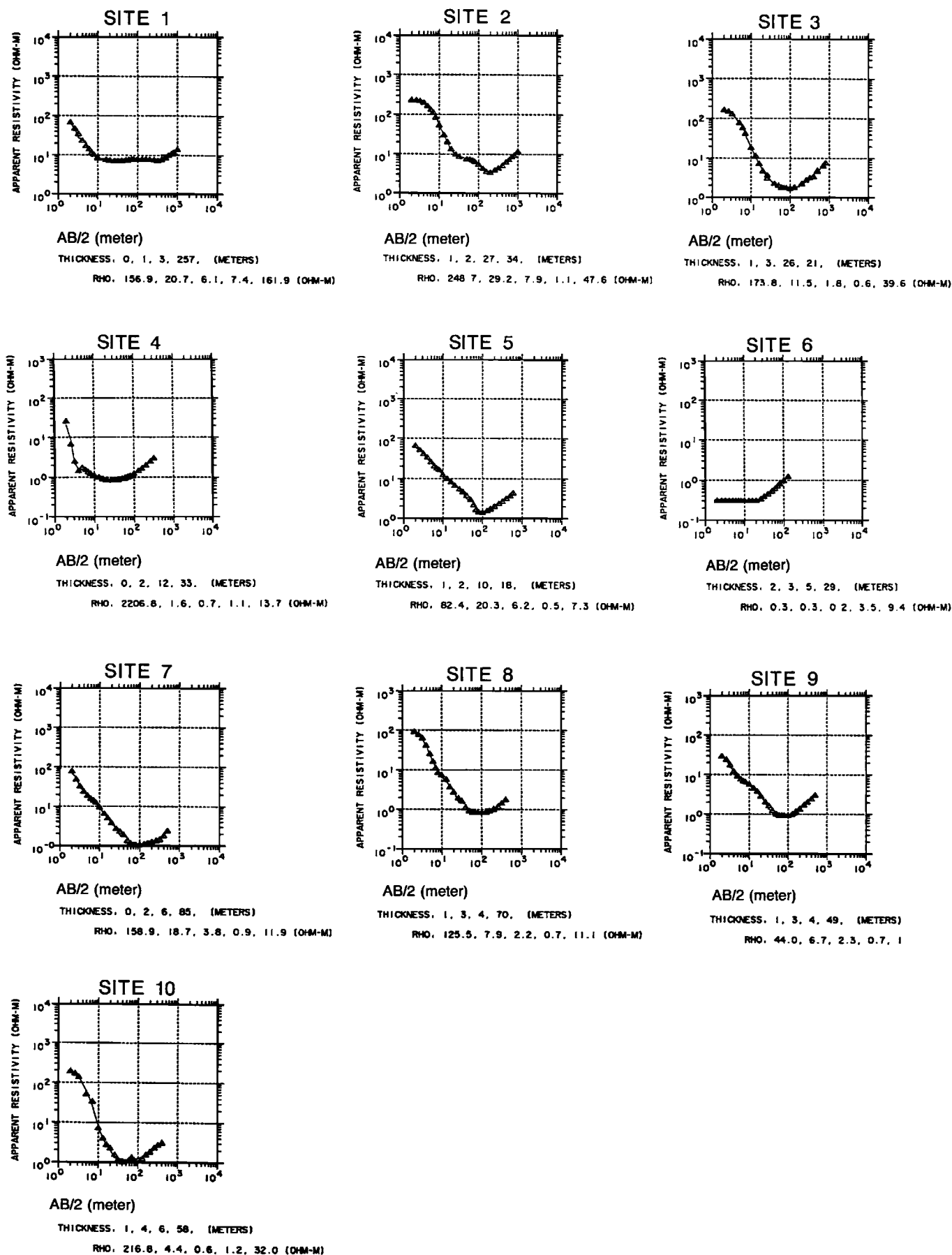


Fig. 67. The dc Schlumberger soundings at AMT sites. Solid curves are responses of inversion model whose parameters are shown. (South Australian Department of Minerals and Energy.)

sion is not the same as interpretation: it is only one step in the process.

There are now a number of general approaches to 1-D inversion in common use, in addition to several 2-D inversion schemes, and still others for joint (1-D) inversion of MT with other types of data. In spite of (or perhaps because of) the extensive literature, aspects of the topic remain sources of confusion and subjects of philosophical discourse. Menke (1984) is probably the most useful practical textbook for general geophysical applications, while the Parker (1983) paper provides a superb overview of the subject, free of the jargon responsible for much of the confusion. Parker deals with the existence of solutions, their uniqueness and construction, and the errors and meaning of the results, by reviewing both his own work and the literature generally. Much of the following discussion is based on that paper. However, Parker assumed that little or no other information was available about the section, whereas we often know a great deal about it from other sources. This knowledge can change both the approach to the problem and the significance of the results, as noted.

At the other extreme to Parker's approach are papers giving practical approximate construction methodologies, without regard to accuracy or dependability (Niblett and Sayn Wittgenstein 1960; Bostick, 1977; Goldberg and Rotstein, 1982). They reflect the range of needs which exists among users. Parker asks whether a result is possible, whether it is unique, what range of solutions is allowed with the prevailing noise, and how well each feature is resolved. However, many practitioners are content just to have a model that roughly fits the data.

Inversion algorithms differ in the types of models they use. In some, the earth is made up of a stack of

uniform layers which might represent a sedimentary section. In others, the variation of resistivity with depth is continuous, as might be expected in a geothermal section, for example. Inversion methods also differ in their ease of use, in the decisions that must be made along the way, and in the nature and amount of information they yield. Even the data used differs: most 1-D inversion chooses one of the modes of  $\rho_a$ , say  $\rho_{xy}$ , and inverts to that data set with or without its phase. Others use the real and imaginary parts of the corresponding impedance element  $Z_{xy}$  or of  $c_{xy}$ , where  $c_{xy} = Z_{xy}/i\omega\mu$ . Real and imaginary parts of  $Z$  and  $c$  are thought to be better behaved statistically than apparent resistivity and phase. Berdichevsky and Dmitriev (1976), and Ranganayaki (1984) used  $(\rho_{xy} + \rho_{yx})/2$ , avoiding some of the difficulties of scatter, particularly due to noisy rotation angles.

However the major differences seem to be in what is asked of the inversion and in the existence and use of prior information. Exploration problems sometimes require a simple "yes" or "no" answer to questions such as "does a buried conductor exist here?", or "is the conductor shallower at point A than at point B?", in order to establish a site for drilling. In other applications we would like to map stratigraphic facies variation and require quantitative estimates of resistivity in a particular horizon, complete with confidence intervals for those estimates. In many situations we may know the depths to particular interfaces very accurately, as from a reflection seismic survey or a well. We may also know resistivity from well logs or core measurements. In other cases, of course, we have no prior information. Where information is available, some existing codes permit incorporating the data as constraints on inversion so that other, unknown parameters can be better determined.

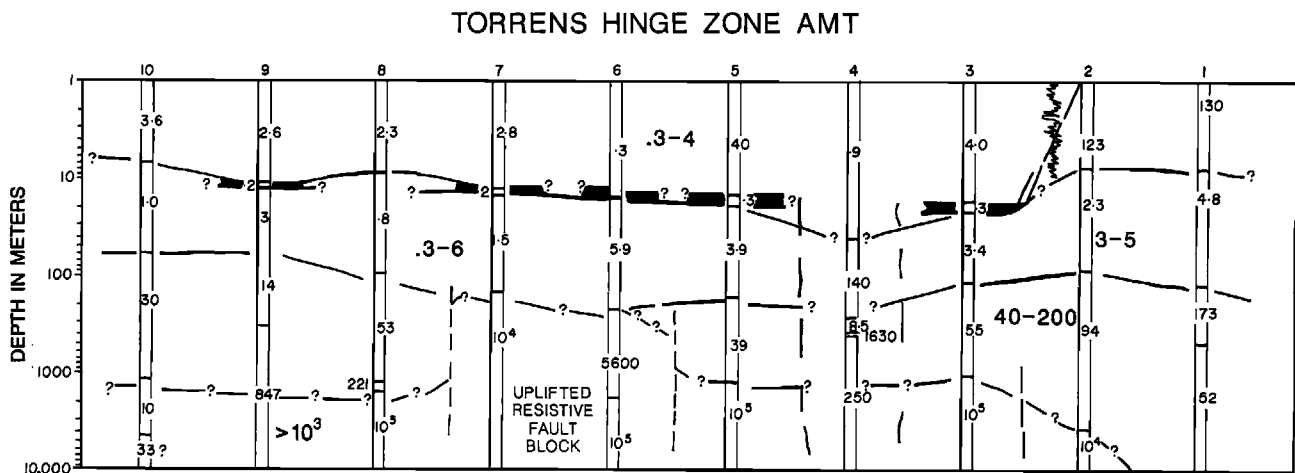


Fig. 68. Resistivity cross section made by stitching together 1-D inversion.

Except for direct curve matching, the earliest and the simplest of the 1-D MT inversion methods was first described in Niblett and Sayn Wittgenstein (1960), and was later independently put forward in Bostick (1977) and in Vanyan et al. (1980). Jones (1983) showed that the three formulations are the same. The  $\rho_a(T)$  curve (where  $T$  is the period) is transformed to  $\rho_B(h)$  (where  $h$  is a depth) by

$$\rho_B(h) = \rho_a(T) \frac{1 + m(T)}{1 - m(T)}$$

where  $h = \sqrt{T\rho_a(T)/2\pi\mu}$  and  $m(T)$  is the slope of the  $\log(\rho_a) - \log(T)$  curve,

$$m(T) = \frac{d[\log\rho_a(T)]}{d[\log(T)]}.$$

Jones' paper includes a brief history of this and related methods, including modifications by Weidelt and the KLM Group (Weidelt et al., 1980) and by Goldberg and Rotstein (1982). The main modification is to use equation (10) to replace the slope  $m(T)$  by a phase term, i.e.

$$\rho_B(T) = \rho_a(T) \left( \frac{\pi}{2\phi(T)} - 1 \right),$$

where  $\phi(T)$  is the phase. Results are not quite the same in the two cases because of the approximation in equation (10). The important attractions of this Niblett-Bostick (N-B) inversion are that the computation involved is trivial, and that assuming an initial model is not necessary. Because computation can be done with pencil and paper if necessary, it is widely used in the field, even though results are qualitative. That is, if the input is a  $\rho_a$  curve calculated for some particular model, the N-B inversion will not generally return the same model. However, common practice uses N-B output both for data presentation and as input to more quantitative inversion programs. An example of a result is included in Figure 50.

Quantitative inversion requires considerably more computation because the relationship between  $\rho_a(f)$  and  $\rho(z)$  is not linear. That is, doubling the value of  $\rho(z)$  at some depth can increase, decrease, or have no detectable affect at all on  $\rho_a(f)$  at a particular frequency. The characteristic of quantitative inversion that distinguishes it from N-B inversion is that, given a theoretical  $\rho_a$  curve for some model, the quantitative algorithm will return that input model as closely as desired, subject to computer roundoff.

Parker (1983) discusses the existence and uniqueness of (1-D) solutions for a given data set. Obviously there will be data sets which are not sensible: they may be irregular or discontinuous (possibly on account of

faulty equipment or incorrect processing), whereas we know that every forward model curve must be smooth and continuous. He shows that *every* valid 1-D apparent resistivity curve can be generated by a particular integral, but that not all such curves have an inverse solution. In the case of numerically sampled and truncated data, Parker showed that, if any solution exists, the best fitting one will consist of a set of highly conductive thin sheets separated by large air gaps, and is extremely unlikely geologically. No geologically acceptable model will give a better fit to the data. He also shows how to decide, on a statistical basis, whether a particular noisy data set is compatible with any 1-D model. This serves as a guide when it is clearly necessary to discard some data points.

As regards the uniqueness of a solution, it was shown earlier (Tikhonov, 1965; Bailey, 1970) that a solution is unique only if it fits perfectly to a continuous data set at all frequencies. Thus in the practical case there are no mathematically unique solutions, although it may be that, in a geological sense, all allowable solutions are equivalent.

It is now generally accepted that inversion involves two steps, finding a model to fit the data and then assessing its significance. Assessment has both a statistical and a geological part: the statistician and theoretician are concerned with the range of other models that could fit the *data* equally well, without regard for geological realities. The practitioner, who may be concerned with explaining the subtleties of basin growth or may have to plan a multi-million dollar well, must worry about both.

As noted, two general kinds of models are used, discrete and continuous, and in some algorithms they can be combined. The discrete models are piles of layers, each of which is uniform. Some algorithms allow the layers to be anisotropic (Abramovici, 1974). A major justification for using layered models in many cases is their correspondence to geological conditions. In the usual case (e.g., Weidelt, 1972; Jupp and Vozoff 1975) the layers can take on any finite positive thicknesses and resistivities. In fact the parameters used are  $\log(\rho_i)$  and  $\log(t_i)$ , which are always positive.

In most inversion algorithms, it is necessary to assume some starting model, including the number of layers. (A brief discussion of exceptions follows.) The algorithms assume that the parameters of the answer are close to those of the starting model. For that model they calculate  $\rho_a(f_i)$  and  $\phi(f_i)$  at each frequency  $f_i$  where there are data. The calculated values will generally be somewhat different from the measured values, and the differences are taken to be fitting errors  $\epsilon_i$ . The partial derivatives of each calculated value with respect to each parameter  $\rho_j$  and  $t_j$  are also calculated, either analytically or by differencing, at

each frequency. Using the notation of Raiche et al. (1985)

$$\epsilon = \mathbf{J}\delta\mathbf{P},$$

where  $\epsilon$  is the vector of errors,  $\mathbf{J}$  is the matrix of partial derivatives,  $\mathbf{P}$  is the vector of (logarithms of) the parameters, and we want to find the parameter changes  $\delta\mathbf{P}$ . When they have been found, these changes are added to the initial parameter values to give a modified model. The process is repeated until no further sensible changes are being made, as judged by the size of the fitting errors relative to measurement errors, or by the relative changes being made in the parameters.

Many different methods have been used for solving the equations. I have used the singular value decomposition (SVD) (Golub and Reinsch 1970), which is fast, allows satisfactory control over the inversion process, and provides the information necessary to assess the resulting model. In particular, we have found that (a) the results are relatively insensitive to the starting model (b) if the final model is inappropriate it is very obvious in the plot of observed and model  $\rho_a$ , at which point a different starting model can be used, (c) the fitting error diminishes as the number of layers is increased, rapidly at first and then slowly, but the parameter values become unstable as the number of parameters increases. Fischer and Le Quang (1982) describe this behavior, and Raiche et al. (1985) suggest the use of the APRE statistic to determine the optimum number of layers.

In this procedure it is clear that some of the parameters of the model may have very little effect on the apparent resistivities. For example, a layer may be too deep to affect the results for the range of frequencies under consideration, or it may be too thin. It is also well known that, for a sufficiently resistive layer in a conductive section, changing its resistivity will have no effect on  $\rho_a$  or phase. This is an unimportant parameter. When inverting to noisy data it is essential that such unimportant parameters are not allowed to change during inversion. This destabilizing effect of unimportant parameters is one of the fundamental difficulties of EM inversion. The problem is said to be poorly posed. Both the SVD and Marquardt schemes incorporate features to control the changes in unimportant parameters.

Another practical difficulty is properly accounting for data noise in inversion and, in particular, in attributing confidence limits to the final parameter values which take the actual noise conditions into consideration. Clearly there can be more confidence in the output value of any parameter when the noise levels are low than when they are high. The method, as

described in Jupp and Vozoff (1975), allows for a variance estimate to be input with each data point. Output confidence limits thus directly reflect the size and location of noise in the data. The sample standard deviation measured in the field is used routinely for this purpose. However in some circumstances it is necessary to arbitrarily reject data points which are clearly outliers, because of their distorting effect when using least squares minimization. Parker (1983, pg. 11) provides a rational objective criterion for rejecting outliers.

Several published methods describe procedures to transform directly from the observed data to a model without minimization, essentially by 'layer-stripping' from the top. (Considering that each data point yields a thin layer, the N-B scheme can be said to operate in the same way.) Parker (1983) shows his results for a COPROD (Jones, A. G., pers. comm.) data set. The method is said to be computationally simple and flexible, and results can be made to fit the observations as closely as desired.

The fitting of smooth resistivity-depth functions is described in Oldenburg (1979), Parker (1983), Coen et al. (1983), and Constable et al. (1987). In the last reference, model smoothness (specified in terms of average slope or average second derivative over the full depth of the section) is minimized along with fitting error and the balance between them is determined by an arbitrary weighting coefficient. The smooth resistivity-depth function is approximated by a large number of thin layers. Because of this, the model response can be made to fit the observed data as closely as desired, but of course the closer the required fit, the more irregular the model. By varying the weighting coefficient, it is possible to obtain any fitting error desired, but with a corresponding change in model roughness. Model roughness was measured either by

$$R_1 = \int (dm/dz)^2 dz$$

or

$$R_2 = \int (d^2m/dz^2)^2 dz.$$

Their objective function to be minimized had the form

$$U = R + \lambda[\epsilon^2 - \epsilon_0^2],$$

where  $\epsilon_0$  is the desired sum squared fitting error,  $\epsilon$  is the error with the current model, and  $\lambda$  is the arbitrary weighting coefficient. The procedure used by Constable et al. (1987) was to sweep through a range of coefficients until they obtained the desired fit. Because of the nonlinearity it was necessary to iterate from a starting model (which could take on just a constant value).

Parker's (1983) approach to the continuous function problem is somewhat more elegant, and involves functions which are truly continuous rather than consisting of many thin layers. He chooses to use an analytical scheme, which requires that the noisy and incomplete field data be converted to a smooth, continuous impedance function. His contribution was to find a smoothing scheme which is both well suited to the kind of data and convenient to manipulate.

To compare the results of five very different approaches applied to a common data set, which Parker presents in his paper is thought provoking. These results are summarized in Figure 69. In this connection, some effort has gone into techniques to find all acceptable models which fit the data by randomly changing the parameters of layered models and keeping those which fit the measurements to within some chosen sum squared error (Monte Carlo search). The conclusions were that the results were unbiased but very expensive on account of the vast number of forward models which had to be generated, especially if the model had more than a few layers. In the end, some external guidance was required to choose between results.

We are then left with the question of the significance of the data, as noted earlier. A wide choice of possible interpretations is available for the same data set, and we must apply some outside constraints if we are to make sense of them. It is necessary to know something about the geological history and style of the location.

Without such information we have no basis to choose between these various pictures, or still others which might be produced in the future. Thus some other data (seismic, gravity, magnetic, or borehole) is necessary. However given such information, the MT data are clearly capable of describing the electrical conductivity structure in useful detail. It is an empirical fact that computer inversion is routinely used for many practical purposes. Considering the pragmatic nature of explorationists, this can only be so because it works. Hence in these cases, at least, an appropriate model family has been found.

Having established that an output model has the right form, what can be said about the specific parameter values which were returned by a particular program for a specific data set? Backus and Gilbert (1970) showed that, for continuous conductivity-depth variation, it is possible to state with confidence a kind of average conductivity over a range of depth. If the conductivity-depth function we get is  $\sigma(z)$ , then at depth  $z_0$  we can confidently state the average value

$$\sigma_{\text{avg}}(z_0) = \int_0^\infty \sigma(z) A(z, z_0) dz.$$

If  $A(z)$  were a delta function  $A(z, z_0) = \delta(z, z_0)$ , then  $\sigma_{\text{avg}}(z_0) = \sigma(z_0)$  and we would have perfect resolution.  $A(z, z_0)$  is a window through which the model has to be viewed. It is a smooth function with its peak near ( $z = z_0$ ) and a width which depends on the model and

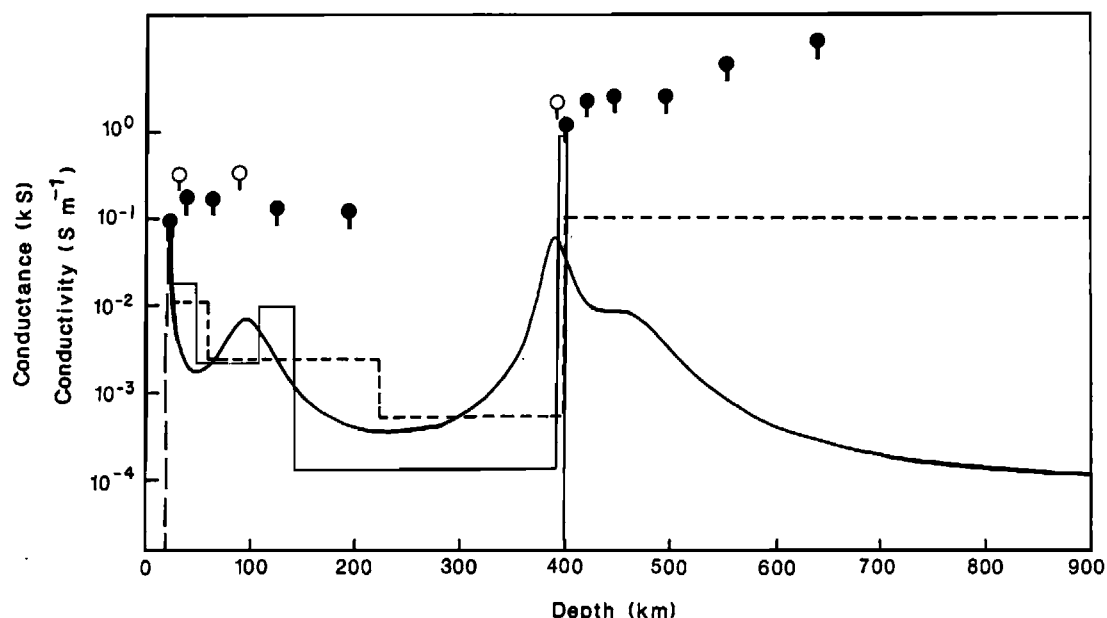


Fig. 69. Five of the inversion results obtained by Parker from a COPROD data set and his smoothed version of it. Circles indicate the conductances of delta function sheets (in kS). Open circles gave the best fit to the data. (After Parker, 1983.)

on the frequency range of the measurements. The width of  $A$  generally increases with depth, which is recognized when we present our results on a log depth scale. Error in the measurements introduces error in  $\sigma_{\text{avg}}$ .

As mentioned the results of inversion can sometimes be improved by fixing parameters known from other measurement which often improves the resolution of some of the other parameters. Also different electrical and EM methods differ in their ability to resolve certain kinds of parameters. For example, inductivity coupled EM methods can resolve the thickness and conductivity of thin conductive layers much better than dc methods can. Conversely, dc methods can resolve the parameters of thin resistive layers much better than is possible by, say, MT. This behavior has been used to advantage to improve the resolution of a complete section by jointly inverting dc and MT data (Jupp and Vozoff 1977; Constable et al. 1987), dc and transient EM data (Raiche et al. 1985) and MT and large scale TEM (LOTEM) data (Hördt, pers. comm., 1990). The improvements obtained by these means can be dramatic.

The same inversion techniques used for layered models can be applied to 2-D models, although the processes are somewhat more time-consuming. Weidelt (1975), and Jupp and Vozoff (1977) used models consisting of rectangular blocks whose resistivities are the variable parameters. Using the TE apparent resistivities and  $H_z$  (Weidelt) or both TE and TM apparent resistivities (Jupp and Vozoff), the algorithms developed for 1-D inversion were used to find best values for block resistivities. A major limitation of these methods is the need to fix block geometry at the start. Pek (1987) modified the Jupp-Vozoff (1977) algorithm to include block boundaries as variables. Travis and Chave (1986) described a program in which the finite element mesh moves in response to high current density gradients. This program should provide better efficiency as well as more accurate results, since current density would be expected to rearrange itself as frequency changes. [A recent book by Crank (1987) deals entirely with the problem of moving boundaries in numerical solutions of partial differential equations.] Another approach is that of Jiracek et al. (1987a), in which a smooth 2-D conductivity section made up of continuous functions is produced. DeGroot-Hedlin and Constable (1990) report on the extension of their Occam inversion to the 2-D geometry. The arguments for smooth or blocked cross sections are directly analogous to those in the 1-D case. Smooth sections seem better suited to problems such as the temperature and fluid distribution in hydrothermal gold and geothermal prospects, whereas the blocks seem a better representation of many other environments.

A major stumbling block to more widespread use of 2-D and 3-D inversion has been the large amount of computing required. Smith (1988) and Oldenburg and Ellis (1990) found that it is possible to use short-cuts which allow major reductions in the computing load. Smith (1988) did approximate inversion one site at a time along a traverse, but using partial derivatives that depend on both vertical and lateral variations in a 2-D model. The resulting single site models are strung together with interpolation between, to make up a new 2-D model. The response of this model is then computed and compared with the observations. The residuals are then used to modify the parameters in the single site inversions. Very satisfactory results are obtained with a few percent of the computing cost of straightforward 2-D inversion.

In the second reference, Oldenburg and Ellis (1990) also use approximate inversion and accurate forward modeling to reduce the computation. Two approaches have been developed. In one, the approximate inversion is applied to both the observations and the model response. The two resulting models are expected to differ, even if the observations and model response are identical, because the model response is computed exactly. The model is then altered on the basis of physical ideas, its response is recomputed exactly, the approximate inversion is repeated, and the models are compared again. This procedure continues iteratively until the differences stabilize.

The difficulties with this approach are that the models become increasingly complicated as iterations proceed, and that there is no obvious objective function to be minimized since one generally does not have a true model for comparison. To enable comparisons to be done on responses while still using approximate inversion, a different procedure was devised in which the data are modified, the modified data are inverted approximately, and the response of the resulting model is computed exactly. These responses are then compared with the observed data. The new difference is then added to the observed data and the process repeated. The sequence was found to converge very rapidly, although regularization was still required in order to keep the models from taking on an oscillatory character.

There seems little doubt that practical 3-D inversion is within reach, given these developments in approximate inversion coupled with simplified forward modeling. The major problem in the future will be collecting enough data, in the right locations.

### Effective Strike Direction

Defining regional trends is an important objective of many geophysical reconnaissance surveys. Estimation

of strike direction is straightforward in many MT surveys through the combination of tensor and tipper rotation angles. However the estimation becomes a problem as soon as any significant three-dimensionality enters. Certainly many cases can be found in which it has not been possible to uniquely specify strike, as illustrated in our example at the start of this section. For example, as discussed further, 3-D statics in (otherwise) 2-D conditions cause offsets which alter the relative values of  $\rho_{xy}$  and  $\rho_{yx}$  and cause the impedance rotation to choose the wrong curve. This can be rectified if a reasonable statics correction can be made. When the three-dimensionality is more profound, the meaning of strike must be reconsidered.

There have been two approaches to solving the problem. Conventional definitions can be applied to 3-D models and their usefulness can be examined graphically. Jones and Vozoff (1978) examined tensor and tipper strike directions for a suite of 3-D models and concluded that, for the limited set of models studied, tensor directions were marginally more stable than tipper directions. However Wannamaker et al. (1984b), for a different model, concluded just the reverse. They also show that tipper directions recover with decreasing frequency in the same way as impedance phase, and so they are expected to be a better interpretive tool.

Gamble et al. (1982) discuss the general problem of defining strike in 3-D geology. They then test the definitions in a set of field data which includes extensive areal coverage. The area is known to be strongly 3-D in places but is expected for geological reasons to have an underlying linear regional grain. By averaging the tipper and impedance strike directions in all frequency bands from all sites, they obtained a direction which was within a few degrees of the expected regional strike. They also concluded that the impedance directions were the more stable of the two. Zhang et al. (1986) use the formalism of Wannamaker et al. (1984a, b) to study data from an area in which there are two strike directions, one superficial and one regional. They show that the local (superficial) can be found by rotating the impedance tensor until the two terms on the diagonal are linearly related (i.e., have the same phase). Similarly, the regional strike direction is defined when the terms in each column have the same phase. Least-squares procedures are given for computing the two directions, which are frequency-dependent. Like other rotation angles derived from  $Z$ , these have a 90 degree ambiguity. These results are in agreement with those of Groom and Bailey (1989) for the same model.

More experience with these new analyses must be acquired, but the best way to fit strike direction, if one exists, may be near a satisfactory solution.

## Statics Compensation

Causes of the statics problem were described in a previous section. Their compensation is an interpretive matter since, by definition, the data themselves are inadequate.

The awareness of static shifts has emerged as experience accumulated with closely spaced sites. The shifts are analogous to the static time delays in reflection seismology. Apparent resistivity curves at closely spaced sites are sometimes found to look very similar, but are displaced vertically along the  $\rho_a$  axis. Alternatively, the TE and TM curves at a single site may look the same except for a vertical displacement. In each case the phases may be indistinguishable. This condition is attributed to superficial features that multiply apparent resistivity curves by a frequency-independent constant. Some attempt must be made to identify and compensate for these or the interpretation of deeper features will be adversely affected.

An analytic demonstration of the effect was developed in Wannamaker et al. (1984b) and was discussed previously. Simply using 1-D inversion in such cases gives model resistivities too large by  $\rho_{ij}/\rho_l$  and depths too great by  $\sqrt{\rho_{ij}/\rho_l}$ . (Refer to the General 3-D Case Section.) When the superficial feature is 2-D with strike in the  $x$  direction, then no surface charge distribution is created and  $P_{xx}$  is zero. Then a 1-D inversion of  $\rho_{xy}$  ( $E \parallel$  to strike) will give results less distorted than the  $\rho_{yx}$  ( $E \perp$ ) component, at least at low frequencies. For example, if  $\rho_a$  is shifted vertically by a factor of two, then the interpreted resistivities will be too large by a factor of two and the thicknesses will be too great by a factor of  $\sqrt{2}$ . Statics in MT are discussed in Adam (1976), Sternberg et al. (1985), Berdichevsky and Dmitriev (1976), Ranganayaki and Madden (1980), Larsen (1977), Jones (1988), and Orange (1989), among others.

A number of solutions to the statics problem have been suggested to date, but none have proven completely satisfactory. For a time, it was common practice simply to shift curves to a common high frequency asymptote based on any defensible assumption. Larsen (1975) developed a rational argument for this approach, using it to correct MT soundings in Hawaii for the effect of the surrounding ocean, and as noted previously, Berdichevsky and Dmitriev (1976) produced semi-analytical dc models to provide guidelines for shifting the curves.

The quantity  $\rho_d$  (or  $\rho_{det}$ ) of Ranganayaki (1984), referred to in the General 3-D Case Section, was introduced by Berdichevsky and Dmitriev (1976) to simplify data presentation. Since the quantity is independent of rotation it avoids data shifts arising from noisy rotation angles, and hence is a kind of smoothing

operator. However, it does not remove statics effects, as can be seen in Figure 37.

Several papers on MT statics corrections given at the Society of Exploration Geophysics meetings in recent years, particularly in connection with CSAMT surveys, tend to follow those approaches used for seismic data. Statics problems will be more severe in CSAMT than in MT because, among other reasons, they can be set up by irregularities at either the transmitter or the receiver. Zonge and Hughes (1991) in the CSAMT chapter in this volume describe several empirical techniques for spatially smoothing those data. Warner et al. (1983), using data from many closely spaced sites, calculated an average of  $\log \rho_a$  at each site and then adjusted each curve so that these averages varied smoothly between sites. The adjustment was justified on the basis that sites were only a few hundred meters apart in a relatively undeformed sedimentary basin. Their results, presented in the form of their N-B inversions, are said to accurately depict known subsurface conditions. Andrieux and Wightman (1984) describe the use of independent electrical information to fix the near surface resistivities. In particular, they suggest the use of shallow resistivity logs, transient EM, or dc resistivity. Sternberg et al. (1982, 1985) discuss both spatial smoothing and the use of independent information. They show an example in which a transient EM central loop sounding was used to shift the MT  $\rho_a$  curve. The inversion of the shifted curve fitted the resistivity log much more closely than that of the original curve. Bloomquist et al. (pers. comm.) used an approach very similar to the one described in Sternberg et al. (1985) involving joint inversion with dc resistivity measurements. This approach can be quite satisfactory provided the region of overlap is isotropic, otherwise the two sets of data will be incompatible and no sensible result can be expected. Kurtz (pers. comm.) in interpreting a regional survey from Vancouver Island, forces depth to a major horizon to vary smoothly and accommodates the data with a smooth variation in the resistivity of that horizon.

The most elaborate of the methods thus far put forward to overcome the problem of static shift is EMAP. Bostick (1985, 1986a, 1986b) and Torres-Verdin (1985) contend that  $E$  field fluctuations due to shallow structure can be filtered out with proper data acquisition and processing. To achieve this filtering, data are collected from contiguous colinear dipoles on lines tens of kilometers long, usually with only one pair of reference orthogonal magnetometers in the neighborhood. Impedances are smoothed spatially at each frequency with low pass filters whose wavelength is of the same order as the skin depth. In examples using both models and real field data, very impressive

improvements were made in the appearance of resistivity depth sections (Bostick, 1986b). Torres-Verdin (1985) demonstrates the rationale of the approach in the case of 3-D resistivity perturbations with very small contrast. Figure 70 is an example, kindly provided by Prof. Bostick, of apparent resistivity-frequency sections in N-B format, before and after EMAP filtering. The data are then interpreted by conventional modeling. Further examples of EMAP results can be found in Word et al. (1986) and Shoe-maker et al. (1986, 1989).

Some major assumptions made in the development of EMAP include the assumptions that the traverse line is perpendicular to strike and that adequate interpretation can be done with only one component (TM) of the  $E$  field, so that its quantitative validity remains an open question. Another limitation of the method is its inability to deal with depths greater than a fraction to the total length of the traverse, so that difficulty would be anticipated with smoothing when the upper part of the section is highly resistive. Investigations of the deep crust and upper mantle, one of the most productive applications of MT, would seem to be out of the question. In its defense, EMAP has in some places greatly improved the quality of the results obtained.

Finally, Bahr (1985, 1987) studies several ways of decoupling local (i.e., upper crustal) and regional (deep crust upper mantle) effects in impedances. One way involves determining magnetic transfer functions at  $S_q$  periods by means of a Geomagnetic Depth Sounding profile between the site of interest and another site which is free of local distortion. If a 2-D conductivity model can then be found which satisfies both the MT impedances and the GDS profile, the problem can be said to be solved. Bahr also suggests a tensor distortion technique based on Wannamaker et al. (1984 a and b) similar to those used in Larsen (1977) and others, in Zhang et al. (1986), and in Groom and Bailey (1989) as discussed above. It is notable that Gamble et al. (1982), Bahr (1985, 1987), Bostick (1986b) and Park and Torres-Verdin (1988) invoke external data (i.e., data other than impedance from a single location) to help solve the problems of statics and strike determination. This is in accord with the conclusion of Groom and Bailey (1989), that there is no internal solution to the statics problem from the MT data at a single location.

Thus at present there are several ways to remove the effects of statics from impedance tensors, and it will be necessary to gain some experience with them in a variety of environments to see what each has to contribute. However the best solution is to collect enough data that interpretation in critical areas can be made with confidence.

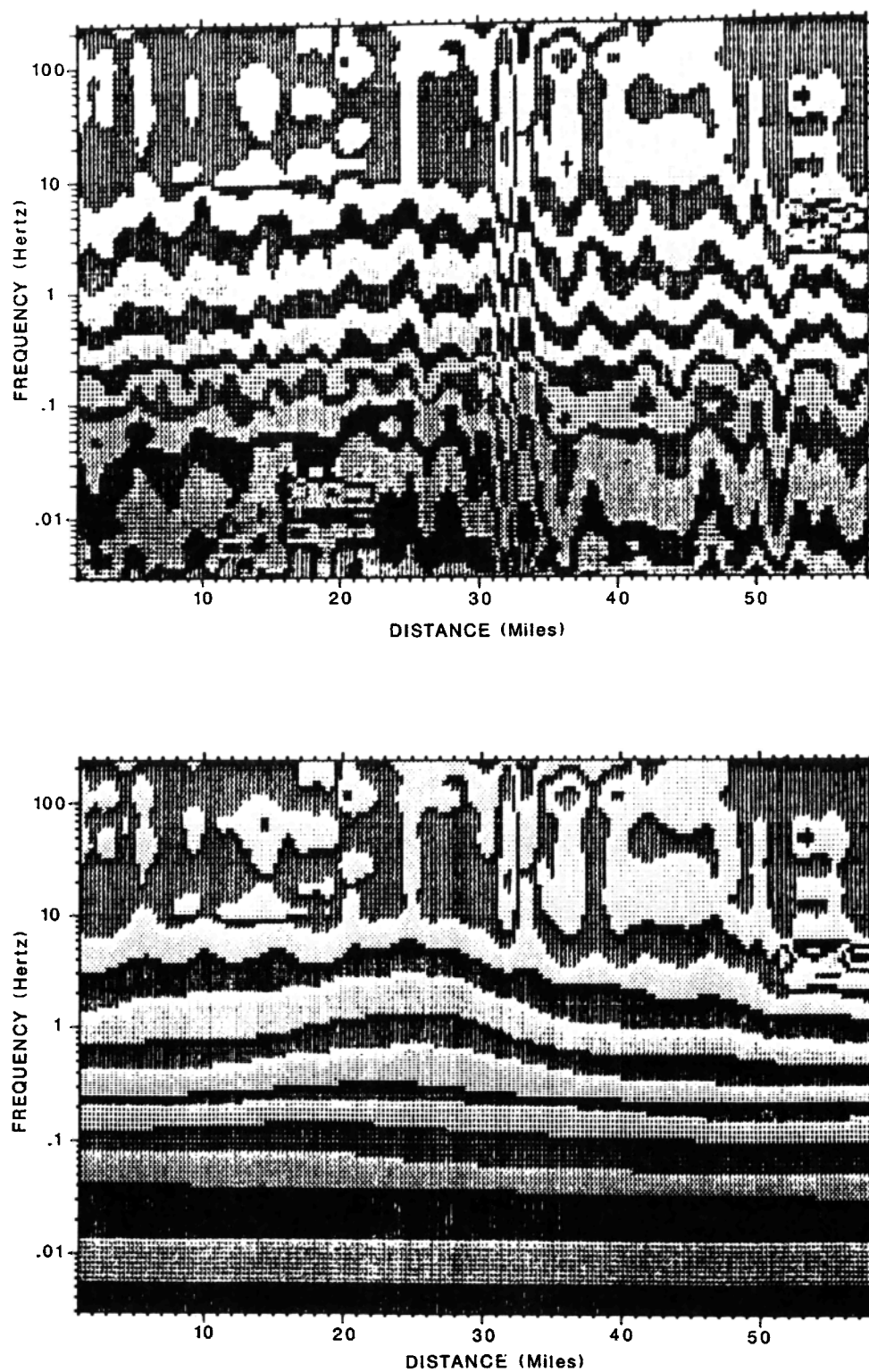


Fig. 70. EMAP data, with raw section above and spatially-filtered section below (Bostick, pers. comm.)

## Case Histories

Unlike the situation a few years ago, there are now a reasonable number of instructive case histories in the literature, and others in preparation, in addition to the one at the start of this section. The Society of Exploration Geophysicists Reprint volume, *Magnetotellurics* (Vozoff, 1986) includes a case histories section. Still others are found in *Geophysics*, *Geophysical Prospecting*, *Exploration Geophysics*, the *Geophysical Journal*, the *American Association of Petroleum Geologists Bulletin*, the *Canadian Journal of Earth Sciences*, in the extended abstracts from the SEG and ASEG meetings, in the geothermal literature, and elsewhere in this volume. As well, numerous nonproprietary surveys have been carried out and are available commercially.

## ACKNOWLEDGMENTS

For their help, of several kinds and often in considerable amounts, I am indebted to Bob Anderson, Dick Bailey, Rod Bashford, Francis Bostick, Ted Bowen, Nigel Edwards, Carlos Flores, Ron Goss, Phil Hallof, Hu Wenbeg, George Jiracek, Alan Jones, Khader Khan, Ron Kurtz, Lou Lanzerotti, Jerry LaTorraca, Liu Song, Frank Morrison, Ed Nichols, Steve Park, Bob Parker, Ken Paulson, Bill Pelton, Risto Pirjola, Dave Redman, Dave Rostoker, Ken Roussel, Joe Sevenhuysen, Doddy Sutarno, Charley Swift, Carlos Torres-Verdin, J. Verø, Wang Jia-ying, Gordon West, Peter Wolfgang, Darrell Word, Eugene Yee and Ken Zonge. The AMT data of the Interpretation Section were collected as part of an AMIRA-sponsored project at Macquarie University. The hospitality provided by the Geophysics Division, Physics Department, University of Toronto made possible a pleasant and productive stay. That of the Geophysics Department, China University of Geosciences, Wuhan, forced me to read and clean up what I'd written in Toronto. Finally, I do sincerely appreciate Misac Nabighian's patience.

## REFERENCES

- Abramovici, F., 1974, The forward magnetotelluric problem for an inhomogeneous and anisotropic structure: *Geophysics*, **39**, 56–68.
- Adam, A., 1976, Geoelectric and geothermal studies (East-Central Europe, Soviet Asia): KAPG Geophysical Monograph, Akad. Kaido, Budapest.
- Andrieux, P., and Wightman, W. E., 1984, The so-called static corrections in magnetotelluric measurements: 54th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded Abstract EM1.1.
- Backus, G. E., and Gilbert, J. F., 1970, Uniqueness in the inversion of inaccurate gross earth data: *Phil Trans. R. Soc. Lond., A*, **266**, 123–193.
- Bahr, K., 1985, Magnetotelluric measurement of the electrical resistivity of the Earth's crust and Upper Mantle in areas with local and regional conductivity anomalies: Ph.D thesis, Univ. of Gottingen.
- , 1987, Interpretation of the magnetotelluric impedance tensor: Regional induction and local telluric distortion: *J. Geophys.*, **62**, 119–127.
- Bahr, K., and Groom, R. W., 1990, Corrections for near surface effects: Tutorial paper, 10th Workshop on EM Induction in the Earth, IAGA WG-12.
- Bailey, R. C., 1970, Inversion of the geomagnetic induction problem: *Proc. Roy. Soc.*, **315**, 185–194.
- , 1977, Electromagnetic induction over the edge of a perfectly conducting ocean: The *H*-polarization case: *Geophys. J. Roy. Astr. Soc.*, **48**, 385–392.
- Bannister, P. R., and Williams, F. J., 1974, Results of the August 1972 Wisconsin Test Facility effective earth conductivity measurements: *J. Geophys. Res.*, **79**, 725–732.
- Bendat, J. S., and Piersol, A. G., 1971, *Random data: Analysis and measurement procedures*: John Wiley & Sons.
- Berdichevsky, M. N., and Dmitriev, V. I., 1976, Basic principles of interpretation of magnetotelluric curves, in *Geoelectric and geothermal studies*: Adam, A., Ed., Akademi Kiado, 165–221.
- Berdichevsky, M. N., Vanyan, L. L., Kuznetsov, V. A., Levadny, V. T., Mandelbaum, M. M., Nechaeva, G. P., Okulesky, B. A., Shilovsky, P. P., and Shpak, I. P., 1980, Geoelectrical model of the Baikal region: *Phys. Earth Planet. Inter.*, **22**, 1–11.
- Bostick, F. X., Jr., 1977, A simple almost exact method of MT analysis: Abstract, in *Workshop on electrical methods in geothermal exploration*, Snowbird, Utah.
- , 1985, *Z-Axis Exploration, Inc., EMAP: Z-Axis, Inc., Unpub. brochure.*
- , 1986a, *EMAP processing procedures*: Unpub. rept.
- , 1986b, *Electromagnetic array profiling (EMAP): 56th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded abstract EM2.1*, 60–61.
- Bozorth, R. M., 1964, *Ferromagnetism*, 2nd Ed.: Van Nostrand.
- Cantwell, T., 1960, Detection and analysis of low frequency magnetotelluric signals: Ph.D. thesis, Mass. Inst. Tech.
- C.C.I.R., 1964, World distribution and characteristics of atmospheric radio noise: *Internat. Telecommunication Union, Rept. 322*, Geneva.
- Carlo, L., Singh, B. P., Rastogi, R. G. and Agarwal, A. K., 1982, The induced effects of geomagnetic variations in the equatorial region: *J. Geophys.*, **51**, 199–205.
- Cavaliere, A., Bailey, R. C., and Desbiens, R., 1986, A magnetotelluric study of the Kapuskasing Uplift: Presented at GAC-MAC-CGU Annual meeting, Ottawa.
- Cevallos, C., 1986, *Magnetotelluric interpretation—another approach*: Ph.D thesis, Macquarie University, Sydney.
- Chave, A. D., Thomson, D. J., and Ander, M. E., 1987, On the robust estimation of power spectra, coherences and transfer functions: *J. Geophys. Res.*, **92B**, 633–648.
- Claerbout, J. F., 1976, *Fundamentals of geophysical data processing*: McGraw-Hill.
- Clarke, J., 1976, Superconducting quantum interference devices for low frequency measurements: in *Superconductor applications: Squids and machines*: Plenum Press, 67–124.
- Clerc, G., and Gilbert, D., 1964, Flux reaction applied to solenoids with permeable cores used for recording rapid variations of the magnetic field: (Note), *Ann. de Geophys.*, **20**, 499–502 (in French).
- Clerc, G., 1971, Contribution to the optimization of sensors designed to measure terrestrial magnetic field variations from ( $10^{-3}$  to  $10^4$  Hz). Thesis, Doctor of Engineering (Physics), Univ. of Paris (in French).
- Coen, S., Quercia, F., and Mackiewicz, M., 1983, Direct inversion of one-dimensional magnetotelluric data: *J. Geophys. Res.*, **88B**, 2407–2412.
- Constable, S. C., Parker, R. L., and Constable, C. G., 1987,

- Occam's inversion: A practical algorithm for generating smooth models from electromagnetic sounding data: *Geophysics*, **52**, 289–300.
- Counil, J. L., Le Mouel, J. L., and Menvielle, M., 1986, Associate and conjugate directions concepts in magnetotellurics: *Ann. Geophysicae*, **4**, B, 2, 115–130.
- Crank, J., 1987, Free and moving boundary problems: Oxford Univ. Press.
- deGroot-Hedlin, C., and Constable, S., 1990, Occam's inversion to generate smooth, two-dimensional models from magnetotelluric data: *Geophysics*, **55**, 1613–1624.
- Dmitriev, V. I. and Berdichevsky, M. N., 1979, The fundamental model of magnetotelluric sounding: *Proc. Inst. Elect. and Electron. Eng. Trans.*, **67**, 1034–1044.
- Edwards, R. N., Wolfgram, P. A., and Judge, A. S., 1988, The ICE-MOSES experiment: Mapping permafrost zones electrically beneath the Beaufort Sea: *Marine Geophys. Res.*, **9**, 265–290.
- Egbert, G. D., and Booker, J. R., 1986, Robust estimation of geomagnetic transfer functions: *Geophys. J. Roy. Astr. Soc.*, **87**, 173–194.
- Eggers, D. E., 1982, An eigenstate formulation of the magnetotelluric impedance tensor: *Geophysics*, **47**, 1204–1214.
- Falco, C. M. and Schuller, I. K., 1981, SQUIDS and their sensitivity for geophysical applications: in Weinstock, H., and Overton, W. C., Jr., Eds., *Squid applications to geophysics*: Soc. Expl. Geophys., 13–18.
- Filloux, J. H., Lilley, F. E. M., Ferguson, I. J., Bindoff, N. L., and Mulhearn, P. J., 1985, The Tasman Project of seafloor magnetotelluric exploration (Extended abstract): *Expl. Geophys.*, **16**, 221–223.
- Fischer, G., and Le Quang, B. V., 1982, Parameter trade-off in one-dimensional magnetotelluric modelling: *J. Geophys.*, **51**, 206–215.
- Fischer, G., and Schnegg, P.-A., 1980, The dispersion relations of the magnetotelluric response and their incidence on the inversion problem: *Geophys. J. Roy. Astr. Soc.*, **62**, 661–673.
- Fischer, G., Schnegg, P.-A., and Usadel, K. D., 1978, Electromagnetic response of an ocean-coast model to E-polarization induction: *Geophys. J. Roy. Astr. Soc.*, **53**, 599–616.
- Flores, C. and Edwards, R. N., 1985, Approximate calculation of low-frequency three-dimensional magnetotelluric responses using a multiple plate model: 55th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded Abstracts.
- Gamble, T. D., Goubau, W. M., and Clarke, J., 1979a, Magnetotellurics with a remote reference: *Geophysics*, **44**, 53–68.
- Gamble, T. D., Goubau, W. M., and Clarke, J., 1979b, Error analysis for remote reference magnetotellurics: *Geophysics*, **44**, 959–968.
- Gamble, T. D., Goubau, W. M., Miracky, R., and Clarke, J., 1982, Magnetotelluric regional strike: *Geophysics*, **47**, 932–937.
- Goldberg, S., and Rotstein, Y., 1982, A simple form of presentation of magnetotelluric data using the Bostick transform: *Geophys. Trans.*, **30**, 211–216.
- Golde, R. H., 1977, Lightning currents and related parameters, in Golde, R. H., Ed., *Lightning*, Vol 1: Academic Press, 309–350.
- Golub, G. H. and Reinsch, C., 1970, Singular value decomposition and least squares solutions: *Num. Math.*, **14**, 403–420.
- Golub, G. H., and Van Loan, C. F., 1983, *Matrix computation*: Johns Hopkins Univ. Press.
- Goubau, W. M., Gamble, T. D., and Clarke, J., 1978, Magnetotelluric data analysis: removal of bias: *Geophysics*, **43**, 1157–1166.
- Goubau, W. M., Maxton, P. M., Koch, R. H., and Clarke, J., 1984, Noise correlation lengths in remote reference magnetotellurics: *Geophysics*, **49**, 433–438.
- Groom, R. W. and Bailey, R. C., 1989, Decomposition of magnetotelluric impedance tensors in the presence of three-dimensional galvanic distortion: *J. Geophys. Res.*, **94B**, 1913–1926.
- Groom, R. W., and Bailey, R. C., 1991, Analytic investigations of the effects of near-surface 3-D galvanic scatterers on MT tensor decompositions: *Geophysics*, **56**, (In press).
- Häkkinen, L. and Pirjola, R., 1986, Calculation of electric and magnetic fields due to an electrojet current system above a layered earth: *Geophysica*, **22**, 31–44.
- Häkkinen, L., Pirjola, R. and Sucksdorff, C., 1990, EISCAT magnetometer cross and theoretical studies connected with the electrojet current system: *Geophysica* (in press).
- Hoehn, G. L., and Warner, B. N., 1983, Magnetotelluric measurements in the Gulf of Mexico at 20 m depth: in Geyer, R., Ed., *CRC Handbook of geophysical exploration at sea*, CRC Press, Inc., 397–416.
- Hohmann, G. W., 1988, Numerical modeling for electromagnetic methods of geophysics: in Nabighian, M. N., Ed., *Electromagnetic methods in applied geophysics—Theory*, Vol. I.
- Holcombe, H. T., 1982, Terrain effects in resistivity and magnetotelluric surveys: Ph.D thesis, Univ. of New Mexico, Albuquerque.
- Holliday, J., 1981, AMT signal problems at 2 kHz to 3 kHz and AMT results from the Griffith area, NSW: B.Sc. (Hons) thesis, Macquarie University.
- Huber, R., 1981, Robust statistics, John Wiley & Sons, Inc.
- Hughes, W. J., and Southwood, D. J., 1976a, The screening of micropulsation signals by the atmosphere and ionosphere: *J. Geophys. Res.*, **81**, 3234–3240.
- Hughes, W. J., and Southwood, D. J., 1976b, An illustration of modification of geomagnetic pulsation structure by the ionosphere: *J. Geophys. Res.*, **81**, 3241–3247.
- Hutton, R., 1965, Equatorial effects: *Radio Sci.*, **69D**, 1169–1177.
- Jackson, J. D., 1975, *Classical Electrodynamics*, 2nd Ed.: John Wiley & Sons, Inc.
- Jaklevic, R. C., Lambe, J., Silver, A. H., and Mercerau, J. E., 1964, Quantum interference effects in Josephson tunneling: *Phys. Rev. Lett.*, **12**, 159–160.
- Jiracek, G. R., 1973, Numerical comparisons of a modified Rayleigh approach with other rough surface EM scattering solutions: *Inst. Elect. and Electron. Eng., Trans. Antennas Propagat.*, **AP-21**, 393–396.
- Jiracek, G. R., 1990, Near-surface and topographic distortions in electromagnetic induction: *Surveys in Geophys.*, **11**, 163–203.
- Jiracek, G. R., Reddig, R. P. and Kojima, R. K., 1989, Application of the Rayleigh-FFT technique to magnetotelluric modelling and correction: *Phys. Earth Planet. Inter.*, **53**, 365–375.
- Jiracek, G. R., Rodi, W. L., and Vanyan, L. L., 1987b, Implications of magnetotelluric modelling for the deep crustal environment in the Rio Grande Rift: *Phys. Earth Planet. Inter.*, **45**, 179–192.
- Jiracek, G. R., Curtis, J., Meile, M. J., and Vozoff, K., 1987a, Static distortion correction in magnetotellurics using Rayleigh-FFT modeling and other geophysical data: Extended abstract, IUGG, Vancouver.
- Jones, A. G., 1983, On the equivalence of the “Niblett” and “Bostick” transformations in the magnetotelluric method: *J. Geophys.*, **53**, 72–73.
- , 1988, Static shift of magnetotelluric data and its removal in a sedimentary basin environment: *Geophysics*, **53**, 967–978.
- Jones, A. G., Chave, A. D., Egbert, G., Auld, D., and Bahr, K., 1989, A comparison of techniques for magnetotelluric response function estimation: *J. Geophys. Res.*, **94**, 14201–14213.
- Jones, F. W., and Price, A. T., 1971, Geomagnetic effects of sloping and shelving discontinuities of earth conductivity: *Geophysics*, **36**, 58–66.

- Jones, F. W., and Vozoff, K., 1978, The calculation of magnetotelluric quantities for three-dimensional conductivity inhomogeneities: *Geophysics*, **43**, 1167–1175.
- Jupp, D. L. B., and Vozoff, K., 1975, Stable iterative methods for the inversion of geophysical data: *Geophys. J. Roy. Astr. Soc.*, **42**, 957–976.
- , 1976, Discussion on 'The magnetotelluric method in the exploration of sedimentary basins', *Geophysics*, **41**, 325–328.
- , 1977, Resolving anisotropy in layered media by joint inversion: *Geophys. Prosp.*, **25**, 460–470.
- Karmann, R., 1975, Optimization of the signal/noise ratio of inductance coil magnetometers for magnetotellurics. Thesis, Doctor of Engineering, Braunschweig Technical University.
- , 1977, Search-coil magnetometers with optimum signal-to-noise ratio. *Acta. Geodaet. Geophys. Mont.*, **12**, 353–357.
- Katz, A. J. and Thompson, A. H., 1987, Prediction of rock electrical conductivity from mercury injection measurements: *J. Geophys. Res.*, **92B**, 599–607.
- Keller, G. V., 1982, Electrical properties of rocks and minerals, in Carmichael, R. S., Ed., *CRC Handbook of physical properties of rocks*, Vol. I: CRC Press, Inc., 217–293.
- Kojima, R. K., 1985, Application of the Rayleigh-FFT technique to magnetotelluric modelling: M.Sc. thesis, San Diego State University.
- Kunetz, G., 1972, Processing and interpretation of magnetotelluric soundings: *Geophysics*, **37**, 1005–1021.
- Labson, V. F., 1985, Geophysical exploration with audio frequency magnetic fields: Ph.D. thesis, Univ. of Calif., Berkeley.
- Labson, V. F., Becker, A., Morrison, H. F., and Conti, U., 1985, Geophysical exploration with audiofrequency natural magnetic fields: *Geophysics*, **50**, 656–664.
- Lanczos, C., 1961, *Linear differential operators*: Van Nostrand.
- Lanzerotti, L. J., and Southwood, D. J., 1979, Hydromagnetic waves, in Lanzerotti, L. J., Kennel, C. F., and Parker, E. N., Eds, *Solar system plasma physics*, Vol. 3: North Holland Publishing Co., 109–135.
- Larsen, J. C., 1975, Low frequency (0.1–6.0 cpd) electromagnetic study of deep mantle electrical conductivity beneath the Hawaiian Islands: *Geophys. J. Roy. Astr. Soc.*, **43**, 17–46.
- , 1977, Removal of local surface conductivity effects from low frequency mantle response curves: *Acta. Geodaet. Geophys. Mont.*, **12**, 183–186.
- LaTorraca, G. A., Madden, T. R., and Korringa, J., 1986, An analysis of the magnetotelluric impedance for three-dimensional conductivity structures: *Geophysics*, **51**, 1819–1829.
- Law, P. F. and Fannin, B. M., 1961, Radiation from a current filament above a homogeneous earth, with application to micropulsations: *J. Geophys. Res.*, **66**, 1049–1059.
- Lee, S., McMechan, G. A., and Aiken, C. L. V., 1987, Phase-field imaging: The electromagnetic equivalent of seismic migration: *Geophysics*, **52**, 678–693.
- Levy, S., Oldenburg, D., and Wang, J., 1986, Subsurface imaging using magnetotelluric data: *Geophysics*, **53**, 104–117.
- Liu Xiaoshan, Fan Sheng, Yang Ling, and Wang Jiaying, 1987, A new magnetotelluric inversion method—Vertical time-distance curve method: *Oil Geophys. Prosp.*, **22**, 445–453. (Chinese with English abstract).
- Madden, T. R., 1971, EMCAL: Two-dimensional network modelling program: Exploration Aids Inc., Bedford, Mass., (unpublished).
- Madden, T. R. and Mackie, R. L., 1989, Three-dimensional magnetotelluric modeling and inversion: *Proc. Inst. Elect. and Electron. Eng.*, **77**, 318–333.
- Madden, T. R., and Nelson, P., 1986, A defense of Cagniard's magnetotelluric method: in Vozoff, K., Ed., *Magnetotelluric methods*, Geophysics Reprint Series No. 5, Soc. Expl. Geophys., 89–102.
- MacIntyre, S. A., 1980, A portable low noise low frequency search coil magnetometer: *Inst. Elect. and Electron. Eng. Trans. Mag.*, **Mag-16**, 5, 761–763.
- Menke, W., 1984, *Geophysical data analysis: Discrete inverse theory*: Academic Press Inc.
- Moore, R. F., 1977, Screening and averaging magnetotelluric data prior to one-dimensional inversion: Record 1977/8, Bureau of Mineral Resources, Australia.
- Morrison, H. F., Conti, U., Labson, V. F., Nichols, E., and Goldstein, N. E., 1984, Field tests of noise in SQUID and induction magnetometers: Tech. Report LBID-901 Lawrence Berkeley Lab. and Univ. of Calif. Berkeley.
- Mozley, E. C., 1982, An investigation of the conductivity distribution in the vicinity of a Cascade volcano: Ph.D. Thesis, Univ. of Calif. Berkeley.
- Niblett, E. R., and Sayn-Wittgenstein, C., 1960, Variation of electrical conductivity with depth by the magnetotelluric method: *Geophysics*, **25**, 998–1008.
- Nichols, E., Morrison, H. F., and Clarke, J., 1988, Signals and noise in magnetotellurics: *J. Geophys. Res.*, **93B**, 13743–13754.
- Ogawa, T., 1982, The lightning current, in Volland, H., Ed., *CRC Handbook of Atmospherics*, Vol. 1: CRC Press, Inc., 23–63.
- Oldenburg, D. W., 1979, One-dimensional inversion of natural source magnetotelluric observations: *Geophysics*, **44**, 1218–1244.
- Oldenburg, D. W. and Ellis, R. G., 1990, Inversion of geophysical data using an approximate inverse mapping: *Geophys. J. Int'l.* (in press).
- Orange, A. S., 1989, Magnetotelluric exploration for hydrocarbons: *Inst. Elect. and Electron. Eng. Proc.*, **77**, 287–317.
- Orr, D., 1984, Magnetospheric hydromagnetic waves: their eigenperiods, amplitudes and phase variations; a tutorial introduction: *J. Geophys.*, **55**, 76–84.
- Park, S. K., 1985, Distortion of magnetotelluric sounding curves by three-dimensional structures: *Geophysics*, **50**, 785–797.
- Park, S. K., Orange, A. S., and Madden, T. R., 1983, Effects of three-dimensional structure on magnetotelluric sounding curves: *Geophysics*, **48**, 1402–1405.
- Park, S. K. and Torres-Verdin, C., 1988, A systematic approach to the interpretation of magnetotelluric data in volcanic environments, with applications to the quest for magma in Long Valley, California: *J. Geophys. Res.*, **93B**, 13265–13284.
- Parker, R. L., 1983, The magnetotelluric inverse problem: *Geophysical Surveys*, **6**, 5–25.
- Parkhomenko, E. I., 1982, Electrical resistivity of minerals and rocks at high temperature and pressure: *Rev. Geophys. Space Phys.*, **20**, 193–218.
- Parkinson, W. D., 1983, *Introduction to geomagnetism*: Scottish Academic Press.
- Pek, J., 1987, Numerical inversion of 2-D MT data by models with variable geometry: *Phys. Earth Planet. Inter.*, **45**, 193–203.
- Pellerin, L. and Hohmann, G. W., 1990, Transient electromagnetic inversion: A remedy for magnetotelluric static shifts: *Geophysics*, **55**, 1242–1250.
- Petiau, G., and Dupis, A., 1980, Noise, temperature coefficient and long time stability of electrodes for telluric observations: *Geophys. Prosp.*, **28**, 792–804.
- Pierce, E. T., 1977, Atmospherics and radio noise, in Golde, R. H., Ed., *Lightning*, Vol 1: Academic Press, 351–384.
- Pirjola, R. and Häkkinen, L., 1990, Electromagnetic field caused by an auroral electrojet current system model: in Kikuchi, H., Ed., *Proc. URSI Symp. on Environmental and Space Electromagnetics*, Springer-Verlag (in press).

- Polk, C., 1982, Schumann resonances, in Volland, H., Ed., CRC Handbook of Atmospheric, Vol 1: CRC Press, Inc., 112-178.
- Potemra, T. A., 1984, Magnetospheric currents: Am. Geophys. Union.
- Quon, C., Vozoff, K., Hoversten, M., Morrison, H. F., and Lee, K.-H., 1979, Localized source effects on magnetotelluric apparent resistivities: J. Geophys., **46**, 291-300.
- Raiche, A. P., Jupp, D. L. B., Rutter, H. and Vozoff, K., 1985, The joint use of coincident loop transient electromagnetic and Schlumberger sounding to resolve layered structures: Geophysics, **50**, 1618-1627.
- Ranganayaki, R. P., 1984, An interpretive analysis of magnetotelluric data: Geophysics, **49**, 1730-1748.
- Ranganayaki, R. P., and Madden, T. R., 1980, Generalized thin sheet analysis in magnetotellurics: Geophys. J. Roy. Astr. Soc., **60**, 445-457.
- Reddig, R. P., 1984, Application of the Rayleigh-FFT technique to topographic corrections in magnetotellurics: M.Sc. thesis, San Diego State Univ.
- Reddy, I. K., Rankin, D., and Phillips, R. J., 1977, Three-dimensional modeling in magnetotelluric and magnetic variational sounding: Geophys. J. Roy. Astr. Soc., **51**, 313-315.
- Rokityansky, I. I., 1961, On the application of the magnetotelluric method to anisotropic and inhomogeneous masses: Izvestia, **11**, 1607-1613.
- Rostoker, G., 1979, Geomagnetic micropulsations: Fund. of Cosmic Phys., **4**, 211-311.
- Samson, J. C., Jacobs, J. A., and Rostoker, G., 1971, Latitude-dependent characteristics of long-period geomagnetic pulsations: J. Geophys. Res., **76**, 3675-3683.
- Samson, J. C. and Rostoker, G., 1972, Latitude-dependent characteristics of high-latitude Pc 4 and Pc 5 micropulsations: J. Geophys. Res., **77**, 6133-6144.
- Sarma, Y. S., Sastry, T. S. and Sarma, S. V. S., 1982, Equatorial effects on nighttime Pc3 pulsations: Indian J. Radio & Space Phys., **11**, 29-32.
- Sastry, T. S., Sarma, Y. S., Sarma, S. V. S., and Sanker Narayan, P. V., 1983, Day-time *Pi* pulsations at equatorial latitudes: J. Atmos. Terr. Phys., **45**, 733-741.
- Schmucker, U., 1971, Interpretation of induction anomalies above nonuniform surface layers: Geophysics, **36**, 156-165.
- Schmucker, U., and Weidelt, P., 1975, Electromagnetic induction in the earth: Lecture Notes, Aarhus University (unpublished).
- Shawan, S. D., 1979, Magnetospheric plasma waves, in Lanzerotti, L. J., Kennel, C. F., and Parker, E. N., Eds, Solar system plasma physics, Vol. 3: North Holland Publishing Co., 211-270.
- Sheriff, R. E., 1984, Encyclopedic dictionary of exploration geophysics: Soc. Expl. Geophys.
- Shoemaker, C. L., Shoham, Y., and Hockey, R. L., 1986, Interpretation of natural source electromagnetic array data: 56th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded abstract EM2.3, 63-65.
- Shoemaker, C. L., Shoham, Y., and Hockey, R. L., 1989, Calibration case study of natural source electromagnetic array data recorded over a well in Oregon: Inst. Elect. and Electron. Eng. Proc., **77**, 334-337.
- Sims, W. E. and Bostick, F. X. Jr., 1969, Methods of magnetotelluric analysis: EGRL Tech. Report 58, Univ. of Texas, Austin.
- Sims, W. E., Bostick, F. X., and Smith, H. W., 1971, The estimation of magnetotelluric impedance tensor elements from measured data: Geophysics, **36**, 938-942.
- Singer, B. Sh., 1990, Allowance for static distortions in magnetotellurics: Review Paper, 10th Workshop on EM Induction in the Earth, Ensenada, Mex., IAGA WG I-2.
- Smith, J. T., 1988, Rapid inversion of multi-dimensional magnetotelluric data: Ph.D. Thesis, Univ. of Washington, Seattle.
- Southwood, D. J. and Hughes, W. J., 1978, Source induced vertical components in geomagnetic pulsation signals: Planet. Space Sci., **26**, 715-720.
- Spaulding, A. D., 1982, Atmospheric noise and its effects on telecommunication systems performance, in Volland, H., Ed., CRC Handbook of Atmospheric, Vol. 1: CRC Press, Inc., 289-328.
- Spies, B. R., 1989, Depth of investigation in electromagnetic sounding methods: Geophysics, **54**, 872-888.
- Spies, B. R., and Eggers, D. E., 1986, The use and misuse of apparent resistivity in electromagnetic methods: Geophysics, **51**, 1462-1471.
- Spitz, S., 1985, The magnetotelluric impedance tensor properties with respect to rotations: Geophysics, **50**, 1610-1617.
- Stanley, W. D., and Tinkler, R. D., 1982, A practical, low-noise coil system for magnetotellurics: Open-file rept., U.S. Geol. Survey.
- Sternberg, B. K., Buller, P. L., Kisabeth, J. L., and Mehreteab, E., 1982, Electrical methods for hydrocarbon exploration: II Magnetotelluric (MT) method, in Proc. Symp. on Unconventional Methods in Exploration for Petroleum and Natural Gas, Dallas, S.M.U. Press.
- Sternberg, B. K., Washburne, J. C., and Anderson, R. G., 1985, Investigation of MT static shift correction methods: 55th Ann. Internat. Mtg. Soc. Expl. Geophys., Extended abstract, 264-267.
- Stinson, K., Levy, S., and Oldenburg, D., 1986, Effects of near-surface conditions on the reflectivity imaging of magnetotelluric data: 56th Ann. Internat. Mtg., Soc. Expl. Geophysics., Expanded abstract EM2.6, 72-74.
- Stratton, J. A., 1941, Electromagnetic theory: McGraw-Hill.
- Sutarno, D. and Vozoff, K., 1989, Robust-M estimation of magnetotelluric impedance tensors: Expl. Geophys., **20**, 383-398.
- Sutarno, D. and Vozoff, K., 1991, An application of regression-M estimation with the Hilbert transform to magnetotelluric data processing: Geophysics, (submitted).
- Swift, C. M., Jr., 1967, A magnetotelluric investigation of an electrical conductivity anomaly in the southwestern United States: Ph.D. thesis, Mass. Inst. of Tech.
- , 1971, Theoretical magnetotelluric and Turam response from two-dimensional inhomogeneities: Geophysics, **36**, 38-52.
- Tikhonov, A. N., 1965, Mathematical basis of the theory of electromagnetic soundings: USSR Comp., Math. & Phys., 201-211.
- Ting, S. C., and Hohmann, G. W., 1981, Integral equation modeling of three-dimensional magnetotelluric response: Geophysics, **46**, 182-197.
- Torres-Verdin, C., 1985, Implications of the Born approximation for the magnetotelluric problem in three-dimensional environments: M.Sc. thesis, Univ. of Texas, Austin.
- Travis, B. J. and Chave, A. D., 1986, A moving finite element model for 2D magnetotelluric analysis: 8th Workshop on Electromagnetic Induction in the Earth and Moon, Neuchatel, Abstract T7.
- Vanyan, L. L., Berdichevsky, M. N., Vasin, N. D., Okulyesskiy, B. A., and Shilovskiy, P. P., 1980, On the normal geoelectric profile: Phys. Solid Earth, **16**, 131-133.
- Volland, H., 1982, Low frequency radio noise, in Volland, H., Ed., CRC Handbook of Atmospheric, Vol. 1: CRC Press, Inc., 179-250.
- Vozoff, K., 1971, The effect of overburden on vertical component anomalies in AFMAG and VLF exploration: Geophysics, **36**, 53-57.
- , 1972, The magnetotelluric method in the exploration of sedimentary basins: Geophysics, **37**, 98-141.
- , 1982, Superconducting gradiometer-magnetometer array for magnetotelluric logging; U.S. Patent No. 4,349,781.
- , 1984, Model study for the proposed magnetotelluric

- (MT) traverse in North India: *Tectonophysics*, **105**, 399–411.
- , 1986, Magnetotelluric methods: *Geophys. Reprint Ser. No. 5*, Soc. Expl. Geophys.
- Vozoff, K., Holcomb, H. L., and Holliday, J., 1982, Audio frequency magnetotellurics: A deep electromagnetic method for groundwater evaluation: Australian Water Resources Council Tech. Paper 76, Macquarie Univ., Sydney.
- Vozoff, K., LeBrocq, L., Moss, D., Zile, M., and Pridmore, D., 1985, Deep transient electromagnetic soundings for petroleum exploration: Final report, NERDDC Proj. 683, Macquarie Univ., Sydney.
- Wait, J. R., 1954, On the relation between telluric currents and the earth's magnetic field: *Geophysics*, **19**, 281–286.
- Wang Jia-ying, Oldenburg, D., and Levy, S., 1985, Seismic interpretation of magnetotelluric sounding data: *Petroleum Geophys. Expl.*, **20**, 65–79 (Chinese with English abstract).
- Wang Jia-ying, Zuo Hai-yan, Fang Shang, and Sun Jian-guo, 1986, Pseudoseismic interpretation to MT: Abstract, ISPCA Conf., Nanjing.
- Wannamaker, P. E., Hohmann, G. W., and San Filippo, W. A., 1984a, Electromagnetic modeling of three-dimensional bodies in layered earths using integral equations: *Geophysics*, **49**, 60–74.
- Wannamaker, P. E., Hohmann, G. W., and Ward, S. H., 1984b, Magnetotelluric responses of three-dimensional bodies in layered earths: *Geophysics*, **49**, 1517–1533.
- Wannamaker, P. E., Stodt, J. A., and Rijo, L., 1986, Two-dimensional topographic responses in magnetotellurics modelled using finite elements: *Geophysics*, **51**, 2131–2144.
- Warner, B. N., Bloomquist, M. G., and Griffith, P. G., 1983, Magnetotelluric interpretations based upon new processing and display techniques: 53rd Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded abstract MT4, 151–154.
- Watt, A. D., 1967, VLF radio engineering: Pergamon Press.
- Weidelt, P., 1972, The inverse problem of geomagnetic induction: *J. Geophys.*, **38**, 257–289.
- , 1975, Inversion of two-dimensional conductivity structures: *Phys. Earth Planet. Inter.*, **10**, 282–291.
- Weidelt, P., Müller, W., Losecke, W., and Knodel, K., 1980, Die Bostick transformation: in Haak, V., and Homilius, J., Eds., *Protokoll über das Kolloquium 'Erdmagnetische Tiefenforschung'*.
- Weinstock, H. and Overton, W. C., Jr., 1981, SQUID applications to geophysics: *Soc. Expl. Geophys.*
- West, G. F., and Edwards, R. N., 1985, A simple parametric model for the electromagnetic response of an anomalous body in a host medium: *Geophysics*, **50**, 2542–2557.
- Whittall, K. P., and Oldenburg, D. W., 1986, Inversion of magnetotelluric data using a practical inverse scattering formulation: *Geophysics*, **51**, 383–395.
- Wight, D. E. and Bostick, F. X., 1981, Cascade decimation—A technique for real time estimation of power spectra: *Proc. Inst. Elect. and Electron. Eng. ICASSP*, 626–629.
- Word, D. R., Smith, H. W., and Bostick, F. X., Jr., 1971, Crustal investigation of the magnetotelluric tensor impedance method, in *The structure and physical properties of the earth's crust*: Am. Geophys. Union, Mono. 14, Washington, D.C., 145–167.
- Word, D. R., Goss, R., and Chambers, D. M., 1986, An EMAP case study: 56th Ann. Internat. Mtg., Soc. Expl. Geophys., Extended abstract EM2.2, 61–63.
- Yamashita, M., Hallof, P. G., and Pelton, W. H., 1985, CSAMT case histories with a multichannel CSAMT system and near-field data correction: 55th Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded abstract MT8.
- Yee, E. and Paulson, K. V., 1987, The canonical decomposition and its relationship to other forms of magnetotelluric impedance tensor analysis: *J. Geophys.*, **61**, 173–189.
- Yee, E. and Paulson, K. V., 1988, Concerning dispersion relations for the magnetotelluric impedance tensor: *Geophys. J.*, **95**, 549–559.
- Young, C. T., 1981, Principles of the Phoenix Geophysics real time remote reference magnetotelluric system, Version 3.2: Phoenix Geophysics, Denver (unpublished).
- Zhang, P., Roberts, R. G., and Pedersen, L. B., 1987, Magnetotelluric strike rules: *Geophysics*, **52**, 267–278.
- Zile, M., Vozoff, K., and Holcombe, L. A., 1983, AMIRA AMT project report: CGER Macquarie Univ., Sydney (unpublished).
- Zimmerman, J. E. and Campbell, W. H., 1975, Tests of cryogenic SQUID for geomagnetic field measurements: *Geophysics*, **40**, 269–284.
- Zonge, K. L. and Hughes, L. J., 1991, Controlled source audio-frequency magnetotellurics: in Nabighian, M. N., Ed., *Electromagnetic methods in applied geophysics*, Vol. II.

## REFERENCES FOR GENERAL READING

- Allan, W., and Poulter, E. M., 1984, The spatial structure of different pulsation types: A review of STARE radar results: *Rev. Geophys. Space Phys.*, **22**, 85–97.
- Bloomquist, M. G., and Warner, B. N., 1983, Magnetotelluric survey of the Pescadito Dome near Laredo, Texas: 53rd Ann. Internat. Mtg., Soc. Expl. Geophys., Expanded abstract RW3.3, 643–646.
- Campbell, W. H., 1965, Equatorial studies of rapid fluctuations in the earth's magnetic field: in de Mendonça, F., Ed., *Proc. 2nd Int'l. Symp. on Equatorial Aeronomy*, Internat. Union Geodesy and Geophys., 495–511.
- Cerv, V., and Pek, J., 1981, Numerical solution of the two-dimensional inverse geomagnetic induction problem: *Studia geoph. et geod.*, **25**, 69–80.
- Clarke, J., Gamble, T. D., and Goubau, W. M., 1978, Squids and magnetotellurics with a remote reference: in Symposium on applications of superconducting quantum interference devices, Univ. of Virginia.
- Clerc, G., 1970, Shallow magnetotelluric prospecting using electromagnetic variations at frequencies over the range between 8 Hz and 16 kHz: *C. R. Acad. Sc., Paris*, **270**, 645–648 (in French).
- Fischer, G., Schnegg, P.-A., Peguiron, M., and LeQuang, B. V., 1981, An analytic one-dimensional magnetotelluric inversion scheme: *Geophys. J. Roy. Astr. Soc.*, **67**, 257–278.
- Grillot, L. R., 1975, Calculation of the magnetotelluric tensor impedance: Analysis of band-limited MT signal pairs: *Geophysics*, **40**, 790–797.
- Jacobs, J. A., 1970, *Geomagnetic micropulsations*: Springer Verlag.
- Levy, B. C., 1985, Layer by layer reconstruction methods for the earth resistivity from direct current measurements: *Inst. Elect. and Electron. Eng. Trans. Geosci. Remote Sensing*, **GE-23**, 841–850.
- Matsushita, S., and Campbell, W. H., 1967, *Physics of geomagnetic phenomena*: Academic Press.
- Parker, R. L., 1977, Understanding inverse theory: *Ann. Rev. Earth Planet. Sci.*, **5**, 35–64.
- Parker, R. L., 1980, The inverse problem of electromagnetic induction: existence and construction of solutions based on incomplete data: *J. Geophys. Res.*, **85B**, 4421–4428.
- Parker, R. L. and Whaler, K. A., 1981, Numerical methods for establishing solutions to the inverse problem of magnetic induction: *J. Geophys. Res.*, **86B**, 9574–9584.
- Rokityansky, I. I., 1982, *Geoelectromagnetic investigation of the earth's crust and mantle*: Springer-Verlag.
- Schmucker, U., 1980, Induktion in geschichteten Halbräumen durch inhomogene Felder, in Haak, V. and Homilius, J., Eds.: *Protokoll über das Kolloquium Elektromagnetische Tiefenforschung*, Berlin, (unpublished).
- Weidelt, P., 1985, Construction of conductance bounds from magnetotelluric impedances: *J. Geophys.*, **57**, 191–206.

